



**IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements**

Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

Amendment 5: Enhancements for Next Generation V2X

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.11bd™-2022

(Amendment to IEEE Std 802.11™-2020
as amended by IEEE Std 802.11ax™-2021,
IEEE Std 802.11ay™-2021,
IEEE Std 802.11ba™-2021,
IEEE Std 802.11™-2020/Cor 1-2022,
and IEEE Std 802.11az™-2022)

IEEE Std 802.11bd™-2022
(Amendment to IEEE Std 802.11™-2020
as amended by IEEE Std 802.11ax™-2021,
IEEE Std 802.11ay™-2021,
IEEE Std 802.11ba™-2021,
IEEE Std 802.11™-2020/Cor 1-2022,
and IEEE Std 802.11az™-2022)

**IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements**

**Part 11: Wireless LAN Medium Access Control
(MAC) and Physical Layer (PHY) Specifications**

**Amendment 5:
Enhancements for Next Generation V2X**

Developed by the
LAN/MAN Standards Committee
of the
IEEE Computer Society

Approved 3 December 2022
IEEE SA Standards Board

Abstract: This amendment defines modifications to both the IEEE 802.11 physical layer (PHY) and the medium access control (MAC) sublayer for next generation vehicle-to-everything (V2X) communication in the 5.9 GHz and 60 GHz frequency bands.

Keywords: amendment, IEEE 802.11™, IEEE 802.11bd™, MAC, medium access control, OCB, outside the context of BSS, PHY, physical layer, V2X, vehicular communication, vehicle-to-everything communication, wireless local area network, WLAN

The Institute of Electrical and Electronics Engineers, Inc.
3 Park Avenue, New York, NY 10016-5997, USA

Copyright © 2023 by the Institute of Electrical and Electronics Engineers, Inc.
All rights reserved. Published 10 March 2023. Printed in the United States of America.

IEEE and 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by the Institute of Electrical and Electronics Engineers, Incorporated.

PDF: ISBN 978-1-5044-9521-9 STD26019
Print: ISBN 978-1-5044-9522-6 STDPD26019

IEEE prohibits discrimination, harassment and bullying.

For more information, visit <http://www.ieee.org/web/aboutus/whatis/policies/p9-26.html>.

No part of this publication may be reproduced in any form, in an electronic retrieval system or otherwise, without the prior written permission of the publisher.

Important Notices and Disclaimers Concerning IEEE Standards Documents

IEEE Standards documents are made available for use subject to important notices and legal disclaimers. These notices and disclaimers, or a reference to this page (<https://standards.ieee.org/ipr/disclaimers.html>), appear in all standards and may be found under the heading “Important Notices and Disclaimers Concerning IEEE Standards Documents.”

Notice and Disclaimer of Liability Concerning the Use of IEEE Standards Documents

IEEE Standards documents are developed within IEEE Societies and subcommittees of IEEE Standards Association (IEEE SA) Board of Governors. IEEE develops its standards through an accredited consensus development process, which brings together volunteers representing varied viewpoints and interests to achieve the final product. IEEE Standards are documents developed by volunteers with scientific, academic, and industry-based expertise in technical working groups. Volunteers are not necessarily members of IEEE or IEEE SA and participate without compensation from IEEE. While IEEE administers the process and establishes rules to promote fairness in the consensus development process, IEEE does not independently evaluate, test, or verify the accuracy of any of the information or the soundness of any judgments contained in its standards.

IEEE makes no warranties or representations concerning its standards, and expressly disclaims all warranties, express or implied, concerning this standard, including but not limited to the warranties of merchantability, fitness for a particular purpose and non-infringement. In addition, IEEE does not warrant or represent that the use of the material contained in its standards is free from patent infringement. IEEE standards documents are supplied “AS IS” and “WITH ALL FAULTS.”

Use of an IEEE standard is wholly voluntary. The existence of an IEEE Standard does not imply that there are no other ways to produce, test, measure, purchase, market, or provide other goods and services related to the scope of the IEEE standard. Furthermore, the viewpoint expressed at the time a standard is approved and issued is subject to change brought about through developments in the state of the art and comments received from users of the standard.

In publishing and making its standards available, IEEE is not suggesting or rendering professional or other services for, or on behalf of, any person or entity, nor is IEEE undertaking to perform any duty owed by any other person or entity to another. Any person utilizing any IEEE Standards document, should rely upon his or her own independent judgment in the exercise of reasonable care in any given circumstances or, as appropriate, seek the advice of a competent professional in determining the appropriateness of a given IEEE standard.

IN NO EVENT SHALL IEEE BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO: THE NEED TO PROCURE SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE PUBLICATION, USE OF, OR RELIANCE UPON ANY STANDARD, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE AND REGARDLESS OF WHETHER SUCH DAMAGE WAS FORESEEABLE.

Translations

The IEEE consensus development process involves the review of documents in English only. In the event that an IEEE standard is translated, only the English version published by IEEE is the approved IEEE standard.

Official statements

A statement, written or oral, that is not processed in accordance with the IEEE SA Standards Board Operations Manual shall not be considered or inferred to be the official position of IEEE or any of its committees and shall not be considered to be, nor be relied upon as, a formal position of IEEE. At lectures, symposia, seminars, or educational courses, an individual presenting information on IEEE standards shall make it clear that the presenter's views should be considered the personal views of that individual rather than the formal position of IEEE, IEEE SA, the Standards Committee, or the Working Group. Statements made by volunteers may not represent the formal position of their employer(s) or affiliation(s).

Comments on standards

Comments for revision of IEEE Standards documents are welcome from any interested party, regardless of membership affiliation with IEEE or IEEE SA. However, **IEEE does not provide interpretations, consulting information, or advice pertaining to IEEE Standards documents.**

Suggestions for changes in documents should be in the form of a proposed change of text, together with appropriate supporting comments. Since IEEE standards represent a consensus of concerned interests, it is important that any responses to comments and questions also receive the concurrence of a balance of interests. For this reason, IEEE and the members of its Societies and subcommittees of the IEEE SA Board of Governors are not able to provide an instant response to comments, or questions except in those cases where the matter has previously been addressed. For the same reason, IEEE does not respond to interpretation requests. Any person who would like to participate in evaluating comments or in revisions to an IEEE standard is welcome to join the relevant IEEE working group. You can indicate interest in a working group using the Interests tab in the Manage Profile & Interests area of the [IEEE SA myProject system](#).¹ An IEEE Account is needed to access the application.

Comments on standards should be submitted using the [Contact Us](#) form.²

Laws and regulations

Users of IEEE Standards documents should consult all applicable laws and regulations. Compliance with the provisions of any IEEE Standards document does not constitute compliance to any applicable regulatory requirements. Implementers of the standard are responsible for observing or referring to the applicable regulatory requirements. IEEE does not, by the publication of its standards, intend to urge action that is not in compliance with applicable laws, and these documents may not be construed as doing so.

Data privacy

Users of IEEE Standards documents should evaluate the standards for considerations of data privacy and data ownership in the context of assessing and using the standards in compliance with applicable laws and regulations.

¹ Available at: <https://development.standards.ieee.org/myproject-web/public/view.html#landing>.

² Available at: <https://standards.ieee.org/content/ieee-standards/en/about/contact/index.html>.

Copyrights

IEEE draft and approved standards are copyrighted by IEEE under US and international copyright laws. They are made available by IEEE and are adopted for a wide variety of both public and private uses. These include both use, by reference, in laws and regulations, and use in private self-regulation, standardization, and the promotion of engineering practices and methods. By making these documents available for use and adoption by public authorities and private users, neither IEEE nor its licensors waive any rights in copyright to the documents.

Photocopies

Subject to payment of the appropriate licensing fees, IEEE will grant users a limited, non-exclusive license to photocopy portions of any individual standard for company or organizational internal use or individual, non-commercial use only. To arrange for payment of licensing fees, please contact Copyright Clearance Center, Customer Service, 222 Rosewood Drive, Danvers, MA 01923 USA; +1 978 750 8400; <https://www.copyright.com/>. Permission to photocopy portions of any individual standard for educational classroom use can also be obtained through the Copyright Clearance Center.

Updating of IEEE Standards documents

Users of IEEE Standards documents should be aware that these documents may be superseded at any time by the issuance of new editions or may be amended from time to time through the issuance of amendments, corrigenda, or errata. An official IEEE document at any point in time consists of the current edition of the document together with any amendments, corrigenda, or errata then in effect.

Every IEEE standard is subjected to review at least every 10 years. When a document is more than 10 years old and has not undergone a revision process, it is reasonable to conclude that its contents, although still of some value, do not wholly reflect the present state of the art. Users are cautioned to check to determine that they have the latest edition of any IEEE standard.

In order to determine whether a given document is the current edition and whether it has been amended through the issuance of amendments, corrigenda, or errata, visit [IEEE Xplore](#) or [contact IEEE](#).³ For more information about the IEEE SA or IEEE's standards development process, visit the IEEE SA Website.

Errata

Errata, if any, for all IEEE standards can be accessed on the [IEEE SA Website](#).⁴ Search for standard number and year of approval to access the web page of the published standard. Errata links are located under the Additional Resources Details section. Errata are also available in [IEEE Xplore](#). Users are encouraged to periodically check for errata.

Patents

IEEE Standards are developed in compliance with the [IEEE SA Patent Policy](#).⁵

Attention is called to the possibility that implementation of this standard may require use of subject matter covered by patent rights. By publication of this standard, no position is taken by the IEEE with respect to the existence or validity of any patent rights in connection therewith. If a patent holder or patent applicant has

³ Available at: <https://ieeexplore.ieee.org/browse/standards/collection/ieee>.

⁴ Available at: <https://standards.ieee.org/standard/index.html>.

⁵ Available at: <https://standards.ieee.org/about/sasb/patcom/materials.html>.

filed a statement of assurance via an Accepted Letter of Assurance, then the statement is listed on the IEEE SA Website at <https://standards.ieee.org/about/sasb/patcom/patents.html>. Letters of Assurance may indicate whether the Submitter is willing or unwilling to grant licenses under patent rights without compensation or under reasonable rates, with reasonable terms and conditions that are demonstrably free of any unfair discrimination to applicants desiring to obtain such licenses.

Essential Patent Claims may exist for which a Letter of Assurance has not been received. The IEEE is not responsible for identifying Essential Patent Claims for which a license may be required, for conducting inquiries into the legal validity or scope of Patents Claims, or determining whether any licensing terms or conditions provided in connection with submission of a Letter of Assurance, if any, or in any licensing agreements are reasonable or non-discriminatory. Users of this standard are expressly advised that determination of the validity of any patent rights, and the risk of infringement of such rights, is entirely their own responsibility. Further information may be obtained from the IEEE Standards Association.

IMPORTANT NOTICE

IEEE Standards do not guarantee or ensure safety, security, health, or environmental protection, or ensure against interference with or from other devices or networks. IEEE Standards development activities consider research and information presented to the standards development group in developing any safety recommendations. Other information about safety practices, changes in technology or technology implementation, or impact by peripheral systems also may be pertinent to safety considerations during implementation of the standard. Implementers and users of IEEE Standards documents are responsible for determining and complying with all appropriate safety, security, environmental, health, and interference protection practices and all applicable laws and regulations.

Participants

At the time this amendment was sent to the IEEE SA Board for approval, the IEEE 802.11 Working Group had the following members:

Dorothy Stanley, *Chair*
Jon W. Rosdahl, *1st Vice Chair*
Robert Stacey, *2nd Vice Chair*
Stephen McCann, *Secretary*

Bo Sun, *IEEE 802.11bd Task Group Chair*
Hongyuan Zhang, *IEEE 802.11bd Task Group Vice Chair*
Joseph S. Levy, *IEEE 802.11bd Task Group Vice Chair*
Yan Zhang, *IEEE 802.11bd Task Group Secretary*
James Lepp, *IEEE 802.11bd Task Group Former Secretary*
Yujin Noh, *IEEE 802.11bd Task Group Technical Editor*
Bahareh Sadeghi, *IEEE 802.11bd Task Group Former Technical Editor*

Rana Abdelaal
Mohamed Abouelseoud
Osama S. Aboulmagd
Tomoko Adachi
Olubukola Adakeja
Shubhodeep Adhikari
Peyush Agarwal
Kosuke Aio
Abdel Karim Ajami
Dmitry Akhmetov
Carlos H. Aldana
Vamsi Amalladinne
Song-Haur An
Amelia Andersdotter
Carol Ansley
Diego Arlandis
Antonio Arregui
Yusuke Asai
Alfred Asterjadhi
Oscar Au
Ziv Avital
Geert A. Awater
Mehmet Aygul
Agustin Badenes
SunHee Baek
Christy Bahn
Eugene Baik
Gabor Bajko
Hari Ram Balakrishnan
Raja Banerjea
Dmitry Bankov
stephane Baron
David Barr
Anuj Batra
Tuncer Baykas
Chris Beg
Firouz Behnamfar
Jianwei Bei
Yaron Ben Arie
Friedbert Berens
Christian Berger

Alan Berkema
Harry Bims
Andreas Bluschke
Lennert Bober
David Boldy
Daniel Borges
Daniel Bravo
Albert Bredewoud
Frank Burkhardt
Radhakrishna Canchi
Rui Cao
Laurent Cariou
William Carney
Edward Carter
Dave Cavalcanti
Gurkan Cepni
Chen-Yi Chang
Matthew Chappell
Kirill Chemrov
Cheng Chen
Evelyn Chen
Xiaogang Chen
You-Wei Chen
Paul Cheng
Phoebe Cheng
Xilin Cheng
George Cherian
James Chiang
Rojan Chitrakar
Lin-Kai Chiu
Shi Baw Chng
Hangyu Cho
Jinsoo Choi
Seungho Choo
Liwen Chu
Jinyoung Chun
Bruce Chung
Chulho Chung
Dana Ciochina
John Coffey
Carlos Cordeiro

Diana Cortes
Claudio da Silva
Dibakar Das
Debashis Dash
Leonard Dauphinee
Rolf J. de Vegt
Antonio DeLaOlivaDelgado
Thomas Derham
Patrice Desmoulin
Rocco Di Taranto
Yanyi Ding
Ali Dogukan
Mingjie Dong
Xiandong Dong
Rui Du
Zhenguo Du
Peter Ecclesine
Richard Edgar
Alecsander Eitan
Ahmed ElSherif
Marc Emmelmann
Vinko Erceg
Serhat Erkucuk
Shuang Fan
Juan Fang
Yonggang Fang
Shuling Feng
David Ferruz
Matthew J. Fischer
Ming Gan
Trivikram Gangur
Ning Gao
Lalit Garg
Thomas Gee
Alireza Ghaderipoor
Chittabrata Ghosh
Sachin Godbole
Bo Gong
Hemamali Gorthi
Fumihide Goto
Niranjan Grandhe
Michael Grigat

Xiangxin Gu	Sundeep Kancherla	Yong Liu
Romain Guignard	Srinivas Kandala	Zhiqun Liu
Jing Guo	HaoHua Kang	Vyacheslav Loginov
Yuchen Guo	Sugbong Kang	Miguel Lopez
Binita Gupta	Sudhir Kasargod	Mikael Lorgeoux
Raghvendra Gupta	Assaf Y. Kasher	Hanqing Lou
Luis Gutierrez	Oren Kedem	Hui-Ling Lou
Muhammad Kumail Haider	Richard H. Kennedy	Kaiying Lu
David Halasz	John Kenney	Liuming Lu
Mark Hamilton	Stuart J. Kerry	Kurt Lumbatis
Xiao Han	Naseem Khan	Chaoming Luo
Zhiqiang Han	Evgeny Khorov	Lily Lv
Thomas Handte	Jeongki Kim	Mikhail Lyakh
Vlado Handziski	Myeong-Jin Kim	Li Ma
Christopher J. Hansen	Sang Gook Kim	Mengyao Ma
Daniel N. Harkins	Sanghyun Kim	Dmitrii Malichenko
Edward Harrison	Yongho Kim	Jouni K. Malinen
Brian D. Hart	Youhan Kim	Hiroshi Mano
Victor Hayes	Akira Kishida	Marcos Martinez Vazquez
Chuanfeng He	Shoichi Kitazawa	Sebastian Max
Jerome Henry	Arik Klein	Daniele Medda
Marco Hernandez	Andrey Klimakov	Morteza Mehrnough
Lili hervieu	Jarkko Knecht	Ezer Melzer
Guido R. Hiertz	Geonjung Ko	Ebubekir Memisoglu
Ryuichi Hirata	Prashant Kota	Dignus-Jan Moelker
Duncan Ho	Michail Koundourakis	Bibhu Mohanty
Oliver Holland	Aleksandra Kozarev	Pooya Monajemi
Ching-Wen Hsiao	Bruce P. Kraemer	Michael Montemurro
Hung-Tao Hsieh	Chung-Ta Ku	Leo Montreuil
Chien-Fang Hsu	Manish Kumar	Juseong Moon
Ostrovsky Hsu	Warren Kumari	Hitoshi Morioka
Chunyu Hu	Chih-Chun Kuo	HiroYuki Motozuka
Mengshi Hu	Aleksey Kureev	Robert Mueller
Shengquan Hu	Massinissa Lalam	Lakshmi Narayana Mukkapati
Chihan Huang	Zhou Lan	Okan Mutgan
Guogang Huang	Leonardo Lanante	Andrew Myles
Lei Huang	James Lansford	Kengo Nagata
Po-Kai Huang	Brice Le Houerou	Gaurang Naik
Qisheng Huang	Hong Won Lee	Hiroki Nakano
Kazi Mohammed Saidul Huq	Hyeong Ho Lee	Junyoung Nam
Sung Hyun Hwang	Nancy Lee	Vamadevan Nambodiri
Ahmed Ibrahim	Wookbong Lee	Nima Namvar
Tetsushi Ikegami	Chris Levesque	Sai Shankar Nandagopalan
Hirohiko Inohiza	Ilya Levitsky	Narengerile Narengerile
Salvador Iranzo	Joseph S. Levy	Sharan Naribole
Insun Jang	Bo Li	Karim Nassiri Toussi
Timothy Jeffries	Jialing Li	Peshal Nayak
Eunsung Jeon	Qinghua Li	Patrice Nezou
Chenhe Ji	Yan Li	Boon Loong Ng
Jia Jia	Yapu Li	An Nguyen
Feng Jiang	Yiqing Li	Hiraku Okada
Jinjing Jiang	Yunbo Li	Hassan Omar
Hanjin Joh	Dong Guk Lim	Stephen Orr
Allan Jones	Hsin-De Lin	Masatomo Ouchi
Vincent Knowles Jones IV	Wei Lin	Satoshi Oyama
Jorge Juarez	Yousi Lin	Basak Ozbakis
Volker Jungnickel	Zinan Lin	Saju Palayur
Ishaque Ashar Kadampot	Erik Lindskog	Stephen Palm
Carl W. Kain	Chenchen Liu	Sheetal Pandey
Naveen K. Kakani	Der-Zheng Liu	Thomas Pare
Manoj Kamath	Jianhan Liu	Eunsung Park
Mahmoud Kamel	Ying Liu	Minyoung Park
		Sungjin Park

Abhishek Patil
 Gaurav Patwardhan
 Lan Peng
 Ronny Peng
 Eldad Perahia
 James E. Petranovich
 Albert Petrick
 Brian Petry
 Charlie Pettersson
 Ron Porat
 Dinakar Prabhakaran
 Henry S. Ptasinski
 Srinath Puducheri
 Rethnakaran Pulikkoonattu
 Rajat Pushkarna
 Emily H. Qi
 Saira Rafique
 Kapil Rai
 Alireza Raissinia
 Enrico-Henrik Rantala
 Vishnu Ratnam
 Oded Redlich
 Kiran Rege
 Dror Regev
 Mor Reich
 Meriam Rezk
 Maximilian Riegel
 Mark Rison
 Joerg Robert
 Craig Rodine
 Benjamin Rolfe
 Sayak Roy
 Kiseon Ryu
 Anirudha Sahoo
 Mohamed Salem
 Hanadi Salman
 Sam Sambasivan
 Stephan Sand
 Ioannis Sarris
 Naotaka Sato
 Takuhiro Sato
 Sigurd Schelstraete
 Martin Schmidhammer
 Benedikt Schweizer
 Andy Scott
 Jonathan Segev
 Sangho Seo
 Joseph Seok
 Yongho Seok
 Nikola Serafimovski
 Ankit Sethi
 Julien Sevin
 Rubayet Shafin
 Amit Shaw
 Andy Shen
 Xiaoman Shen
 Ian Sherlock

Shimi Shilo
 Atsushi Shirakawa
 Tongxin Shu
 Ulrich Sinn
 Graham Kenneth Smith
 Luther Smith
 Ju-Hyung Son
 Hao Song
 Ayush Sood
 Robert Sosack
 Sudhir Srinivasa
 Veena Srivatsa
 Adrian P. Stephens
 Noel Stott
 Paul Strauch
 Hang Su
 Hongjia Su
 Jung Hoon Hoon Suh
 Takenori Sumi
 Li-Hsiang Sun
 Sheng Sun
 Yanjun Sun
 Dennis Sundman
 Frank Suraci
 Szymon Szott
 Shivkumar Tadahal
 Mineo Takai
 Danny Tan
 Yusuke Tanaka
 Rakesh Taori
 Darshak Thakore
 Sidharth Thakur
 Sri Ramya Thota
 Francois Thoumy
 Bin Tian
 Alexander Tolpin
 Alejandro Torrijo
 Solomon B. Trainin
 Tsung-Han Tsai
 Genadiy Tsodik
 Yuki Tsujimaru
 Chunjiang Tu
 Halise Turkmen
 Kiran Uln
 Paul Unterhuber
 Yoshio Urabe
 Richard D. J. Van Nee
 Allert Van Zelst
 Prabodh Varshney
 Daniel Verenzuela
 Lochan Verma
 Sindhu Verma
 Sameer Vermani
 Santiago Vicent Colonques
 Pascal Viger
 Chao Chun Wang
 Hao Wang

Hejun Wang
 Huizhao Wang
 Lei Wang
 Qi Wang
 Shukun Wang
 Steven Qi Wang
 Xiaofei Wang
 Zisheng Wang
 Roy Want
 Lisa Ward
 Dong Wei
 Matthias Wendt
 Menzo M. Wentink
 Leif Wilhelmsson
 Hao Wu
 Kanke Wu
 Tianyu Wu
 Wayne Wu
 John Wullert
 Qing Xia
 Bo Xiao
 Liangxiao Xin
 Yan Xin
 Fangxin Xu
 Weijie Xu
 Yanchao Xu
 Hassan Yaghoobi
 Ryota Yamada
 Aiguo Yan
 Zhongjiang Yan
 Bo Yang
 Jay Yang
 Lin Yang
 Mao Yang
 Rui Yang
 Steve Ts Yang
 Xun Yang
 Kazuto Yano
 James Yee
 Peter Yee
 Yongjiang Yi
 Takahiro Yokoyama
 Su Khiong Yong
 Homin Yoo
 Jeonghwan Yoon
 Yuki Yoshikawa
 Christopher Young
 Jian Yu
 Fangchao Yuan
 Salah Eddine Zegrar
 Ruochen Zeng
 Jiayi Zhang
 Jiayin Zhang
 Lei Zhou
 Pei Zhou
 Renlong Zhou
 Juan Carlos Zuniga
 Zhisong Zuo

The following members of the individual balloting committee voted on this revision. Balloters may have voted for approval, disapproval, or abstention.

Osama S. Aboulmagd	David Hunter	Bansi Patel
Tomoko Adachi	Tetsushi Ikegami	Abhishek Patil
Iwan Adhicandra	Yasuhiko Inoue	Arumugam Paventhan
Carlos H. Aldana	Raj Jain	Albert Petrick
Thomas Alexander	SangKwon Jeong	David Piehler
Song-Haur An	Pranav Jha	Clinton Powell
Amelia Andersdotter	Joe Natharoj Juisai	Venkatesha Prasad
Carol Ansley	Carl W. Kain	Kapil Rai
Butch Anton	Sugbong Kang	Maximilian Riegel
Arthur Arrington	Piotr Karocki	Benjamin Rolfe
Alfred Asterjadhi	John Kenney	Jon W. Rosdahl
Kwok Shum Au	Stuart J. Kerry	Stephan Sand
Oscar Au	Evgeny Khorov	Naotaka Sato
Tuncer Baykas	Yongbum Kim	Michael Seaman
Friedbert Berens	Youhan Kim	Jonathan Segev
Harry Bims	Shoichi Kitazawa	Robert Sosack
Vern Brethour	Jarkko Knecht	Robert Stacey
Koti Reddy Butukuri	Jan Kruys	Dorothy Stanley
William Byrd	Yasushi Kudoh	Noel Stott
Radhakrishna Canchi	Thomas Kurihara	Walter Struppler
Rui Cao	Gavin Lai	Gerald Stueve
Paul Cardinal	James Lansford	Mark Sturza
William Carney	Hyeong Ho Lee	Bo Sun
Pin Chang	James Lepp	Jasja Tijnink
Diana Cortes	Joseph S. Levy	Payam Torab Jahromi
Claudio Da Silva	Jialing Li	Solomon B. Trainin
Di Dieter Dieter	Qinghua Li	Prabodh Varshney
Richard Edgar	Daozhuang Lin	John Vergis
Alecsander Eitan	Jouni K. Malinen	Lochan Verma
Marc Emmelmann	Stephen McCann	Lei Wang
Ramy Fathy	Jonathon McLendon	Xiaofei Wang
Xiang Feng	Michael Montemurro	Lisa Ward
Avraham Freedman	Hiroyuki Motozuka	Stephen Webb
Ming Gan	Ronald Murias	Dong Wei
Devon Gayle	Rick Murphy	Matthias Wendt
James Gilb	Rajesh Murthy	Scott Willy
Rainer Hach	Andrew Myles	Andreas Wolf
Xiang He	Gaurang Naik	Rui Yang
Jerome Henry	N. Kishor Narang	James Yee
Marco Hernandez	Karim Nassiri Toussi	Yu Yuan
Guido R. Hiertz	John Notor	Oren Yuen
Werner Hoelzl	Satoshi Oyama	

When the IEEE SA Standards Board approved this amendment on 3 December 2022, it had the following membership:

David J. Law, *Chair*
Ted Burse, *Vice Chair*
Gary Hoffman, *Past Chair*
Konstantinos Karachalios, *Secretary*

Edward A. Addy	Johnny Daozhuang Lin	Mark Siira
Ramy Ahmed Fathy	Kevin Lu	Dorothy V. Stanley
J. Travis Griffith	Daleep C. Mohla	Lei Wang
Guido R. Hiertz	Andrew Myles	F. Keith Waters
Yousef Kimiagar	Damir Novosel	Karl Weber
Joseph L. Koepfinger*	Annette D. Reilly	Sha Wei
Thomas Koshy	Robby Robson	Philip B. Winston
John D. Kulick	Jon Walter Rosdahl	Daidi Zhong

*Member Emeritus

Introduction

This introduction is not part of IEEE Std 802.11bd-2022, IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications—Amendment 5: Enhancements for Next Generation V2X.

This amendment defines modifications to both the IEEE 802.11 physical layer (PHY) and the medium access control (MAC) sublayer for next generation vehicle-to-everything (V2X) communication in the 5.9 GHz and 60 GHz frequency bands.

Contents

3.	Definitions, acronyms, and abbreviations.....	18
3.2	Definitions specific to IEEE Std 802.11	18
3.4	Abbreviations and acronyms	19
4.	General description	19
4.3	Components of the IEEE Std 802.11 architecture	19
4.3.17	STA transmission of Data frames outside the context of a BSS	19
4.3.17a	Next generation vehicle-to-everything (V2X) (NGV) STA	19
4.3.22	DMG STA.....	20
5.	MAC service definition	23
5.2	MAC data service specification	23
5.2.2a	NGV MAC data service specification	23
5.2.3	MA-UNITDATA.request	23
5.2.3.2	Semantics of the service primitive	23
5.2.4	MA-UNITDATA.indication	25
5.2.4.2	Semantics of the service primitive	25
5.2.5	MA-UNITDATA-STATUS.indication.....	27
5.2.5.2	Semantics of the service primitive	27
6.	Layer management.....	29
6.3	MLME SAP interface	29
6.3.126	Cancel transmissions of MSDUs	29
6.3.126.1	Introduction	29
6.3.126.2	MLME-CANCELTX.request	29
6.3.126.3	MLME-CANCELTX.confirm	29
6.3.127	NGV radio environment measurement.....	30
6.3.127.1	General	30
6.3.127.2	MLME-RADIOENVIRONMENT.request.....	30
6.3.127.3	MLME-RADIOENVIRONMENT.indication	31
6.3.128	DMG operation outside the context of a BSS.....	31
6.3.128.1	General	31
6.3.128.2	MLME-DMG-OCB-START.request.....	31
6.3.128.3	MLME-DMG-OCB-START.confirm.....	32
6.3.128.4	MLME-DMG-OCB-STOP.request.....	33
6.3.128.5	MLME-OCB-DMGDISCOVERY.indication.....	33
6.3.128.6	MLME-OCB-LINKSTATUS.indication	34
9.	Frame formats	36
9.2	MAC frame formats.....	36
9.2.4	Frame fields	36
9.2.4.7	Frame Body field	36
9.2.5	Duration/ID field (QoS STA)	40
9.2.5.1	General	40
9.3	Format of individual frame types.....	40
9.3.4	Extension frames.....	40

9.3.4.2	DMG Beacon	40
9.4	Management and Extension frame body components	41
9.4.1	Fields that are not elements	41
9.4.1.46	DMG Parameters field	41
9.4.2	Elements.....	41
9.4.2.1	General	41
9.4.2.298	Ranging Parameters element.....	42
9.4.2.308	DMG OCB element	42
9.5	Fields used in Management and Extension frame bodies and Control frames	43
9.5.3	Sector Sweep Feedback field.....	43
9.7	Aggregate MPDU (A-MPDU).....	43
9.7.1	A-MPDU format	43
9.7.3	A-MPDU contents	44
10.	MAC sublayer functional description.....	43
10.2	MAC architecture	43
10.2.3	Hybrid coordination function (HCF)	43
10.2.3.2	HCF contention based channel access (EDCA).....	43
10.3	DCF.....	43
10.3.6	Group addressed MPDU transfer procedure.....	43
10.12	A-MPDU operation.....	43
10.12.2	A-MPDU length limit rules	43
10.12.4	A-MPDU aggregation of group addressed Data frames	44
10.13	PPDU duration constraint	44
10.23	HCF.....	44
10.23.2	HCF contention based channel access (EDCA)	44
10.23.2.8	Multiple frame transmission in an EDCA TXOP	44
10.23.2.9	TXOP limits	44
10.25	Block acknowledgment (block ack)	44
10.25.1	Introduction.....	44
10.25.2	Setup and modification of the block ack parameters	45
11.	MLME	47
11.1	Synchronization	47
11.1.3	Maintaining synchronization	47
11.1.3.4	DMG beacon generation before establishment of a BSS.....	47
11.1.4	Acquiring synchronization, scanning	47
11.1.4.1	General	47
11.1.4.7	DMG discovery outside the context of a BSS	47
11.18	STAs communicating Data frames outside the context of a BSS.....	49
11.21	Wireless network management procedures	50
11.21.6	Fine timing measurement (FTM) procedure.....	50
11.21.6.4	Measurement exchange.....	50
11.27	DMG beamformed link and BSS maintenance.....	51
11.27.1	Beamformed link maintenance	51
11.27.1.1	General	51
12.	Security	53
12.13	Preassociation security negotiation.....	53
12.13.1	General.....	53
12.13.3	Key establishment with PASN authentication.....	53

12.13.3.1	Overview	53
12.13.3.2	PASN frame construction and processing	53
17.	Orthogonal frequency division multiplexing (OFDM) PHY specification	55
17.2	OFDM PHY specific service parameter list	55
17.2.2	TXVECTOR parameters.....	55
17.2.2.1	General	55
17.2.2.9	TXVECTOR CH_BANDWIDTH_IN_NON_NGV.....	55
17.2.2.10	TXVECTOR DYN_BANDWIDTH_IN_NON_NGV	55
17.2.3	RXVECTOR parameters	56
17.2.3.1	General	56
17.2.3.9	RXVECTOR CH_BANDWIDTH_IN_NON_NGV	56
17.2.3.10	RXVECTOR DYN_BANDWIDTH_IN_NON_NGV	56
17.3	OFDM PHY	56
17.3.2	PPDU format.....	56
17.3.2.2	Overview of the PPDU encoding process.....	56
17.3.5	DATA field.....	57
17.3.5.5	PHY DATA scrambler and descrambler	57
31.	Next Generation V2X (NGV) MAC specification	61
31.1	Introduction.....	61
31.2	Operation in the 5.9 GHz band	61
31.2.1	Coexistence with non-NGV STAs.....	61
31.2.2	Channel access and transmission methods for 20 MHz OCB transmission	61
31.2.3	A-MSDU operation, A-MPDU operation, and BA operation	63
31.2.4	NON_NGV_10 repetition transmission.....	64
31.2.5	Non-NGV duplication operation	64
31.3	Operation in the 60 GHz band	65
31.3.1	DMG beamforming outside the context of a BSS	65
31.4	NGV ranging.....	66
31.5	NGV MAC data service.....	67
31.6	Radio environment measurement	68
32.	Next Generation V2X (NGV) PHY specification	69
32.1	Introduction.....	69
32.1.1	Introduction to NGV PHY	69
32.1.2	Scope.....	69
32.1.3	NGV PHY functions	70
32.1.3.1	General	70
32.1.3.2	PHY layer management entity (PLME).....	70
32.1.3.3	Service specification method	70
32.1.4	PPDU Formats	70
32.2	NGV PHY service interface	70
32.2.1	Introduction.....	70
32.2.2	TXVECTOR and RXVECTOR.....	71
32.2.3	PHY CONFIG_VECTOR.....	75
32.2.4	Effects of CH_BANDWIDTH parameter on PPDU format.....	76
32.2.5	Support for NON_NGV format	76
32.2.5.1	General	76
32.2.5.2	Support for NON_NGV format	78
32.2.5.3	Repetition transmission of NON_NGV_10 PPDU	79

	32.2.5.4	TXSTATUS parameters.....	79
32.3		NGV PHY.....	80
	32.3.1	Introduction.....	80
	32.3.2	NGV PPDU format.....	80
	32.3.3	Transmitter block diagram.....	81
	32.3.4	Overview of the PPDU encoding process.....	82
	32.3.4.1	General	82
	32.3.4.2	Construction of L-STF	83
	32.3.4.3	Construction of the L-LTF	83
	32.3.4.4	Construction of the L-SIG and RL-SIG	83
	32.3.4.5	Construction of the NGV-SIG and RNGV-SIG	84
	32.3.4.6	Construction of NGV-STF.....	84
	32.3.4.7	Construction of NGV-LTF.....	85
	32.3.4.8	Construction of the Data field in an NGV PPDU	85
	32.3.5	NGV modulation and coding schemes	85
	32.3.6	Timing related parameters	86
	32.3.7	Mathematical description of signals	87
	32.3.7.1	Notation.....	87
	32.3.7.2	Subcarrier indices in use	87
	32.3.7.3	Transmitted signal.....	87
	32.3.7.4	Definition of tone rotation.....	90
	32.3.8	NGV preamble.....	91
	32.3.8.1	Introduction	91
	32.3.8.2	Cyclic shift.....	91
	32.3.8.3	L-STF definition	91
	32.3.8.4	L-LTF definition	92
	32.3.8.5	L-SIG definition	93
	32.3.8.6	RL-SIG definition	94
	32.3.8.7	NGV-SIG definition.....	94
	32.3.8.8	RNGV-SIG definition.....	96
	32.3.8.9	NGV-STF definition	96
	32.3.8.10	NGV-LTF definition	97
	32.3.9	Data field.....	101
	32.3.9.1	General	101
	32.3.9.2	SERVICE field.....	101
	32.3.9.3	Scrambler	102
	32.3.9.4	Coding.....	102
	32.3.9.5	Stream parser.....	102
	32.3.9.6	Constellation mapping	102
	32.3.9.7	Pilot subcarriers.....	102
	32.3.9.8	OFDM modulation.....	103
	32.3.9.9	Midambles.....	105
	32.3.9.10	Non-NGV duplicate transmission.....	105
	32.3.10	SU-MIMO.....	106
	32.3.11	Transmit specification.....	106
	32.3.11.1	Transmit spectrum mask	106
	32.3.11.2	Spectral flatness	107
	32.3.11.3	Transmit center frequency and symbol clock frequency tolerance.....	108
	32.3.11.4	Modulation accuracy	108
	32.3.12	Receiver specification.....	110
	32.3.12.1	General	110
	32.3.12.2	Receiver minimum input sensitivity	111
	32.3.12.3	Adjacent channel rejection.....	111
	32.3.12.4	Nonadjacent channel rejection	113

32.3.12.5	Receiver maximum input level	113
32.3.12.6	CCA sensitivity	113
32.3.13	NGV transmit procedure.....	114
32.3.14	NGV receive procedure	116
32.3.15	Regulatory requirements.....	120
32.3.16	NGV ranging NDP.....	120
32.4	NGV PLME	121
32.4.1	PLME-SAP	121
32.4.2	PHY MIB.....	122
32.4.3	TXTIME and PSDU_LENGTH calculation.....	122
32.4.4	NGV PHY	123
32.5	Parameters for NGV-MCSs	123
Annex B (normative) Protocol Implementation Conformance Statement (PICS) proforma.....		129
B.2	Abbreviations and special symbols.....	129
B.2.2	General abbreviations for Item and Support columns.....	129
B.4	PICS proforma—IEEE Std 802.11-2020.....	129
B.4.3	IUT configuration.....	129
B.4.24	DMG features	130
B.4.24.1	DMG MAC features.....	130
B.4.38	NGV features.....	130
B.4.38.1	NGV MAC features	130
B.4.38.2	NGV PHY features	131
B.4.38.3	NGV extended MAC service features.....	132
Annex C (normative) ASN.1 encoding of the MAC and PHY MIB		133
C.3	MIB detail	133
Annex D (normative) Regulatory references.....		145
D.2	Radio performance specifications.....	145
D.2.3	Transmit spectrum mask	145
Annex P (informative) Location and Time Difference accuracy test.....		147
P.3	Differential Distance Computation using Fine Timing Measurement frames.....	147

Tables

Table 5-1—PPDU format values	24
Table 5-2—Channel width values	25
Table 5-3—RSSI values	26
Table 9-25—Maximum data unit sizes (in octets) and durations (in microseconds)	37
Table 9-45—DMG Beacon frame body	40
Table 9-69—The BSS Type subfield when the Discovery mode field is 1	41
Table 9-92—Element IDs	41
Table 9-322a—Format and Bandwidth subfield	42
Table 17-1—TXVECTOR parameters	55
Table 17-2—RXVECTOR parameters	56
Table 17-7—Contents of the first 7 bits of the scrambling sequence	58
Table 17-8a—TXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV values	59
Table 17-9a—RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV values	59
Table 17-10a—DYN_BANDWIDTH_IN_NON_NGV values	59
Table 31-1—Maximum NGV MPDU length	63
Table 32-1—TXVECTOR and RXVECTOR parameters	71
Table 32-2— Interpretation of FORMAT, NON_NGV_MODULATION, and CH_BANDWIDTH parameters	76
Table 32-3—Mapping of the NGV PHY parameters for non-NGV operation	78
Table 32-4—TXSTATUS parameters	79
Table 32-5—Fields of the NGV PPDU	80
Table 32-6—Timing related constants	86
Table 32-7—Frequently used parameters	87
Table 32-8—Tone scaling factor and guard interval duration values for PHY fields	90
Table 32-9—CH_BANDWIDTH	90
Table 32-10—Fields in the NGV-SIG field	95
Table 32-11—Number of NGV-LTFs required for different numbers of spatial streams	97
Table 32-12—SERVICE field	101
Table 32-13—Spectrum mask data for 20 MHz channel spacing	107
Table 32-14—Maximum transmit spectral flatness deviations	108
Table 32-15—Allowed relative constellation error versus constellation size and coding rate	109
Table 32-16—Receiver minimum input level sensitivity	111
Table 32-17—Minimum required adjacent and nonadjacent channel rejection level	112
Table 32-18—Optional enhanced minimum required adjacent and nonadjacent channel rejection level ..	112
Table 32-19—NGV PHY MIB attributes	121
Table 32-20—NGV PHY characteristics	123
Table 32-21—NGV-MCSs for 10 MHz, NSS = 1	124
Table 32-22—NGV-MCSs for 10 MHz, NSS = 2	124
Table 32-23—NGV-MCSs for 20 MHz, NSS = 1	125
Table 32-24—NGV-MCSs for 20 MHz, NSS = 2	125

Figures

Figure 9-79—Clustering Control field format if the Discovery Mode is 1	40
Figure 9-788fm—DMG OCB element format	42
Figure 9-788fn—DMG OCB Parameters field format	42
Figure 9-848—SSW Feedback field format when transmitted as part of an ISS	43
Figure 9-849a—SSW Feedback field format when not transmitted as part of an ISS and the EDMG Extension Flag subfield is 0	43
Figure 11-10a—DMG discovery outside the context of a BSS.....	48
Figure 11-10b—Beamforming training during the DMG discovery outside the context of a BSS	49
Figure 17-7—Data scrambler	57
Figure 32-1—PHY interaction on transmit for various PPDU formats.....	77
Figure 32-2—PHY interaction on receive for various PPDU formats	77
Figure 32-3—PHY-CONFIG and CCA interaction with Clause 17 and Clause 32.....	78
Figure 32-4—Example of NON_NGV_10 repetition transmission with N_PPDU_REP = 2	79
Figure 32-5—NGV PPDU format	80
Figure 32-6—Transmitter block diagram for the L-SIG, RL-SIG, NGV-SIG, and RNGV-SIG fields for an NGV PPDU	82
Figure 32-7—Transmitter block diagram for the Data field of an NGV transmission.....	82
Figure 32-8—Timing boundaries for NGV PPDU fields.....	88
Figure 32-9—Generation of NGV-LTF symbols	99
Figure 32-10—Generation of NGV-LTF-1x symbols.....	99
Figure 32-11—Generation of NGV-LTF-2x-Repeat symbols	100
Figure 32-12—NGV PPDU with midamble and randomization procedure.....	105
Figure 32-13—PHY transmit procedure for NGV transmission	114
Figure 32-14—PHY transmit state machine.....	116
Figure 32-15—PHY receive procedure for NGV transmission.....	117
Figure 32-16—PHY receive state machine	118
Figure 32-17—NGV ranging NDP format	120
Figure 32-18—Example of NGV-LTF with NSS = 2 and LTF_REP = 1	121

**IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements**

**Part 11: Wireless LAN Medium Access Control
(MAC) and Physical Layer (PHY) Specifications**

**Amendment 5:
Enhancements for Next Generation V2X**

(This amendment is based on IEEE Std 802.11™-2020 as amended by IEEE Std 802.11ax™-2021, IEEE Std 802.11ay™-2021, IEEE Std 802.11ba™-2021, IEEE Std 802.11™-2020/Cor 1-2022, and IEEE Std 802.11az™-2022.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strike through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editorial instructions, change markings and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.⁶

⁶ Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

3. Definitions, acronyms, and abbreviations

3.2 Definitions specific to IEEE Std 802.11

Change the definition for “dynamic bandwidth operation” in 3.2 as follows:

dynamic bandwidth operation: A feature of a very high throughput (VHT) station (STA) or a next generation vehicle-to-everything (NGV) STA in which the request-to-send/clear-to-send (RTS/CTS) exchange, using non-high-throughput (non-HT) duplicate physical layer (PHY) protocol data units (PPDUs) or non-NGV duplicate PPDUs, negotiates a potentially reduced channel width (compared to the channel width indicated by the RTS) for subsequent transmissions within the current transmission opportunity (TXOP).

Change the definition for “bandwidth signaling transmitter address (TA)” (as amended by IEEE Std 802.11ax-2021) in 3.2 as follows:

bandwidth signaling transmitter address (TA): A TA that is used by a very high throughput (VHT) station (STA), ~~or a high-efficiency (HE) STA, or a next generation vehicle-to-everything (NGV) STA~~ to indicate the presence of additional signaling related to the bandwidth to be used in subsequent transmission in an enhanced distributed channel access (EDCA) transmission opportunity (TXOP). It is the IEEE medium access control (MAC) individual address of the transmitting VHT STA but with the Individual/Group bit equal to 1.

Insert the following definitions in 3.2 in alphabetical order:

non-next generation vehicle-to-everything (V2X) (non-NGV) duplicate: A transmission format of the physical layer (PHY) that duplicates a 10 MHz non-NGV transmission in two adjacent 10 MHz channels and allows a station (STA) communicating outside the context of a basic service set (BSS)(OCB) on either of the 10 MHz channels to receive the transmission.

non-next generation vehicle-to-everything (V2X) (non-NGV) physical layer (PHY) protocol data unit (PPDU): A PPDU with 10 MHz channel width that is transmitted outside the context of a basic service set (BSS) (OCB) in the 5.9 GHz band and that is not a next generation V2X (NGV) PPDU.

non-next generation vehicle-to-everything (V2X) (non-NGV) station (STA): A STA that transmits or receives non-NGV physical layer (PHY) protocol data units (PPDUs) and that is not able to transmit next generation V2X (NGV) PPDUs.

next generation vehicle-to-everything (V2X) (NGV) physical layer (PHY) protocol data unit (PPDU): A Clause 32 PPDU with the TXVECTOR parameter FORMAT equal to NGV or NON_NGV_10.

outside the context of a basic service set (OCB) primary channel: A 10 MHz channel that is designated by a higher layer (via medium access control (MAC) sublayer management entity (MLME) primitives and/or management information base (MIB) parameters). The OCB primary channel and OCB secondary channel are contiguous and together form a 20 MHz channel for transmission of 20 MHz next generation vehicle-to-everything (V2X) (NGV) physical layer (PHY) protocol data units (PPDUs).

outside the context of a basic service set (OCB) secondary channel: A 10 MHz channel that is designated by a higher layer (via medium access control (MAC) sublayer management entity (MLME) primitives and/or management information base (MIB) parameters). The OCB secondary channel and the OCB primary channel are contiguous and together form a 20 MHz channel for the transmission of 20 MHz next generation vehicle-to-everything (V2X) (NGV) physical layer (PHY) protocol data units (PPDUs).

3.4 Abbreviations and acronyms

Insert the following abbreviations in 3.4 in alphabetical order:

NGV	next generation vehicle-to-everything (V2X)
V2X	vehicle-to-everything

4. General description

4.3 Components of the IEEE Std 802.11 architecture

4.3.17 STA transmission of Data frames outside the context of a BSS

Insert the following paragraphs after the fourth paragraph of 4.3.17 (beginning “Communication of Data frames ...”):

A STA with dot11OCBActivated equal to true might operate as a DMG STA. A DMG STA with dot11DMGOCBActivated equal to true supports the MAC and MLME functions defined in 31.3 in addition to the MAC functions defined in Clause 10 and the MLME functions defined in Clause 11 for DMG STAs. A DMG STA operating OCB might support EDMG features.

A STA whose MIB does not include the dot11DMGOCBActivated attribute operates as if the attribute is false.

Insert 4.3.17a after 4.3.17 as follows:

4.3.17a Next generation vehicle-to-everything (V2X) (NGV) STA

A next generation vehicle-to-everything (NGV) STA provides MAC and PHY features as defined in Clause 31 and Clause 32. An NGV STA supports at least twice the throughput and at least 3 dB better receiver minimum input sensitivity than a non-NGV STA when operating OCB in high mobility channel environments in the 5.9 GHz band as defined in E.2.3 and E.2.4. The double throughput and higher receiver minimum input sensitivity might not be supported simultaneously. In an NGV STA dot11NGVOptionImplemented is true. In a non-NGV STA with dot11NGVOptionImplemented, the attribute is false. A STA whose MIB does not include the dot11NGVOptionImplemented attribute operates as if the attribute is false.

An NGV STA might be co-located with a DMG STA for which dot11DMGOCBActivated is true in the 60 GHz frequency band (57 GHz to 71 GHz) as defined in E.1. The NGV STA might assist the DMG STA in performing DMG discovery OCB by communicating with a peer NGV STA to exchange the higher layer information that contains information of a peer DMG STA with which the peer NGV STA is co-located. The information is provided from/to a DMG STA through the MLME SAP interface. The protocol to exchange this higher layer information is outside the scope of this standard.

An NGV STA supports interoperability, coexistence, backward compatibility, and fairness in contending for the medium with non-NGV STAs when operating OCB in the 5.9 GHz band. An NGV STA is also a non-NGV STA that supports transmission and reception of non-NGV PHY PPDU when operating OCB in the 5.9 GHz band.

An NGV STA is capable of transmitting and receiving frames OCB as specified in 11.18.

The main PHY features of an NGV STA that are not present in a non-NGV STA are the following:

- Mandatory support for the 10 MHz NGV PPDU format and optional support for the 20 MHz NGV PPDU and 20 MHz non-NGV duplicate PPDU formats
- For the 20 MHz and 10 MHz NGV PPDU, mandatory support for LDPC coding, midambles, BPSK-DCM, and 256-QAM
- Mandatory support for transmit spectral mask C2 by a STA that transmits a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU
- Mandatory support for repetitive NON_NGV_10 PPDU

- Mandatory support for coexistence with non-NGV STAs operating in the same channel
- Mandatory support for transmit spectral mask C by a STA that transmits a 10 MHz NGV PPDU and belongs to transmit power class C
- Optional support for transmission and reception of SU MIMO with two spatial streams

Additional NGV STA PHY features are:

- Mandatory support for a single spatial stream
- Mandatory support for Clause 17 10 MHz PPDU
- Optional support for Classes A, B, and D of spectrum mask requirement for 10 MHz bandwidth

The main MAC features of an NGV STA that are not present in a non-NGV STA are the following:

- Mandatory support for coexistence with non-NGV STAs operating in the same channel
- Mandatory support for extended MAC service interface to provide higher layers with the ability to control NGV transmissions and receive status regarding NGV receptions and the radio environment
- Mandatory support for NGV capability indication for non-NGV PPDU encoded in the Duration/ID field of the MAC header
- Mandatory support for block ack
- Mandatory support for reception of frame aggregation when communicating OCB
- Optional support for transmission of frame aggregation when communicating OCB
- Optional support for 20 MHz OCB communication
- For optional 20 MHz OCB communication an NGV STA supports 20 MHz channel access with 10 MHz primary and 10 MHz secondary channels
- Optional support for 20 MHz non-NGV duplicate operation

Additional NGV STA MAC features are:

- Mandatory support for 10 MHz OCB communication

An NGV STA might support positioning based on NGV ranging, which consists of a subset of fine timing measurement (FTM) functionalities as defined in 11.21.6 and P.3:

- Fine timing Measurement procedure negotiation and termination for non-TB ranging as defined in 11.21.6.3 and non-TB ranging as described in 11.21.6.4.4
- Differential distance computation as detailed in P.3

An NGV STA might be co-located with a STA operating outside the 5.9 GHz band that supports the fine timing measurement procedure as defined in 11.21.6.

4.3.22 DMG STA

Change the first paragraph of 4.3.22 as follows:

The IEEE 802.11 DMG STA provides PHY and MAC features that can support a throughput of 1 Gb/s and greater, as measured at the MAC data service access point (SAP). A DMG STA supports DMG features as identified in Clause 10, Clause 11, and Clause 20. Optionally, a DMG STA with dot11DMGOCBActivated equal to true supports features as identified in 31.3. A DMG STA operates in a DMG BSS and supports transmission and reception of frames that are compliant with PHY specifications as defined in Clause 20. A DMG STA is also a QoS STA. The basic channel access of a DMG STA (see 10.39) allows it to operate in an Infrastructure BSS, in an IBSS, and in a PBSS. Certain DMG features such as service period allocation are available only to DMG STAs that are associated with an AP or with a PCP, while other DMG features such as EDCA operation in a PBSS and DMG STAs operating OCB do not require association. A DMG

STA supports beamforming (BF) as described in 10.42 and 20.9 and GCM encryption as described in 12.5.5.

Change the third paragraph of 4.3.22 (as amended by IEEE Std 802.11ay-2021) as follows:

A DMG STA supports MAC features that provide channel access in an environment in which transmissions use a directional antenna pattern. The MAC entity of a DMG STA provides frame aggregation, block ~~acknowledgment~~ acknowledgment, service periods, contention based access periods, DMG protected period, AP or PCP clustering, dynamic channel time management (see 10.39.7, 10.39.8, and 10.39.9), reverse direction, spatial sharing, beamforming, discovery assistance, TDD channel access, ~~and fast session transfer in a multi-band device~~, transmission of Data frames OCB (see 11.18), and DMG discovery OCB (see 11.1.4.7). A DMG STA is not a mesh STA. A DMG STA does not use any of the following: HCCA, power save multi-poll (PSMP), TDLS, GCR.

5. MAC service definition

5.2 MAC data service specification

Insert 5.2.2a after 5.2.2 as follows:

5.2.2a NGV MAC data service specification

An NGV STA coordinates with higher layer ITS entities (e.g., IEEE 1609 [B22] family of standards) to provide wireless communications services via the LLC sublayer. The MA-UNITDATA primitives contain the radio environment request vector to facilitate the coordination.

5.2.3 MA-UNITDATA.request

5.2.3.2 Semantics of the service primitive

Change the first paragraph of 5.2.3.2 as follows:

The parameters of the primitive are as follows:

```
MA-UNITDATA.request(  
    source address,  
    destination address,  
    routing information,  
    data,  
    priority,  
    drop eligible,  
    service class,  
    station vector,  
    MSDU format,  
    radio environment request vector  
)
```

Insert the following at the end of 5.2.3.2:

The radio environment request vector parameter contains information that allows higher layer entities to configure the PPDU format, encoding, and MPDU handling for NGV transmission. This parameter is present when dot11NGVOptionImplemented is true or dot11NONNGVRadioEnvironmentImplemented is true, and absent otherwise.

The radio environment request vector parameter contains the following members pertaining to the transmission of the MPDU that contains the MSDU associated with the request containing the vector:

- PPDU format (non-NGV/NGV)
- data rate/NGV-MCS
- number of spatial streams
- permitted aggregation
- number of repetitions
- expiry time
- channel
- primary channel
- channel width

- fallback enabled
- transmit power spectral density

The PPDU format member indicates the format of the PPDU as defined in Table 5-1 that is used to transmit the MSDU.

Table 5-1—PPDU format values

PPDU format value	Description
0	Non-NGV PPDU
1	NGV PPDU
2	The PPDU format decided by the MAC layer

The data rate/NGV-MCS member indicates the data rate of the PPDU carrying the MSDU if the PPDU format has value 0. The data rate can be either 0 (decided by MAC layer), 3 Mb/s, 4.5 Mb/s, 6 Mb/s, 9 Mb/s, 12 Mb/s, 18 Mb/s, 24 Mb/s, or 27 Mb/s as defined in Table 17-4. If the PPDU format has value 1, the data rate/NGV-MCS member indicates the NGV-MCS index as defined in 32.5 with an exception that the value 14 means that the actual MCS is decided by the MAC layer. If the PPDU format has value 2, the data rate/NGV-MCS member is set to value 14 indicating that the actual data-rate/MCS is decided by the MAC layer.

The number of spatial streams member with value 1 or 2 indicates the number of spatial streams being used to transmit the PPDU carrying the MSDU. The number of spatial streams member with value 0 means that the number of spatial streams is decided by the MAC layer. The permitted aggregation member indicates whether A-MPDU aggregation can be applied to the MSDU. The value 0 means that aggregation is not applied, the value 1 means that aggregation can be applied, while the value 2 means that it is up to the MAC layer to decide whether the aggregation can be applied.

The number of repetitions member indicates the maximal number of repetitions that can be used to transmit the MSDU in a broadcast MPDU of a non-NGV PPDU in a 10 MHz channel width where the values can be in the range 0 to 3 or 15. The value 15 means that it is up to the MAC layer to decide whether the repetition is applied to the MSDU. This parameter is reserved when the MSDU is not in a broadcast MPDU in 10 MHz channel width.

The expiry time member indicates the time in milliseconds until the MSDU is discarded if still not transmitted, where the value 0 means that it is up to the MAC layer to decide the expiry time of the MSDU.

NOTE 1—When an MSDU reaches its expiry time and is discarded, the CW state variable of the EDCAF associated with that MSDU is unchanged.

The channel member indicates the channel where the MSDU is transmitted in the 5.9 GHz band as defined in E.2.3 and E.2.4 with the exception for value 0, which means that the MAC layer selects the channel.

The primary channel indicates the primary 10 MHz channel where the MSDU is transmitted, where the value 0 means that the MAC layer selects the primary channel. This member is present when dot11NGVOptionImplemented is true and when the channel width member is 20 MHz and is absent otherwise.

The channel width member indicates either 10 MHz or 20 MHz channel width used to transmit the MSDU as defined in Table 5-2. When the channel width indicates 20 MHz, the primary channel member indicates the OCB primary channel.

NOTE 2—An NGV STA transmitting in a 20 MHz channel maintains a single primary channel at any given time.

Table 5-2—Channel width values

Channel width value	Description
0	10 MHz
1	20 MHz
2	The channel width is decided by the MAC layer

The fallback enabled member indicates whether the transmission of a 10 MHz PPDU in the OCB primary channel is allowed by an NGV STA while the NGV STA performs channel access to transmit a 20 MHz NGV PPDU. The value 0 means that the fallback to 10 MHz can be applied to the MSDU, the value 1 means that the fallback to 10 MHz cannot be applied to the MSDU, while the value 3 means that it is up to the MAC layer to decide whether the fallback is applied to the MSDU. This member is present when dot11NGVOptionImplemented is true and when the channel width member is 20 MHz and is absent otherwise.

The transmit power spectral density member indicates the combined transmit power at the transmit antenna connector of all the antennas used to transmit the MSDU in units of dBm/10 MHz. The transmit power is described with a resolution of 1 dB, with values in the range 0 to 60 representing –20 dBm/10 MHz to 40 dBm/10 MHz, respectively. Values from 61 to 126 are reserved. The value 127 means that it is up to the MAC layer to decide the transmit power of the MSDU.

5.2.4 MA-UNITDATA.indication

5.2.4.2 Semantics of the service primitive

Change the first paragraph of 5.2.4.2 as follows:

The parameters of the primitive are as follows:

```
MA-UNITDATA.indication(
    source address,
    destination address,
    routing information,
    data,
    reception status,
    priority,
    drop eligible,
    service class,
    station vector,
    MSDU format,
    radio environment status vector
)
```

Insert the following at the end of 5.2.4.2:

The radio environment status vector parameter provides information to higher layer entities about the current radio environment and the most recent reception. This parameter is present when dot11NGVOptionImplemented is true or dot11NONNGVRadioEnvironmentImplemented is true, and absent otherwise.

The radio environment status vector parameter contains the following members pertaining to the reception of the MPDU that contained the MSDU associated with the indication containing the vector:

- PPDU format (non-NGV/NGV)
- data rate/NGV-MCS
- aggregation
- channel
- primary channel
- channel width
- RSSI

The PPDU format member indicates the format of the received PPDU that carries the MSDU as defined in Table 5-1.

The data rate/NGV-MCS member indicates the data rate of the PPDU carrying the MSDU if the PPDU format has value 0. The data rate can be one of 3 Mb/s, 4.5 Mb/s, 6 Mb/s, 9 Mb/s, 12 Mb/s, 18 Mb/s, 24 Mb/s, or 27 Mb/s as defined in Table 17-4. If the PPDU format has value 1, the data rate/NGV-MCS member indicates the NGV-MCS index as defined in 32.5.

The aggregation member indicates whether A-MPDU aggregation is applied to the received MSDU. The value 0 means that aggregation is not applied, while the value 1 means that aggregation is applied.

The channel member indicates the channel where the MSDU is transmitted in the 5.9 GHz band as defined in E.2.3 and E.2.4 with the exception for value 0, which means that the MAC layer selects the channel.

The primary channel member indicates the primary 10 MHz channel where the MSDU is received.

The channel width member indicates either 10 MHz or 20 MHz channel width of the PPDU carrying the received MSDU as defined in Table 5-2. When the channel width indicates 20 MHz, the primary channel parameter indicates the OCB primary channel.

The RSSI parameter indicates the receive signal power, measured at the STA's antenna connector and averaged over the antennas, for the NGV portion of the received NGV PPDU and is defined in Table 5-3.

Table 5-3—RSSI values

RSSI	Description
0 to 90	The RSSI, in units of dBm, is $-113 + F_{val}$, where F_{val} is the subfield value
91 to 127	Reserved

5.2.5 MA-UNITDATA-STATUS.indication

5.2.5.2 Semantics of the service primitive

Change the first paragraph of 5.2.5.2 as follows:

The parameters of the primitive are as follows:

```
MA-UNITDATA-STATUS.indication(  
    source address,  
    destination address,  
    transmission status,  
    provided priority,  
    provided service class,  
    radio environment request vector  
)
```

Insert the following at the end of 5.2.5.2:

The radio environment request vector parameter (see 5.2.3.2) contains information that is used by the MAC entity to report the used format, encoding, and MPDU handling for NGV transmission. This parameter is present when dot11NGVOptionImplemented is true or dot11NONNGVRadioEnvironmentImplemented is true, and absent otherwise.

6. Layer management

6.3 MLME SAP interface

Insert 6.3.126, 6.3.127, and 6.3.128 after 6.3.125 (inserted by IEEE Std 802.11ba-2021) as follows:

6.3.126 Cancel transmissions of MSDUs

6.3.126.1 Introduction

This primitive allows higher layer entities (e.g., entities based on the IEEE 1609 [B22] family of standards) to cancel transmission of MSDUs that were previously sent to the STA and are still in the MAC entity's transmit queue.

6.3.126.2 MLME-CANCELTX.request

6.3.126.2.1 Function

Requests cancellation of transmission of all queued MSDUs belonging to the specified access category.

6.3.126.2.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-CANCELTX.request(  
    AccessCategoryIndex  
)
```

Name	Type	Valid range	Description
AccessCategoryIndex	Integer	0–4	Specifies the access category index of the MSDUs that are to be removed from the transmit queue. Index 0–3 is the value of the access category in Table 9-283 and index 4 indicates all access categories.

6.3.126.2.3 When generated

This primitive is generated by the SME when the SME receives a request (from a higher layer entity) to remove MSDUs from the transmit queue.

6.3.126.2.4 Effect of receipt

The receipt of this primitive by the MAC entity causes the MAC entity to remove MSDUs of the specified access category index from the transmit queue.

6.3.126.3 MLME-CANCELTX.confirm

6.3.126.3.1 Function

This primitive reports completion of the removal of MSDUs from the transmit queue.

6.3.126.3.2 Semantics of the service primitive

This primitive has no parameters.

6.3.126.3.3 When generated

The primitive is generated by the MLME to inform the SME that an MLME-CANCELTX.request primitive has been completed, so that the SME can inform the requesting entity (a higher layer entity).

6.3.126.3.4 Effect of receipt

The SME is notified that the MAC entity has removed MSDUs of the specified access category from the transmit queue.

6.3.127 NGV radio environment measurement

6.3.127.1 General

This primitive provides a report on the STA's radio environment.

6.3.127.2 MLME-RADIOENVIRONMENT.request

6.3.127.2.1 Function

This primitive is used to request the status information on the state of the STA's radio environment.

6.3.127.2.2 Semantics of the service primitive

The primitive parameters are as follows:

MLME-RADIOENVIRONMENT.request (
RadioEnvironmentMeasurementPeriod,
StationMeasurementPeriod
)

Name	Type	Valid range	Description
RadioEnvironmentMeasurementPeriod	Integer	0 or 100–1000	Specifies the amount of time in milliseconds during which the NGV STA measures the channel busy percentage measurement. When equal to 0, measurements are not made.
StationMeasurementPeriod	Integer	0 or 100–1000	Specifies the amount of time in milliseconds during which the NGV STA measures the number of neighboring STAs and the number of neighboring NGV STAs. When equal to 0, measurements are not made.

6.3.127.2.3 When generated

The primitive is generated when a higher layer entity wants to acquire the status information on the state of the STA's radio environment.

6.3.127.3 MLME-RADIOENVIRONMENT.indication

6.3.127.3.1 Function

This primitive provides status information on the state of the STA's radio environment.

6.3.127.3.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-RADIOENVIRONMENT.indication(
    StationCount,
    NGVCapabilityPercentage,
    ChannelBusyPercentage
)
```

Name	Type	Valid range	Description
StationCount	Integer	0–65 535	Indicates the number of unique individual station MAC addresses detected during the most recent measurement period of duration dot11RadioEnvironmentMeasurementPeriod.
NGVCapabilityPercentage	Integer	0–100	Indicates the percentage of the stations indicated in StationCount whose transmissions contain indication of NGV capability. When StationCount is equal to 0, the NGVCapabilityPercentage shall be set to 0.
ChannelBusyPercentage	Integer	0–100	The percentage of time the channel was busy during the most recent measurement of duration dot11RadioEnvironmentMeasurementPeriod. The channel is busy when the NGV STA is either transmitting or its clear channel assessment function indicates the channel is busy. It is calculated as follows: $\frac{100 \times \text{duration as indicated channel busy}}{\text{dot11RadioEnvironmentMeasurementPeriod}}$

6.3.127.3.3 When generated

The primitive is generated by the MAC entity every dot11RadioEnvironmentMeasurementPeriod if dot11NGVOptionImplemented is true.

6.3.128 DMG operation outside the context of a BSS

6.3.128.1 General

The following primitives support DMG operation outside the context of a BSS.

6.3.128.2 MLME-DMG-OCB-START.request

6.3.128.2.1 Function

This primitive requests the MAC entity to initiate DMG operation outside the context of a BSS.

6.3.128.2.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-DMG-OCB-START.request(  

    Channel number,  

    Discovery beacon,  

    DMG parameters,  

    DMG capabilities,  

    EDMG capabilities,  

    Unsolicited Block Ack Extension,  

    VendorSpecificInfo  

)
```

Name	Type	Valid range	Description
Channel number	Integer	Selected from the valid channel range for the appropriate PHY and carrier set.	Specifies a channel number to perform discovery for peer STAs outside the context of a BSS.
Discovery beacon	Boolean	true, false	Indicates whether the STA is required to transmit DMG Beacon frames to discover peer STAs.
DMG parameters	As defined in frame format	As defined in 9.3.4.2	Specifies the parameters to be advertised to peer STAs.
DMG capabilities	As defined in frame format	As defined in 9.4.2.127	Specifies the parameters to be advertised to peer STAs.
EDMG capabilities	As defined in frame format	As defined in 9.4.2.265	Specifies the parameters to be advertised to peer STAs. Optionally present if dot11EDMGOptionImplemented is true and is absent otherwise.
Unsolicited Block Ack Extension	Unsolicited Block Ack Extension element	As defined in 9.4.2.279	Specifies the parameters within the Unsolicited Block Ack Extension element that are supported by the MAC entity. Optionally present if dot11UnsolicitedBAActivated is true and is absent otherwise.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.25	Zero or more elements.

6.3.128.2.3 When generated

The primitive is generated by the SME for a STA to initiate DMG operation outside the context of a BSS.

6.3.128.2.4 Effect of receipt

The primitive initiates a discovery procedure and data transmission outside the context of a BSS.

6.3.128.3 MLME-DMG-OCB-START.confirm

6.3.128.3.1 Function

This primitive reports the result of the initiation of a DMG operation outside the context of a BSS.

6.3.128.3.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-DMG-OCB-START.confirm(  
    ResultCode,  
    VendorSpecificInfo  
)
```

Name	Type	Valid range	Description
ResultCode	Enumeration	SUCCESS, NOT_SUPPORTED	Indicates the result of the MLME-DMG-OCB-START.request primitive.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.25	Zero or more elements.

6.3.128.3.3 When generated

The primitive is generated by the MLME as a result of an MLME-DMG-OCB-START.request primitive to initiate a discovery procedure and data transmission outside the context of a BSS.

6.3.128.3.4 Effect of receipt

The SME is notified of the result of the initiation of a DMG operation outside the context of a BSS.

6.3.128.4 MLME-DMG-OCB-STOP.request

6.3.128.4.1 Function

This primitive requests the MAC entity to stop the DMG operation outside the context of a BSS previously started by using an MLME-DMG-OCB-START.request.

6.3.128.4.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-DMG-OCB-STOP.request()
```

6.3.128.4.3 When generated

This primitive is generated by the SME to terminate discovery procedure and data transmission operating outside the context of a BSS by the MAC entity. The MLME-DMG-OCB-STOP.request primitive shall be generated only after the SME receives an MLME-DMG-OCB-START.confirm primitive with result code equal to SUCCESS.

6.3.128.4.4 Effect of receipt

This request terminates the DMG operation outside the context of a BSS when the current frame exchange sequence is completed.

6.3.128.5 MLME-OCB-DMGDISCOVERY.indication

6.3.128.5.1 Function

This primitive indicates discovery of peer DMG STAs outside the context of a BSS.

6.3.128.5.2 Semantics of the service primitive

The primitive parameters are as follows:

MLME-OCB-DMGDISCOVERY.indication(
 PeerInfoSet,
 VendorSpecificInfo
)

Name	Type	Valid range	Description
PeerInfoSet	Set of PeerInfo vectors as defined below	N/A	The PeerInfoSet is returned to indicate the results of the discovery request outside the context of a BSS as described in 11.1.4.7. It is a set containing zero or more instances of a PeerInfo vector.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.25	Zero or more elements.

Each PeerInfo vector consists of the parameters shown in the following table, in which the term peer STA refers to the STA that transmitted the received DMG Beacon frame, SSW frame, SSW Feedback frame, or SSW Ack frame.

Name	Type	Valid range	Description
PeerSTAAddress	MAC address	Any valid individual MAC address	Specifies the MAC address of the discovered peer STA.
Sector sweep	As defined in frame format	As defined in 9.3.4.2	The values from the Sector Sweep field from the DMG Beacon frame, else null.
Sector sweep feedback	As defined in frame format	As defined in 9.5.3	The values from the Sector Sweep field from the SSW, SSW Feedback, or SSW Ack frame, else null.
DMG parameters	As defined in frame format	As defined in 9.3.4.2	The values from the DMG Parameters field from the DMG Beacon frame, else null.
DMG capabilities	As defined in frame format	As defined in 9.4.2.127	The values from the DMG Capabilities element if such an element was present in the DMG Beacon frame, else null.
EDMG capabilities	As defined in frame format	As defined in 9.4.2.265	Specifies the parameters within the EDMG Capabilities element that are supported by the MAC entity.

6.3.128.5.3 When generated

This primitive is generated by the MLME when the MAC entity successfully completes the beamforming training with the discovered peer MAC entities.

6.3.128.5.4 Effect of receipt

The SME is notified of the discovered peer MAC entities.

6.3.128.6 MLME-OCB-LINKSTATUS.indication

6.3.128.6.1 Function

This primitive indicates the status change of beamformed link with a peer STA.

6.3.128.6.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-OCB-LINKSTATUS.indication(
    PeerSTAAddress,
    BeamLinkStatus,
    VendorSpecificInfo
)
```

Name	Type	Valid range	Description
PeerSTAAddress	MAC address	Any valid individual MAC address	The MAC address of the STA that is the peer STA in the beamformed link being reported on.
BeamLinkStatus	Enumeration	ESTABLISHED, TIMER_EXPIRED, NOT_ESTABLISHED	Indicates the status of beamformed link as described in 11.27.1.1.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.25	Zero or more elements.

6.3.128.6.3 When generated

This primitive is generated by the MLME when the MAC entity detects the expiration of a beam link maintenance timer or any status change on a beamformed link with peer MAC entity.

6.3.128.6.4 Effect of receipt

The SME is notified of the status of the beamformed link and beam link maintenance timer with the peer MAC entity.

9. Frame formats

9.2 MAC frame formats

9.2.4 Frame fields

9.2.4.7 Frame Body field

9.2.4.7.1 General

Change Table 9-25 (as amended by IEEE Std 802.11ay-2021) as shown on the following three pages:

Table 9-25—Maximum data unit sizes (in octets) and durations (in microseconds)

	Non-HT non-VHT non-HE non-SIG non-DMG PPDU and non-HT duplicate PPDU	HT PPDU	VHT PPDU	HE PPDU	SIG PPDU	DMG PPDU	EDMG PPDU	<u>NGV PPDU</u>
MMPDU size	2304	2304	See NOTE 1	See NOTE 1	See NOTE 1	2304	2304	<u>2304</u>
MSDU size	2304	2304	2304	2304	2304	Without SAR agreement: for the basic A-MSDU format, it is equal to the value of A-MSDU size minus 14, or minus 2 for the short A-MSDU format, if the MPDU Limit subfield of the Extended MPDU Capability field of the DMG Capabilities element is valid; otherwise, it is equal to 7920. With SAR agreement: see NOTE 8.	Without SAR agreement: for the basic A-MSDU format, it is equal to the value of A-MSDU size minus 14, or minus 2 for the short A-MSDU format, if the MPDU Limit subfield of the Extended MPDU Capability field of the DMG Capabilities element is valid; otherwise, it is equal to 7920. With SAR agreement: see NOTE 8.	<u>2304</u>
A-MSDU size	3839 or 4065 (see NOTE 2) (HT STA, see also Table 9-184), or N/A (non-HT STA, see also 10.11)	3839 or 7935 (see also Table 9-184)	See NOTE 3	2.4 GHz band: 3839 or 7935 (see also Table 9-184) Otherwise: see NOTE 3	See NOTE 3	Without SAR agreement: indirectly limited by the value of the MPDU Limit subfield in the Extended MPDU Capability field of the DMG Capabilities element, if the subfield is valid; otherwise, it is equal to 7935. With SAR agreement: see NOTE 8.	Without SAR agreement: indirectly limited by the value of the MPDU Limit subfield in the Extended MPDU Capability field of the DMG Capabilities element, if the subfield is valid; otherwise, it is equal to 7935. With SAR agreement: see NOTE 8.	<u>7935</u>

Table 9-25—Maximum data unit sizes (in octets) and durations (in microseconds) (continued)

	Non-HT non-VHT non-HE non-SIG non-DMG PPDU and non-HT duplicate PPDU	HT PPDU	VHT PPDU	HE PPDU	SIG PPDU	DMG PPDU	EDMG PPDU	<u>NGV PPDU</u>
MPDU size	See NOTE 4	See NOTE 5	3895 or 7991 or 11 454 (see also Table 9-271)	2.4 GHz band; see NOTE 5 Otherwise: 3895 or 7991 or 11 454 (see also Table 9-271) See NOTE 7	3895 or 7991 (see also Table 9-300)	The value indicated in the MPDU Limit subfield of the Extended MPDU Capability field of the DMG Capabilities element if the subfield is valid; otherwise, as in NOTE 5.	The value indicated in the MPDU Limit subfield of the Extended MPDU Capability field of the DMG Capabilities element if the subfield is valid; otherwise, as in NOTE 5.	<u>7991</u>
PSDU size	2^{12-1} (see Table 15-5, Table 16-4, Table 17-21, Table 18-5)	2^{16-1} (see Table 19-25)	4 692 480 ($\sim 2^{22.16}$) (see Table 21-28)	6 500 631 ($\sim 2^{22.63}$) (see Table 27-54)	797 160 ($\sim 2^{19.60}$) (see Table 23-40)	2^{18-1} (see Table 20-30)	2^{22-1} (see Table 28-12 and Table 28-19)	<u>244 800</u>
PPDU duration	See NOTE 6	5484 (HT_MF, see 10.27.4) or 10 000 (HT_GF, see Table 19-25)	5484 (see Table 21-28)	5484 (see Table 27-54)	27 840 (see Table 23-40)	2000 (see Table 20-30)	2000 (see Table 20-30)	<u>10 968</u>

Table 9-25—Maximum data unit sizes (in octets) and durations (in microseconds) (continued)

Non-HT non-VHT non-HE non-SIG non-DMG PPDU and non-HT duplicate PPDU	HT PPDU	VHT PPDU	HE PPDU	SIG PPDU	DMG PPDU	EDMG PPDU	<u>NGV</u> <u>PPDU</u>
NOTE 1—No direct constraint on the maximum MMPDU size; indirectly constrained by the maximum MPDU size (see 9.3.3.1).							
NOTE 2—Indirect constraint from the maximum PSDU size: 212–1 octets minus the minimum QoS Data frame overhead (26 octets for the MAC header and 4 octets for the FCS).							
NOTE 3—No direct constraint on the maximum A-MSDU size; indirectly constrained by the maximum MPDU size.							
NOTE 4—No direct constraint on the maximum MPDU size; indirectly constrained by the maximum MSDU/MMPDU or (for HT STAs only) A-MSDU size.							
NOTE 5—No direct constraint on the maximum MPDU size; indirectly constrained by the maximum A-MSDU size.							
NOTE 6—No direct constraint on the maximum duration, but an L_LENGTH value above 2332 might not be supported by some receivers (see NOTE 2 in 10.27.4).							
NOTE 7—The maximum MPDU size might be greater than the size declared as supported by the recipient if the MPDU is an HE Compressed Beamforming/CQI frame.							
NOTE 8—No direct constraint on the maximum MSDU or A-MSDU size; indirectly constrained by the maximum PSDU size. Each MPDU in an A-MPDU of the PSDU that contains the MSDU or A-MSDU generates an overhead of MPDU Header (26 bytes), FCS (4 bytes), GCMP Header (8 bytes), MIC (16 bytes), and MPDU delimiter (4 bytes).							

9.2.5 Duration/ID field (QoS STA)

9.2.5.1 General

Insert the following paragraph at the end of 9.2.5.1:

The value in the Duration/ID field in a frame transmitted by an NGV STA is further defined in 31.2.1.

9.3 Format of individual frame types

9.3.4 Extension frames

9.3.4.2 DMG Beacon

Insert a new row above the last row of Table 9-45 as follows (unchanged rows not shown):

Table 9-45—DMG Beacon frame body

Order	Information	Notes
...		
64	DMG OCB	This element is present if dot11DMGOCBActivated is true; otherwise not present.
...		

Replace Figure 9-79 with the following figure (with a new column for B48):

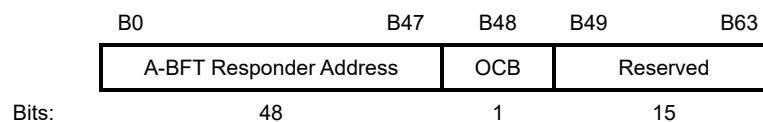


Figure 9-79—Clustering Control field format if the Discovery Mode is 1

Change the last paragraph of 9.3.4.2 as follows:

The A-BFT Responder Address subfield contains the MAC address of the STA that is allowed to transmit during the A-BFT, if present, that follows the BTI. If all bits of the A-BFT Responder Address subfield are equal to 1 and the OCB subfield is equal to 1, any STA is allowed to transmit during the A-BFT, if present.

Insert the following paragraph at the end of 9.3.4.2:

The OCB subfield is set to 1 to indicate the support of communication outside the context of a BSS by the STA transmitting the DMG Beacon frame.

9.4 Management and Extension frame body components

9.4.1 Fields that are not elements

9.4.1.46 DMG Parameters field

Change Table 9-69 as follows:

Table 9-69—The BSS Type subfield when the Discovery mode field is 1

Subfield value	Responding STA role	Applicable BSS types
3	AP	Infrastructure BSS
2	PCP	PBSS
1	Non-AP STA	PBSS, IBSS, <u>OCB</u>
0	Any	Infrastructure BSS, PBSS, IBSS, <u>OCB</u>
<u>NOTE—OCB is not a BSS Type, but rather an indication of communication outside the context of a BSS.</u>		

9.4.2 Elements

9.4.2.1 General

Insert two rows above the last reserved row and change the last reserved row in Table 9-92 (as amended by IEEE Std 802.11az-2022) as follows (unchanged rows not shown):

Table 9-92—Element IDs

Element	Element ID	Element ID Extension	Extensible	Fragmentable
...				
Reserved	255	106–114		
DMG OCB (see 9.4.2.308)	255	115	Yes	No
Reserved	255	106–255 <u>116–255</u>		
...				

9.4.2.298 Ranging Parameters element

Insert two new rows above the last row, change the last row, and insert a table note at the end of Table 9-322a (as inserted by IEEE Std 802.11az-2022) as follows (unchanged rows not shown):

Table 9-322a—Format and Bandwidth subfield

Field value	Format	Bandwidth
...		
6	NGV	10
7	NGV	20
6-63 8-63	Reserved	Reserved
NOTE—Values 6 and 7 are supported only when dot11NGVOptionImplemented is true.		

Insert 9.4.2.308 after 9.4.2.307 (as inserted by IEEE Std 802.11az-2022) as follows:

9.4.2.308 DMG OCB element

The format of the DMG OCB element is shown in Figure 9-788fm.

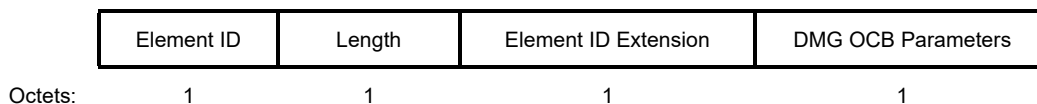


Figure 9-788fm—DMG OCB element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The DMG OCB Parameters field is defined in Figure 9-788fn.

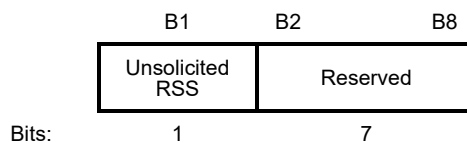


Figure 9-788fn—DMG OCB Parameters field format

The Unsolicited RSS subfield is defined in 9.5.5.

9.5 Fields used in Management and Extension frame bodies and Control frames

9.5.3 Sector Sweep Feedback field

Replace Figure 9-848 (as amended by IEEE Std 802.11ay-2021) with the following figure (with a new column for B17):

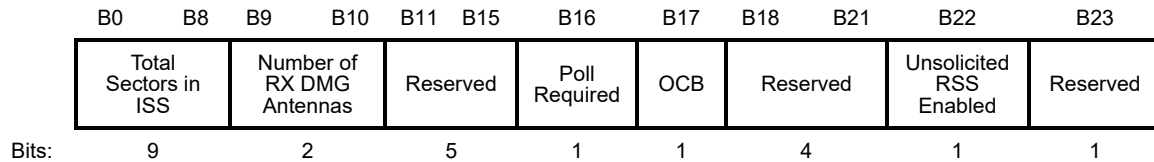


Figure 9-848—SSW Feedback field format when transmitted as part of an ISS

Replace Figure 9-849a (as inserted by IEEE Std 802.11ay-2021) with the following figure (with a new column for B17):

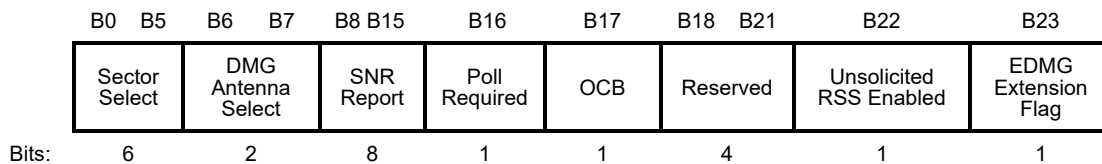


Figure 9-849a—SSW Feedback field format when not transmitted as part of an ISS and the EDMG Extension Flag subfield is 0

Insert the following paragraph at the end of 9.5.3:

The OCB subfield is set to 1 if dot11DMGOCBAActivated is true, and is set to 0 otherwise. If equal to 1, this subfield indicates that the STA is operating outside the context of a BSS.

9.7 Aggregate MPDU (A-MPDU)

9.7.1 A-MPDU format

Change the third paragraph of 9.7.1 as follows:

In an HT-~~or~~₂ DMG₂ or NGV PPDU, the final A-MPDU subframe is not padded. In a VHT, EDMG, or S1G PPDU, padding is added as described below.

9.7.3 A-MPDU contents

Change the first paragraph of 9.7.3 as follows:

In a non-DMG PPDU, an A-MPDU is a sequence of A-MPDU subframes carried in a single PPDU with one of the following combinations of RXVECTOR or TXVECTOR parameter values:

- The FORMAT parameter set to VHT
- The FORMAT parameter set to HT_MF or HT_GF and the AGGREGATION parameter set to 1
- The FORMAT parameter set to S1G, S1G_DUP_1M, or S1G_DUP_2M and the AGGREGATION parameter set to 1
- The FORMAT parameter set to HE_SU, HE_MU, HE_TB, or HE_ER_SU
- The FORMAT parameter set to NGV

10. MAC sublayer functional description

10.2 MAC architecture

10.2.3 Hybrid coordination function (HCF)

10.2.3.2 HCF contention based channel access (EDCA)

Change the second paragraph of 10.2.3.2 as follows:

For each AC an enhanced variant of the DCF, called an *enhanced distributed channel access function (EDCAF)*, contends for TXOPs using a set of EDCA parameters. When communicating Data frames outside the context of a BSS (`dot11OCBActivated` is true), the EDCA parameters are the corresponding default values or are as set by the SME in `dot11EDCATable` (except for TXOP limits, which shall be when the TXOP limits are set to 0 for each AC as specified in 10.23.2.9). For a STA operating OCB the STA's transmit queue for an AC may be cleared by the invocation of the MLME-CANCELTX.request primitive (see 6.3.126). For a non-AP STA communicating within a non-mesh QoS BSS, the EDCA parameters used are from the EDCA Parameter Set element or (for a non-AP STA prior to associating with an AP of an infrastructure BSS, a mesh STA, or a STA that operates OCB) from the default values for the parameters. The parameters used by the EDCAF to control its operation are defined by `dot11QAPEDCATable` at the AP and by `dot11EDCATable` at the non-AP STA.

10.3 DCF

10.3.6 Group addressed MPDU transfer procedure

Change the fourth paragraph of 10.3.6 as follows:

A DMG STA may transmit a copy of the same group addressed MPDU using different antenna configurations. This might be needed to provide a quasi-omni coverage or to enable transmission by an MCS that is higher than MCS 0. A DMG STA with `dot11DMGOCBActivated` equal to true may transmit copies of the same group addressed MPDU using different antenna configurations to provide sufficient coverage to communicate with peer STAs with which the DMG STA is maintaining beam links and may omit transmissions that use an antenna configuration over which a beamformed link is not maintained. If multiple copies of a group addressed MPDU with a To DS subfield equal to 0 are transmitted, the STA shall not transmit a different frame before the completion of the transmission of all copies of the group addressed MPDU.

10.12 A-MPDU operation

10.12.2 A-MPDU length limit rules

Insert the following paragraph and note at the end of 10.12.2:

An NGV STA may transmit an A-MPDU without capability exchange as specified in 31.2.3. An NGV STA shall support reception of an A-MPDU as specified in 31.2.3.

NOTE—An NGV STA does not use the HT Capabilities element or the VHT Capabilities element to establish the maximum length of an A-MPDU, as an NGV STA only operates OCB and need not exchange these elements.

10.12.4 A-MPDU aggregation of group addressed Data frames

Change the first paragraph as follows:

A STA that is ~~neither not one of an AP, nor a mesh STA, or an NGV STA~~ shall not transmit an A-MPDU containing an MPDU with an RA that is a group address.

10.13 PPDU duration constraint

Insert the following paragraph at the end of 10.13:

A STA shall not transmit an NGV PPDU that has a duration (as determined by the PHY-TXTIME.confirm primitive defined in 6.5.6) that is greater than aPPDUMaxTime defined in Table 32-20.

10.23 HCF

10.23.2 HCF contention based channel access (EDCA)

10.23.2.8 Multiple frame transmission in an EDCA TXOP

Insert two new items after the item beginning “A Ranging NDP Announcement frame followed after SIFS by an HE Ranging NDP...” (inserted by IEEE Std 802.11az-2022) at the end of the dashed list after the first paragraph of 10.23.2.8 as follows:

- A Ranging NDP Announcement frame followed after SIFS by an NGV NDP followed after SIFS by another NGV NDP followed after SIFS by an LMR frame.
- A Ranging NDP Announcement frame followed after SIFS by an NGV NDP followed after SIFS by another NGV NDP followed after SIFS by an LMR frame followed after SIFS by another LMR frame.

10.23.2.9 TXOP limits

Change the fourth paragraph of 10.23.2.9 as follows:

When dot11OCBAActivated is true, dot11NGVOptionImplemented is false or not present, and dot11DMGOCBAActivated is false or not present, the TXOP limits-limit shall be 0 for each AC.

10.25 Block acknowledgment (block ack)

10.25.1 Introduction

Change the second paragraph of 10.25.1 (as amended by IEEE Std 802.11ay-2021) as follows:

The block ack mechanism is initialized by an exchange of ADDBA Request/Response frames, except for GLK-GCR block ack and block ack between two NGV STAs, or by using the unsolicited block ack extension mechanism. After initialization, blocks of QoS Data frames may be transmitted from the originator to the recipient. A block may be started within a polled TXOP, within an SP, or by winning EDCA contention. The number of frames in the block is limited, and the amount of state that is to be kept by the recipient is bounded. The MPDUs within the block of frames are acknowledged by a BlockAck frame, which is requested by a BlockAckReq frame.

10.25.2 Setup and modification of the block ack parameters

Insert the following paragraph at the end of 10.25.2:

An NGV STA does not initialize the block ack mechanism as defined in this subclause; a block ack agreement with parameters defined in 31.2.3 is always in place.

11. MLME

11.1 Synchronization

11.1.3 Maintaining synchronization

11.1.3.4 DMG beacon generation before establishment of a BSS

Change the now ninth paragraph of 11.1.3.4 (after insertion of a paragraph by IEEE Std 802.11ay-2021) as follows:

A STA ~~with dot11DMGOCBActivated equal to false~~ that is transmitting DMG Beacon frames with the Discovery Mode field equal to 1 should cease transmitting these beacons when it has received a DMG Beacon frame from another STA, or when it has received acknowledgment of a transmitted Probe Response frame. If a BSS is not initialized as a result of the channel scanning, the STA can resume transmitting DMG Beacon frames with the Discovery Mode field equal to 1.

Insert a new paragraph after the now ninth paragraph of 11.1.3.4 as follows:

A STA with dot11DMGOCBActivated equal to true that is transmitting DMG Beacon frames with the Discovery Mode field equal to 1 may continue transmitting these beacons for discovery as described in 11.1.4.7 even if it has received a DMG Beacon frame from another STA.

11.1.4 Acquiring synchronization, scanning

11.1.4.1 General

Insert the following at the end of 11.1.4.1:

Upon receipt of the MLME-DMG-OCB-START.request primitive, a DMG STA with dot11DMGOCBActivated equal to true shall start a discovery procedure outside the context of a BSS as described in 11.1.4.7 and start operation outside the context of a BSS as described in 31.3.

Insert 11.1.4.7 after 11.1.4.6 as follows:

11.1.4.7 DMG discovery outside the context of a BSS

This subclause applies to a DMG STA when dot11DMGOCBActivated is true.

Upon receipt of the MLME-DMG-OCB-START.request primitive, a DMG STA continuously discovers new peer STAs. If the discovery beacon parameter is equal to true, the STA shall start transmitting DMG Beacon frames with the Discovery Mode field set to 1, the OCB subfield in the Clustering Control field set to 1, and the Beacon Interval field set to a random value as described in 11.1.3.4. If the discovery beacon parameter is equal to false, the DMG STA shall not transmit DMG Beacon frames.

When the STA receives one or more SSW frames with the OCB subfield equal to 1 during an A-BFT or DTI and completes SLS with the peer STA, and the address of the peer STA is an address that is newly discovered, the STA shall issue an MLME-OCB-DMGDISCOVERY.indication with the PeerInfoSet parameter including the PeerInfo vector defined in 6.3.128.5 for the peer STA that transmitted the SSW frame.

When the STA receives one or more DMG Beacon frames with the OCB subfield equal to 1 from a peer STA, and the address of the peer STA is an address that is newly discovered, the STA shall perform an SLS

with the OCB subfield set to 1 in transmitted SSW frames during the A-BFT following the DMG Beacon frames or the DTI. If the SLS is completed, the STA shall issue an MLME-OCB-DMGDISCOVERY.indication with the PeerInfoSet parameter including the PeerInfo for the peer STA that transmitted the DMG Beacon frame.

When the STA completes SLS with a peer STA that transmitted an SSW frame with the OCB subfield equal to 1 or a DMG Beacon frame with the OCB subfield equal to 1, and the address of the peer STA is not an address that is newly detected, the STA may issue an MLME-OCB-DMGDISCOVERY.indication with the PeerInfoSet parameter including the PeerInfo regarding the peer STA.

Upon receipt of the MLME-DMG-OCB-STOP.request primitive, a DMG STA shall terminate transmission of DMG Beacon frames and cease discovery of peer STAs.

Figure 11-10a illustrates an example of the DMG discovery outside the context of a BSS, in which the discovery beacon parameter is equal to true in the MLME-DMG-OCB-START.request primitive for both STAs.

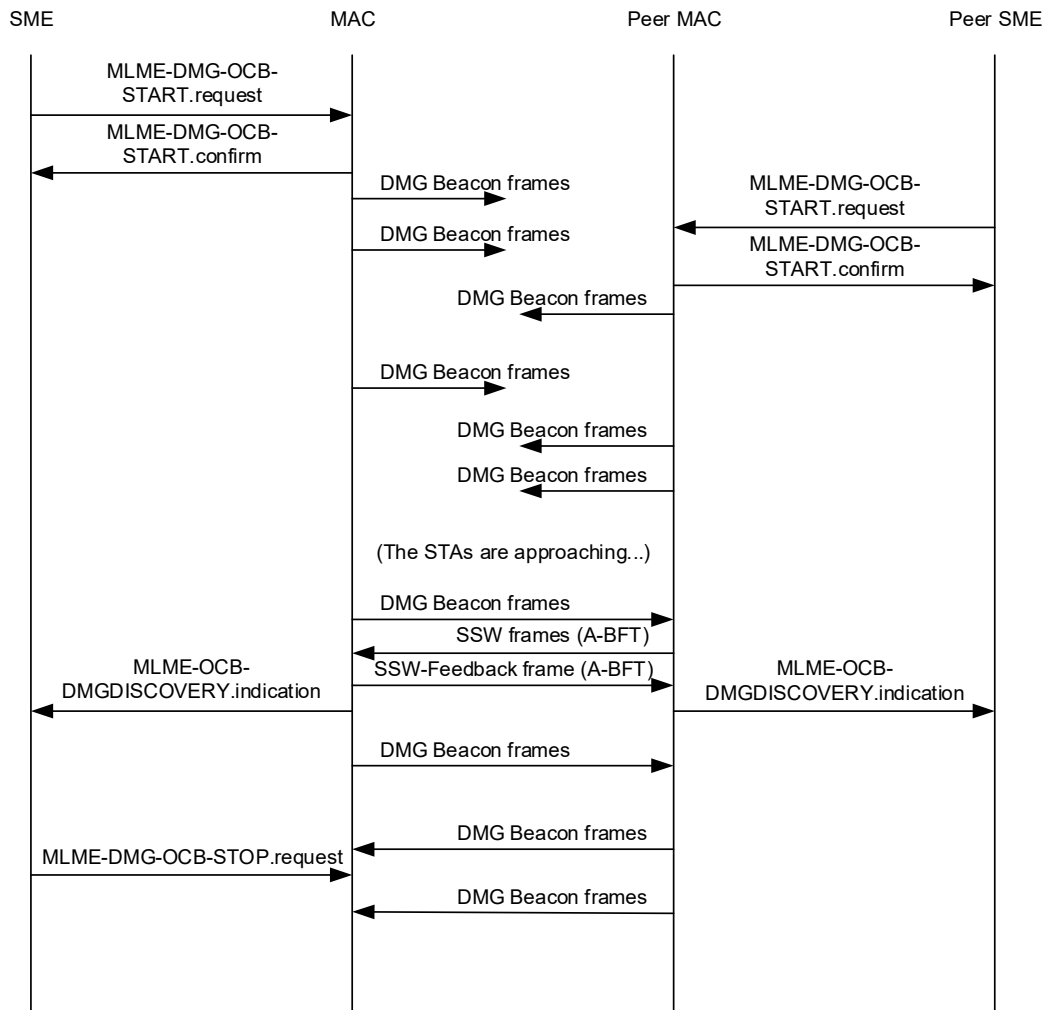


Figure 11-10a—DMG discovery outside the context of a BSS

Figure 11-10b illustrates an example of beamforming training during the DMG discovery outside the context of a BSS, in which the MAC address of the peer STA is informed over a higher layer and included in the MLME-BF-TRAINING.request primitive.

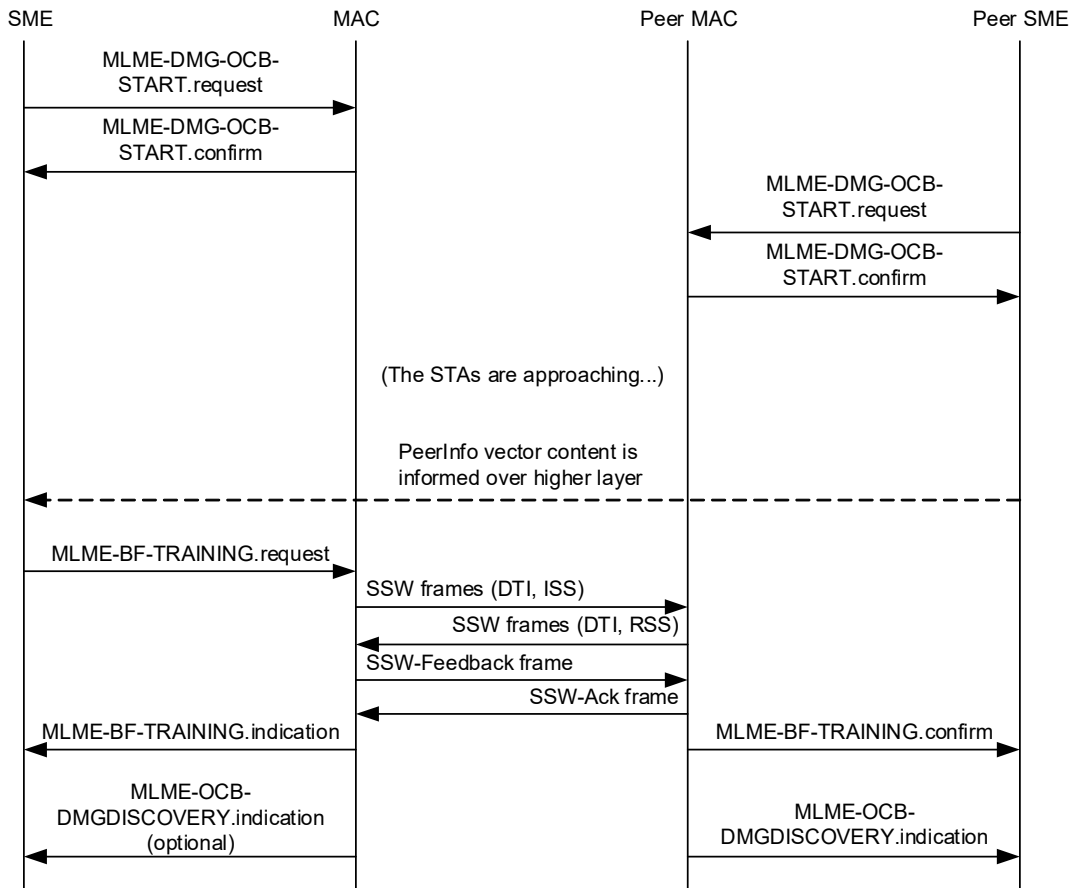


Figure 11-10b—Beamforming training during the DMG discovery outside the context of a BSS

11.18 STAs communicating Data frames outside the context of a BSS

Change item b) in the lettered list after the first paragraph of 11.18 as follows:

- b) The STA may send Control frames, except those of subtype PS-Poll and CF-End. If the STA is a DMG STA for which dot11DMGOCBActivated is true, in addition to not sending Control frames of subtype PS-Poll and CF-End, the STA shall not send Control frames of subtype Poll, SPR, Grant, Grant Ack, Sector Ack, Block Ack Schedule, and TDD Beamforming.

Insert a new item e) at the end of the lettered list after the first paragraph of 11.18 as follows:

- e) If the STA is a DMG STA for which dot11DMGOCBActivated is true, the STA may send DMG Beacon frames. A STA sending DMG Beacon frames shall set the Discovery Mode field to 1, and set the CBAP Only and CBAP Source subfields in the DMG Parameters field to 1 and 0, respectively.

11.21 Wireless network management procedures

11.21.6 Fine timing measurement (FTM) procedure

11.21.6.4 Measurement exchange

11.21.6.4.4 Non-TB ranging measurement exchange

11.21.6.4.4.2 Measurement sounding phase of non-TB ranging

Change the first paragraph of 11.21.6.4.4.2 (as inserted by IEEE Std 802.11az-2022) as follows:

An ISTA shall initiate a non-TB ranging measurement instance by transmitting a Ranging NDP Announcement frame addressed to the RSTA, followed by an I2R NDP SIFS after. In response to the correctly received Ranging NDP Announcement frame addressed to itself, the RSTA shall transmit an R2I NDP; see Figure 11-37i. I2R NDP and R2I NDP refer to NGV Ranging NDPs when dot11NGVOptionImplemented is true and HE ranging NDPs otherwise. The measurement-reporting phase consists of an LMR frame, which is a Location Measurement Report as defined in 9.6.7.49.

Change the 11th paragraph of 11.21.6.4.4.2 (as inserted by IEEE Std 802.11az-2022) as follows:

Accordingly:

- An ISTA transmitting an I2R NDP shall set the TXVECTOR parameter CH_BANDWIDTH to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding Ranging NDP Announcement frame.
- An RSTA transmitting an R2I NDP shall set the TXVECTOR parameter CH_BANDWIDTH to the bandwidth of the Ranging NDP Announcement frame and/or the I2R NDP; which are obtained from the RXVECTOR parameter CH_BANDWIDTH of the Ranging NDP Announcement frame or I2R NDP, respectively. For the Ranging NDP Announcement frame, when not received in an HE/VHT/HT/NGV PPDU: from the RXVECTOR parameter CH_BANDWIDTH_IN_NON_HT when the Ranging NDP Announcement frame is received in a non-HT duplicate PPDU and is 20 MHz when the Ranging NDP Announcement frame is received in a non-HT PPDU.

11.21.6.4.6 Transmission of a ranging NDP

Insert the following new paragraphs at the end of 11.21.6.4.6 (as inserted by IEEE Std 802.11az-2022):

An RSTA transmitting an NGV ranging NDP to an ISTA shall set the TXVECTOR parameters as follows:

- The FORMAT parameter is set to NGV.
- The CH_BANDWIDTH parameter is set to the same value as the RXVECTOR parameter CH_BANDWIDTH in the preceding I2R NDP.
- The NGV-LTF-TYPE parameter is set to 1 to indicate NGV_LTF_2X is used in the NGV ranging NDP.
- The NUM_SS parameter is set to the same value as the R2I N_STS in the STA Info field in the preceding Ranging NDP Announcement frame.
- The APEP_LENGTH parameter is set to 0.
- The LTF_REP parameter is set to the same value as the R2I LTF_REP in the STA Info field in the preceding Ranging NDP Announcement frame.

An ISTA transmitting an NGV ranging NDP to an RSTA shall set the TXVECTOR parameters as follows:

- The FORMAT parameter is set to NGV.

- The CH_BANDWIDTH parameter is set to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding Ranging NDP Announcement frame.
- The NGV-LTF-TYPE parameter is set to 1 to indicate NGV_LTF_2X is used in the NGV ranging NDP.
- The NUM_SS parameter is set to the same value as the I2R N_STS in the STA Info field in the preceding Ranging NDP Announcement frame.
- The APEP_LENGTH parameter is set to 0.
- The LTF_REP parameter is set to the same value as the I2R LTF_REP in the STA Info field in the preceding Ranging NDP Announcement frame.

11.27 DMG beamformed link and BSS maintenance

11.27.1 Beamformed link maintenance

11.27.1.1 General

Insert a new paragraph and note after the ninth paragraph (beginning “The recipient DMG STA shall ...”) of 11.27.1.1 (as re-numbered by IEEE Std 802.11ay-2021) as follows:

When a DMG STA with dot11DMGOCBActivated equal to true establishes a beamformed link with a peer STA, the STA shall generate an MLME-OCB-LINKSTATUS.indication with the BeamLinkStatus parameter set to ESTABLISHED or an MLME-OCB-DMGDISCOVERY.indication to inform the SME of the establishment of the beamformed link with the peer STA.

NOTE—When the beamformed link was established as a result of the discovery and the STA generated an MLME-OCB-DMGDISCOVERY.indication primitive as specified in 11.1.4.7, the STA might omit generation of an MLME-OCB-LINKSTATUS.indication.

Insert a new paragraph after the now 15th paragraph (beginning “Any time after dot11BeamLinkMaintenancetime has elapsed, the initiator ...”) of 11.27.1.1 (as re-numbered by IEEE Std 802.11ay-2021) as follows:

Upon expiration of dot11BeamLinkMaintenancetime, a DMG STA for which dot11DMGOCBActivated is true shall generate an MLME-OCB-LINKSTATUS.indication primitive with BeamLinkStatus parameter set to EXPIRED to inform the SME of the expiration of the beam link maintenance timer with the peer STA.

12. Security

12.13 Preassociation security negotiation

12.13.1 General

Change the second paragraph of 12.13.1 (as inserted by IEEE Std 802.11az-2022) as follows (with the following dashed list unchanged):

PASN is primarily intended for use in infrastructure networks for a STA and an AP to establish a PTKSA using a three-message authentication frame exchange. It may also be used by two STAs each co-located with an NGV STA (see 31.4). Some salient aspects of this protocol are:

12.13.3 Key establishment with PASN authentication

12.13.3.1 Overview

Change the first paragraph of 12.13.3.1 (as inserted by IEEE Std 802.11az-2022) as follows:

This subclause defines the procedure for establishing a PTKSA and the corresponding shared keys between a PASN capable STA and AP or two PASN capable STAs each co-located with an NGV STA (see 31.4). The PASN frame sequence used is depicted in Figure 12-56. It consists of three Authentication frames with the Authentication Algorithm Number field (9.4.1.1) set to 7 (PASN authentication) and the corresponding Authentication Transaction Sequence Number field (9.4.1.2) set to 1, 2, and 3, respectively.

Insert a note in Figure 12-56 as follows:

NOTE—Beacon is only present when the authenticator is an AP and not otherwise. For example when an authenticator is a non-AP STA co-located with an NGV STA, beacons are not present (see 31.4).

12.13.3.2 PASN frame construction and processing

Change the first item in the dashed list after the first paragraph of 12.13.3.2 (as inserted by IEEE Std 802.11az-2022) as follows:

- Base AKMP from among AKMPs advertised by the AP or provisioned by a higher layer (applicable for STAs co-located with NGV STAs (see 31.4)) if RSNA authentication is desired. Otherwise, if `dot11NoAuthPASNActivated` is true, Base AKMP chosen is the PASN AKMP, indicating that PTKSA is to be established without mutual authentication, that is, without a corresponding PMKSA.

17. Orthogonal frequency division multiplexing (OFDM) PHY specification

17.2 OFDM PHY specific service parameter list

17.2.2 TXVECTOR parameters

17.2.2.1 General

Insert two new rows at the end of Table 17-1 as follows (unchanged rows not shown):

Table 17-1—TXVECTOR parameters

Parameter	Associated primitive	Value
...		
CH_BANDWIDTH_IN_NON_NGV	PHY-TXSTART.request (TXVECTOR)	If present, CBW10 or CBW20
DYN_BANDWIDTH_IN_NON_NGV	PHY-TXSTART.request (TXVECTOR)	If present, Static or Dynamic

Insert 17.2.2.9 and 17.2.2.10 after 17.2.2.8 as follows:

17.2.2.9 TXVECTOR CH_BANDWIDTH_IN_NON_NGV

If present, this parameter is used to modify the first 7 bits of the scrambling sequence to indicate the bandwidth of the non-NGV duplicate PPDU.

NOTE—The CH_BANDWIDTH_IN_NON_NGV parameter is not present when the frame is transmitted by a non-NGV STA. The CH_BANDWIDTH_IN_NON_NGV parameter is not present when the frame is transmitted by an NGV STA and a non-NGV STA is one of the intended recipients (including group addressed transmissions).

17.2.2.10 TXVECTOR DYN_BANDWIDTH_IN_NON_NGV

If present, this parameter is used to modify the first 7 bits of the scrambling sequence to indicate if the transmitter is capable of Static or Dynamic bandwidth operation. If DYN_BANDWIDTH_IN_NON_NGV is present, then CH_BANDWIDTH_IN_NON_NGV is also present.

NOTE—The DYN_BANDWIDTH_IN_NON_NGV parameter is not present when the frame is transmitted by a non-NGV STA. The DYN_BANDWIDTH_IN_NON_NGV parameter is not present when the frame is transmitted by an NGV STA and a non-NGV STA is one of the intended recipients (including group addressed transmissions).

17.2.3 RXVECTOR parameters

17.2.3.1 General

Insert two new rows at the end of Table 17-2 as follows (unchanged rows not shown):

Table 17-2—RXVECTOR parameters

Parameter	Associated primitive	Value
...		
CH_BANDWIDTH_IN_NON_NGV	PHY-RXSTART.request (RXVECTOR)	If present, CBW10 or CBW20
DYN_BANDWIDTH_IN_NON_NGV	PHY-RXSTART.request (RXVECTOR)	If present, Static or Dynamic
...		

Insert 17.2.3.9 and 17.2.3.10 after 17.2.3.8 as follows:

17.2.3.9 RXVECTOR CH_BANDWIDTH_IN_NON_NGV

If present and valid, this parameter indicates the bandwidth of the non-NGV duplicate PPDU. This parameter is used by the MAC only when valid (see 10.3.2.9 and 10.6.6.6).

NOTE—The CH_BANDWIDTH_IN_NON_NGV parameter is not present when the frame is received by a non-NGV STA (see 10.6.12).

17.2.3.10 RXVECTOR DYN_BANDWIDTH_IN_NON_NGV

If present and valid, this parameter indicates whether the transmitter is capable of Static or Dynamic bandwidth operation. This parameter is used by the MAC only when valid (see 10.3.2.9 and 10.6.6.6). If DYN_BANDWIDTH_IN_NON_NGV is present, then CH_BANDWIDTH_IN_NON_NGV is also present.

NOTE—The DYN_BANDWIDTH_IN_NON_NGV parameter is not present when the frame is received by a non-NGV STA (see 10.6.12).

17.3 OFDM PHY

17.3.2 PPDU format

17.3.2.2 Overview of the PPDU encoding process

Change item e) in the lettered list in 17.3.2.2 as follows:

- e) If neither of the TXVECTOR parameters CH_BANDWIDTH_IN_NON_HT and CH_BANDWIDTH_IN_NON_NGV is present, initiate the scrambler with a pseudorandom nonzero seed and generate a scrambling sequence. If the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT or CH_BANDWIDTH_IN_NON_NGV is present, construct the first 7 bits of the scrambling sequence from CH_BANDWIDTH_IN_NON_HT or CH_BANDWIDTH_IN_NON_NGV, DYN_BANDWIDTH_IN_NON_HT (if present) or DYN_BANDWIDTH_IN_NON_NGV (if present), and a pseudorandom integer constrained such

that the first 7 bits of the scrambling sequence are not all 0s; then set the scrambler state to these 7 bits and generate the remainder of the scrambling sequence. XOR the scrambling sequence with the extended string of data bits. Refer to 17.3.5.5 for details.

17.3.5 DATA field

17.3.5.5 PHY DATA scrambler and descrambler

Replace Figure 17-7 with the following figure:

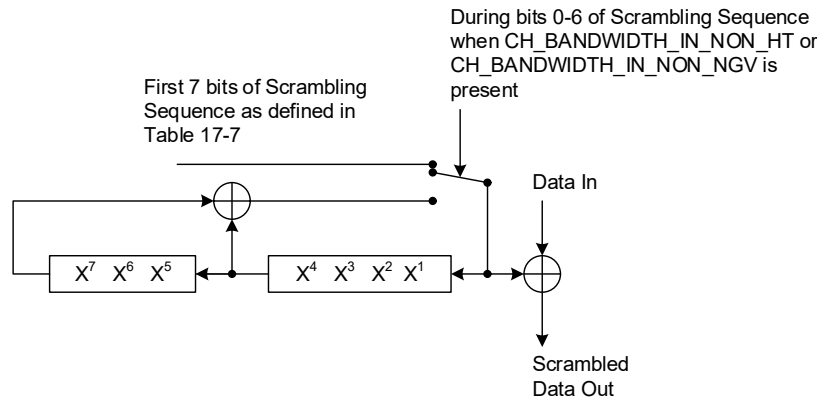


Figure 17-7—Data scrambler

Change the second and third paragraphs of 17.3.5.5 as follows:

The same scrambler is used to scramble transmit data and to descramble receive data. If neither of the TXVECTOR parameters CH_BANDWIDTH_IN_NON_HT and CH_BANDWIDTH_IN_NON_NGV is not present, when transmitting, the initial state of the scrambler shall be set to a pseudorandom nonzero state. If the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT or CH_BANDWIDTH_IN_NON_NGV is present;

- The first 7 bits of the scrambling sequence shall be set as shown in Table 17-7 (with field values defined in Table 17-8 ~~and, Table 17-8a, Table 17-10, and Table 17-10a~~) and shall be also used to initialize the state of the scrambler.
- The scrambler with this initialization shall generate the remainder (i.e., after the first 7 bits) of the scrambling sequence as shown in Figure 17-7.
- CH_BANDWIDTH_IN_NON_HT is transmitted LSB first. For example, if CBW80 has a value of 2, which is 10 in binary representation, then B5=0 and B6=1.
- CH_BANDWIDTH_IN_NON_NGV is transmitted LSB first.

If the TXVECTOR parameter SCRAMBLER_RESET is ~~set equal~~ to RESET_SCRAMBLER and dot11MACPrivacyActivated is true, the initial state of the scrambler shall be set to a nonzero random value not based on the scrambler value at the end of the last transmitted PPDU, before changes based on CH_BANDWIDTH_IN_NON_HT or CH_BANDWIDTH_IN_NON_NGV defined above are applied.

Change Table 17-7 as follows:

Table 17-7—Contents of the first 7 bits of the scrambling sequence

Parameter	Condition	First 7 bits of scrambling sequence				
		B0	B3	B4	B5	B6
						
TXVECTOR	CH_BANDWIDTH_IN_NON_HT is present and DYN_BANDWIDTH_IN_NON_HT is not present in TXVECTOR; or CH_BANDWIDTH_IN_NON_NGV is present and DYN_BANDWIDTH_IN_NON_NGV is not present in TXVECTOR	5-bit pseudorandom nonzero integer if CH_BANDWIDTH_IN_NON_HT (if present) equals CBW20 or CH_BANDWIDTH_IN_NON_NGV (if present) equals CBW10; and a 5-bit pseudorandom integer otherwise			CH_BANDWIDTH_IN_NON_HT (if present) or CH_BANDWIDTH_IN_NON_NGV (if present)	
TXVECTOR	CH_BANDWIDTH_IN_NON_HT is present and DYN_BANDWIDTH_IN_NON_HT is present in TXVECTOR; or CH_BANDWIDTH_IN_NON_NGV is present and DYN_BANDWIDTH_IN_NON_NGV is present in TXVECTOR	4-bit pseudorandom nonzero integer if CH_BANDWIDTH_IN_NON_HT (if present) equals CBW20 and DYN_BANDWIDTH_IN_NON_HT (if present) equals Static, or CH_BANDWIDTH_IN_NON_NGV (if present) equals CBW10 and DYN_BANDWIDTH_IN_NON_NGV (if present) equals Static; and a 4-bit pseudorandom integer otherwise		DYN_BANDWIDTH_IN_NON_HT (if present) or DYN_BANDWIDTH_IN_NON_NGV (if present)		
RXVECTOR	CH_BANDWIDTH_IN_NON_HT and DYN_BANDWIDTH_IN_NON_HT are present in RXVECTOR; or CH_BANDWIDTH_IN_NON_NGV and DYN_BANDWIDTH_IN_NON_NGV are present in RXVECTOR	—		DYN_BANDWIDTH_IN_NON_HT (if present) or DYN_BANDWIDTH_IN_NON_NGV (if present)		CH_BANDWIDTH_IN_NON_HT_INDICATOR or CH_BANDWIDTH_IN_NON_NGV_INDICATOR (see Table 17-9 and Table 17-9a).

Insert Table 17-8a and a new paragraph after Table 17-8 as follows:

Table 17-8a—TXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV values

Enumerated value	Value
CBW10	0
CBW20	1

During reception by an NGV STA, the RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV shall be determined from selected bits in the scrambling sequence as shown in Table 17-7 and Table 17-9a. During reception by an NGV STA, the RXVECTOR parameter DYN_BANDWIDTH_IN_NON_NGV shall be set to selected bits in the scrambling sequence as shown in Table 17-7 and Table 17-10a. The fields shall be interpreted as being sent LSB-first.

Insert Table 17-9a after Table 17-9 as follows:

Table 17-9a—RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV values

CH_BANDWIDTH_IN_NON_NGV_INDICATOR field of first 7 bits of scrambling sequence	RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV
0	CBW10
1	CBW20

Insert Table 17-10a after Table 17-10 as follows:

Table 17-10a—DYN_BANDWIDTH_IN_NON_NGV values

Enumerated value	Value
Static	0
Dynamic	1

Change NOTE 2 and NOTE 3 in 17.3.5.5 as follows:

NOTE 2—The receiving PHY cannot determine whether the CH_BANDWIDTH_IN_NON_HT, ~~and~~ DYN_BANDWIDTH_IN_NON_HT, CH_BANDWIDTH_IN_NON_NGV, and DYN_BANDWIDTH_IN_NON_NGV parameters were present in the TXVECTOR of the transmitting PHY; therefore, the receiving PHY in a VHT STA always includes values for the CH_BANDWIDTH_IN_NON_HT and DYN_BANDWIDTH_IN_NON_HT parameters in the RXVECTOR if the PPDU is a non-HT PPDU; ~~the receiving PHY in an NGV STA always includes values for the~~ CH_BANDWIDTH_IN_NON_NGV and DYN_BANDWIDTH_IN_NON_NGV parameters in the RXVECTOR if the PPDU is a non-NGV PPDU. It is the responsibility of the MAC to determine the validity of the RXVECTOR parameters CH_BANDWIDTH_IN_NON_HT, ~~and~~ DYN_BANDWIDTH_IN_NON_HT, CH_BANDWIDTH_IN_NON_NGV, and DYN_BANDWIDTH_IN_NON_NGV.

NOTE 3—The receiving PHY cannot determine whether the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT or CH_BANDWIDTH_IN_NON_NG_V was present, but it does not matter since descrambling the DATA field is the same either way.

Insert new Clause 31 and Clause 32 after Clause 30 (inserted by IEEE Std 802.11ba-2021) as follows:

31. Next Generation V2X (NGV) MAC specification

31.1 Introduction

An NGV STA supports the MAC and MLME functions defined in 31.2 in addition to the MAC functions defined in Clause 10 and the MLME functions defined in Clause 11 for STAs communicating Data frames OCB.

If dot11DMGOCBAActivated is true, a DMG STA shall support the features defined in 31.3, the MAC functions defined in Clause 10, and the MLME functions defined in Clause 11 for DMG or EDMG STAs communicating Data frames OCB.

31.2 Operation in the 5.9 GHz band

31.2.1 Coexistence with non-NGV STAs

In order to enable coexistence and interoperability with non-NGV STAs, if dot11NGVOptionImplemented is true then dot11OCBAActivated shall be true. Moreover, the non-NGV PPDU transmitted by NGV STAs carry an indication that informs the receiving NGV STAs that the transmitter is an NGV STA. This indication is carried in the Duration/ID field of the MAC header.

NOTE 1—An NGV STA that wishes to communicate OCB with one or more neighboring non-NGV STAs is recommended to use the non-NGV PPDU format.

When an NGV STA transmits an Ack frame solicited by an individually addressed Management or QoS Data frame in a non-NGV PPDU and the Duration field value specified in 9.2.5.7 (Setting for control response frames) is 0, the Duration/ID field in the Ack frame shall be set to 2. When an NGV STA transmits an individually addressed Management or QoS Data frame in a non-NGV PPDU as the last frame of a TXOP, the Duration/ID field of the Management or QoS Data frame shall be set to the sum of 4, SIFS, and the transmission time of the responding Ack frame as defined in 10.6.

NOTE 2—An NGV STA can transmit either a non-NGV PPDU or an NGV PPDU. The value of the Duration field carried in a non-NGV PPDU indicates to a receiving NGV STA whether the transmitter is an NGV STA or a non-NGV STA.

When an NGV STA transmits a group addressed frame in a non-NGV PPDU, the Duration/ID field in the group addressed frame shall be set to 2.

An NGV STA determines that a received non-NGV PPDU was transmitted by an NGV STA if the non-NGV PPDU contains one of the following frames:

- An Ack or BlockAck frame with Duration/ID field equal to 2.
- An individually addressed frame with Duration/ID field equal to the sum of 4, SIFS, and the transmission time of the responding Ack or BlockAck frame as defined in 10.6.
- A group addressed frame with Duration/ID field equal to 2.

31.2.2 Channel access and transmission methods for 20 MHz OCB transmission

An NGV 20 MHz channel consists of two contiguous 10 MHz channels: the OCB primary channel and the OCB secondary channel.

The OCB primary channel is designated by the primary channel parameter of the radio environment vector in the MA-UNITDATA.request primitive (see 5.2.3).

An NGV STA transmitting a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU shall contend for the medium using EDCA as defined in 10.2.3.2 and 10.3.2 based on the medium sensing results of two contiguous 10 MHz channels, OCB primary channel and OCB secondary channel in an NGV 20 MHz channel. An NGV STA performing 20 MHz channel access determines that the 20 MHz medium is idle if all of the following conditions are met:

- The CS mechanism (see 10.3.2.1) indicates that the OCB primary channel is idle.
- The PHY-CCA.indication primitive indicates that the OCB secondary channel is idle.
- The virtual CS mechanism (see 10.3.2.1) indicates that the OCB secondary channel is idle, if dot11VirtualCSonOCBSecondaryImplemented is true.

Otherwise, the NGV STA performing 20 MHz channel access determines that the 20 MHz medium is busy.

NOTE—An NGV STA transmitting a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU follows rules defined in 31.2.2 and does not follow rules defined in 11.15.9.

If either the medium of the OCB primary channel or the OCB secondary channel is determined to be busy, an NGV STA shall invoke the random backoff procedure as described in 10.23.2.2.

When the random backoff procedure is invoked, an NGV STA transmitting a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU shall perform as described in 10.23.2.4, with the exception of the definition of the slot boundaries. In this subclause, the slot boundaries are defined as follows:

- a) Following $\text{AIFSN[AC]} \times \text{aSlotTime} - \text{aRXTxTurnaroundTime}$ of idle medium on the 20 MHz channel after SIFS (not necessarily idle medium during the SIFS) after the last busy medium on the antenna that was the result of reception of a frame with a correct FCS.
- b) Following $\text{EIFS} - \text{DIFS} + \text{AIFSN[AC]} \times \text{aSlotTime} + \text{aSIFSTime} - \text{aRXTxTurnaroundTime}$ of idle medium on the 20 MHz channel after the last indicated busy medium as determined by the physical CS mechanism that was the result of a frame reception that has resulted in FCS error, or of a frame reception that has resulted in a PHY-RXEND.indication (RXERROR) primitive where the value of RXERROR is not NoError.
- c) When any other EDCAF at this STA transmitted a frame requiring immediate acknowledgment, the earlier of:
 - 1) The end of the AckTimeout interval timed from the PHY-TXEND.confirm primitive, followed by $\text{AIFSN[AC]} \times \text{aSlotTime} + \text{aSIFSTime} - \text{aRXTxTurnaroundTime}$ of idle medium on the 20 MHz channel.
 - 2) The end of the first $\text{AIFSN[AC]} \times \text{aSlotTime} - \text{aRXTxTurnaroundTime}$ of idle medium on the 20 MHz channel after SIFS (not necessarily medium idle during the SIFS, the start of the SIFS implied by the length in the PHY header of the previous frame) when a PHY-RXEND.indication primitive occurs as specified in 10.3.2.11.
- d) Following $\text{AIFSN[AC]} \times \text{aSlotTime} - \text{aRXTxTurnaroundTime}$ of idle medium on the 20 MHz channel after SIFS (not necessarily medium idle during the SIFS) after the last busy medium on the antenna that was the result of transmission of a frame for any EDCAF and that did not require an acknowledgment and after the expiration of the TXNAV timer if nonzero.
- e) Following $\text{AIFSN[AC]} \times \text{aSlotTime} + \text{aSIFSTime} - \text{aRXTxTurnaroundTime}$ of idle medium on the 20 MHz channel after the last indicated busy medium as indicated by the CS mechanism on the OCB primary channel that is not covered by a) to d).
- f) Following $\text{EIFS} - \text{DIFS} + \text{AIFSN[AC]} \times \text{aSlotTime} + \text{aSIFSTime} - \text{aRXTxTurnaroundTime}$ of idle medium on the 20 MHz channel after the last indicated busy medium as indicated by the CCA mechanism on the OCB secondary channel that is not covered by a) to d), when the duration of channel busy is not known.

- g) Following $\text{AIFSN[AC]} \times \text{aSlotTime} + \text{aSIFSTime} - \text{aRxTxTurnaroundTime}$ of idle medium on the 20 MHz channel after the last indicated busy medium as indicated by the CS mechanism on the OCB secondary channel that is not covered by a) to d), when the duration of channel busy is known.
- h) Following aSlotTime of idle medium on the 20 MHz channel, which occurs immediately after any of these conditions, a) to g), is met for the EDCAF.

If an NGV STA is unable to transmit a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU because the medium of the OCB secondary channel is busy and the fallback enabled member of the radio environment request vector so allows, the STA may instead use the medium access procedure to transmit a 10 MHz PPDU on the OCB primary channel. The medium of the OCB secondary channel is busy if a `PHY-CCA.indication(BUSY, {secondary})` primitive occurs or the virtual CS indicates that the medium is busy on the OCB secondary channel, if `dot11VirtualCSonOCBSecondaryImplemented` is true.

31.2.3 A-MSDU operation, A-MPDU operation, and BA operation

NGV STA frame aggregation parameters are fixed and are defined in this clause. This provides the ability to transmit an A-MPDU and an A-MSDU outside the context of a BSS (OCB). An NGV STA may transmit an A-MPDU or A-MSDU and shall support the reception of an A-MSDU or A-MPDU with aggregation parameters as defined in this clause.

An NGV STA shall support receiving an MPDU of length equal to or less than the maximum NGV MPDU length in Table 31-1. An NGV STA shall not transmit an MPDU with a length greater than the maximum NGV MPDU length shown in Table 31-1.

Table 31-1—Maximum NGV MPDU length

NGV-MCS	1SS 10 MHz (octets)	2SS 10 MHz (octets)	1SS 20 MHz (octets)	2SS 20 MHz (octets)
0	4524	7991	7991	7991
1	7991	7991	7991	7991
2	7991	7991	7991	7991
3	7991	7991	7991	7991
4	7991	7991	7991	7991
5	7991	7991	7991	7991
6	7991	7991	7991	7991
7	7991	7991	7991	7991
8	7991	7991	7991	7991
9	N/A	N/A	7991	7991
15	2192	4524	4386	7991

NOTE 1—The maximum MPDU length is determined based on a maximum PPDU duration of 10 968 μs .

NOTE 2—The maximum PPDU duration may be reduced by regulatory authorities in some regulatory domains. If the maximum PPDU duration is less than 10 968 μs , the maximum MPDU length may be limited to a lower value for some MCSs.

An NGV STA shall transmit adjacent MPDUs in an A-MPDU with a minimum of 2 μs between the MPDUs.

An NGV STA shall support block ack and does so without exchanging capabilities or creating a block ack agreement. An NGV STA shall have the following block ack configuration:

- A-MSDU is supported
- Block ack policy is HT-immediate block ack
- No timeout
- The number of buffers available is 32
- Each buffer is capable of holding 7935 octets (the maximum size of an A-MSDU)
- The block ack frame is a Compressed BlockAck frame (see 9.3.1.8.1 and 9.3.1.8.2).

31.2.4 NON_NGV_10 repetition transmission

The NON_NGV_10 repetition transmission mode supports OCB broadcast service to both NGV STAs and non-NGV STAs with improved frame reception reliability.

This mode allows an NGV STA to repeat one NON_NGV_10 PPDU, defined in 32.1.4, multiple times with a time gap of SIFS between every two transmissions.

For a group addressed MPDU containing only group addressed MPDUs, the number of repetitions of the NON_NGV_10 PPDU is decided by a higher layer and indicated by the number of repetitions element of the radio environment request vector (5.2.3) in the MAC SAP. Otherwise, the number of repetitions of the NON_NGV_10 PPDU is fixed to 0 by the MAC. The MAC sets the number of repetitions, N_PPDU_REP, via the PHY service interface using the PHY-TXSTART.request(TXVECTOR) primitive, as described in Table 32-1.

NOTE—The Duration setting of the Duration/ID field in NON_NGV_10 repetition transmission does not follow the rules in 9.2.5. An NGV STA that transmits the frames in NON_NGV_10 repetition transmission mode follows the rules in 10.23.2.9.

31.2.5 Non-NGV duplication operation

A 20 MHz NGV STA may transmit RTS frames in non-NGV duplicate PPDU to protect a 20 MHz NGV PPDU.

A 20 MHz NGV STA that is addressed by an RTS frame in a non-NGV or non-NGV duplicate PPDU that has a bandwidth signaling TA and that has the RXVECTOR parameter DYN_BANDWIDTH_IN_NON_NGV equal to Static behaves as follows:

- If the NAV indicates idle and CCA indicates idle for the secondary 10 MHz channel when the RTS frame's RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV is equal to CBW20, then the STA shall respond with a CTS frame carried in a non-NGV duplicate PPDU after a SIFS. The CTS frame's TXVECTOR parameters CH_BANDWIDTH and CH_BANDWIDTH_IN_NON_NGV shall be set to CBW20.
- If the NAV indicates idle when the RTS frame's RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV is equal to CBW10, then the STA shall respond with a CTS frame carried in a non-NGV PPDU after a SIFS. The CTS frame's TXVECTOR parameters CH_BANDWIDTH and CH_BANDWIDTH_IN_NON_NGV shall be set to CBW10.
- Otherwise, the STA shall not respond with a CTS frame.

A 20 MHz NGV STA that is addressed by an RTS frame in a non-NGV or non-NGV duplicate PPDU that has a bandwidth signaling TA and that has the RXVECTOR parameter DYN_BANDWIDTH_IN_NON_NGV equal to Dynamic behaves as follows:

- If the NAV indicates idle and the CCA indicates idle for the secondary 10 MHz channel when the RTS frame's RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV is equal to CBW20,

then the 20 MHz NGV STA shall respond with a CTS frame in a non-NGV duplicate PPDU after a SIFS. The CTS frame's TXVECTOR parameters CH_BANDWIDTH and CH_BANDWIDTH_IN_NON_NGV shall be set to CBW20.

- If the NAV indicates idle and the CCA indicates busy for the secondary 10 MHz channel when the RTS frame's RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV is equal to CBW20, then the 20 MHz NGV STA shall respond with a CTS frame in a non-NGV PPDU after a SIFS. The CTS frame's TXVECTOR parameters CH_BANDWIDTH and CH_BANDWIDTH_IN_NON_NGV shall be set to CBW10.
- If the NAV indicates idle when the RTS frame's RXVECTOR parameter CH_BANDWIDTH_IN_NON_NGV is equal to CBW10, then the 20 MHz NGV STA shall respond with a CTS frame in a non-NGV PPDU after a SIFS. The CTS frame's TXVECTOR parameters CH_BANDWIDTH and CH_BANDWIDTH_IN_NON_NGV shall be set to CBW10.
- Otherwise, the STA shall not respond with a CTS frame.

If a 20 MHz NGV STA receives a frame that solicits a response and is carried in a 20 MHz NGV PPDU, it should transmit the response in a 20 MHz non-NGV duplicate PPDU.

31.3 Operation in the 60 GHz band

31.3.1 DMG beamforming outside the context of a BSS

A DMG STA with dot11DMGOCBActivated equal to true may transmit DMG Beacon frames as described in 10.42.4 outside the context of a BSS. The DMG STA shall set the Discovery Mode field and OCB subfield in the Clustering Control field to 1 in each of the DMG Beacon frames when the STA performs beamforming training with the DMG Beacon frame outside of the context of a BSS. The STA may set all the bits in the A-BFT Responder Address subfield to 1 in a DMG Beacon frame to allow any STAs to transmit SSW or Short SSW frames during A-BFT that follows the BTI in which the DMG Beacon frame is transmitted, or set the A-BFT Responder Address subfield to an individual address to indicate the STA that is allowed to transmit during the A-BFT. A DMG OCB element is optionally present in a DMG Beacon frame to indicate the STA transmitting the DMG Beacon frame supports optional features used during communication outside the context of a BSS.

When a DMG STA with dot11DMGOCBActivated equal to true receives a DMG beacon frame with the OCB subfield equal to 1, the STA may perform beamforming training as described in 10.42.5. When the DMG STA transmits SSW frames during the A-BFT after a BTI in which the STA received a DMG beacon frame with the OCB subfield equal to 1, the STA shall set the OCB subfield to 1 in the SSW frames transmitted during the A-BFT.

If a responder DMG STA that receives a DMG beacon frame with the OCB subfield equal to 1 has successfully completed an SLS or a BRP with the initiator STA that transmitted the DMG Beacon frame recently, the responder STA should not transmit SSW frames during the A-BFT following the BTI to avoid collisions during the A-BFT.

A DMG STA with dot11DMGOCBActivated equal to true may transmit SSW frames as described in 10.42.6 outside the context of a BSS. The DMG STA shall set the OCB subfield to 1 in the SSW frames during beamforming training outside of the context of a BSS.

A DMG STA with dot11DMGOCBActivated equal to true shall support Beamformed link maintenance procedure as described in 11.27.1.1.

31.4 NGV ranging

An NGV STA may support NGV ranging. NGV ranging is a subset of fine timing measurement (FTM) functionalities defined in 11.21.6; the supported functionalities are as follows:

- Fine timing measurement procedures, including FTM procedure negotiation and FTM session termination for non-TB ranging as defined in 11.21.6.3 and in 11.21.6.6, with the modifications specified in this subclause.
- Non-TB ranging as described in 11.21.6.4.4 with the modifications specified in this subclause. PHY security is not specified for NGV STAs.
- NGV STAs may support differential distance computation as detailed in P.3.

Additionally, if an NGV STA is co-located with a STA that supports Fine Timing Measurement (FTM) functionalities as defined in 11.21.6, it may support Fine Timing Measurement procedure outside the 5.9 GHz frequency band.

The capability discovery for whether an NGV STA supports ranging as specified in this subclause is out of the scope of IEEE Std 802.11 and is expected to be conducted by higher layers. The higher layer exchanges are over the 5.9 GHz frequency band. The higher layer negotiations are expected to identify the following:

- Whether an NGV STA supports non-TB ranging in the 5.9 GHz band.
- Whether an NGV STA is co-located with a STA that supports ranging on IEEE 802.11 bands outside the 5.9 GHz band and if so the type of ranging: non-TB ranging and/or EDCA based ranging (see 11.21.6.1.1 and 11.21.6.1.2).
- For an NGV STA co-located with a STA that supports ranging on 802.11 bands outside the 5.9 GHz band, whether the co-located STA supports PASN, and if so, the contents of the PASN Parameters element, or alternatively the security context required for exchange of protected frames.
- The channel number for ranging exchanges between peer STAs.
- Role of STA: ISTA or RSTA.
- LMR Feedback policy.

NOTE—The channel number is not limited to the 5.9 GHz band.

The fine timing measurement procedure negotiation is performed by an NGV STA as specified in 11.21.6.3 with the following differences:

- For ranging in the 5.9 GHz band, in the Ranging Parameters element included in the IFTMR frame:
 - Status indication subfield and Value subfield are reserved.
 - Max R2I STS > 80 MHz subfield is reserved.
 - Max I2R STS > 80 MHz subfield is reserved.
 - BSS Color Information subfield is reserved.
- For ranging in the 5.9 GHz band, a non-TB ranging measurement exchange is used by NGV STAs as defined in 11.21.6.4.4 with the following changes:
 - An NGV ranging NDP is transmitted instead of an HE ranging NDP.
 - A Ranging NDP Announcement frame (9.3.1.19) is transmitted. The following differences in the subfields of the STA Info fields of the Ranging NDP Announcement frame apply compared to 9.3.1.19:
 - LTF Offset is set to 0.
 - R2I N_STS subfield is set to the number of spatial streams of the R2I NGV ranging NDP minus 1.
 - R2I Rep subfield is set to 0 if the NGV-LTF in the R2I NGV ranging NDP is not repeated and is set to 1 if the NGV-LTF in the R2I NGV ranging NDP is repeated.

- I2R N_STS subfield is set to the number of spatial streams of the I2R NGV ranging NDP minus 1.
- I2R Rep subfield is set to 0 if the NGV-LTF in the I2R NGV ranging NDP is not repeated and is set to 1 if the NGV-LTF in the I2R NGV ranging NDP is repeated.
- The I2R N_STS and I2R Rep subfields are used to indicate the following I2R NGV ranging NDP's NGV-LTF configuration, 32.3.16, while the R2I N_STS and R2I Rep subfields indicate the NGV-LTF configuration of the R2I NGV ranging NDP sent in response by the RSTA, see 11.21.6.4.4.
- When the TXVECTOR parameter CH_BANDWIDTH is CBW10, the I2R NDP Tx Power subfield indicates the combined average power per 10 MHz bandwidth referenced to the antenna connector, of all antennas used to transmit the following I2R NDP.

For ranging on IEEE Std 802.11 bands outside the 5.9 GHz band, whether PASN is required is indicated by higher layers. Two STAs co-located with NGV STAs may establish PASN authentication if so indicated by higher layers using the procedures as defined in 12.13 with the following change: the exchanges between ISTA and RSTA are done by two STAs each co-located with an NGV STA. Alternatively, the security association between the two STAs may be established by higher layer exchanges in the 5.9 GHz band.

31.5 NGV MAC data service

An NGV STA shall follow the rules in 10.2.7 and the additional rules as defined in this subclause.

When a member in radio environment request vector represents “selection within MAC sublayer”, the NGV STA shall select the related member value by itself. Otherwise the following rules shall be applied:

- When transmitting an MPDU that encapsulates an MSDU, an NGV STA shall use the PPDU format indicated by the PPDU format member of the radio environment request vector related to the MSDU.
- When transmitting an MPDU that encapsulates an MSDU, an NGV STA shall use the data rate/NGV-MCS indicated by the data rate/NGV-MCS member of the radio environment request vector related to the MSDU in the initial transmission of the MPDU. In the retransmission of the MPDU, the data rate/NGV-MCS shall be no more than data rate/NGV-MCS indicated by the data rate/NGV-MCS member of the radio environment request vector related to the MSDU.
- When transmitting an MPDU that encapsulates an MSDU, an NGV STA shall use the number of spatial streams indicated by the number of spatial streams member of the radio environment request vector related to the MSDU in the initial transmission of the MPDU. In the retransmission of the MPDU, the number of spatial streams shall be no more than the number of spatial streams indicated by the number of spatial streams member of the radio environment request vector related to the MSDU.
- An NGV STA shall transmit a frame that encapsulates an MSDU in an A-MPDU if the permitted aggregation member of the radio environment request vector related to the MSDU is equal to 1. Otherwise the NGV STA shall not aggregate the frame in an A-MPDU.
- An NGV STA shall set the lifetime of an MSDU to the value of expiry time member of the radio environment request vector related to the MSDU.
- An NGV STA shall transmit an MPDU that encapsulates an MSDU in the channel defined by the channel member, primary channel member, and channel width member of the radio environment request vector related to the MSDU with the following exception:
 - If the channel width indicates 20 MHz channel width and the fallback enabled parameter indicates 1, the channel can be 10 MHz channel.

- An NGV STA shall transmit an MPDU that encapsulates an MSDU with the transmit power indicated by the transmit power spectral density member of the radio environment request vector related to the MSDU.

NOTE—The higher layer sets the parameters in the radio environment request vector based on the NGV STA's capabilities.

When reporting a received MSDU to a higher layer, an NGV STA shall report the radio environment status vector of the received MSDU as defined in 5.2.4.

31.6 Radio environment measurement

An NGV STA provides periodic reports on the radio environment.

If requested by a higher layer, an NGV STA with `dot11RadioEnvironmentMeasurementPeriod` not equal to 0 shall measure the channel busy percentage and report the measurement results to a higher layer as defined in 6.3.127.

If requested by a higher layer, an NGV STA with `dot11StationMeasurementPeriod` not equal to 0 shall measure the number of neighboring STAs (non-NGV STAs and NGV-STAs) and the number of neighboring NGV STAs, and report these measurement results to a higher layer as defined in 6.3.127.

An NGV STA with both `dot11StationMeasurementPeriod` and `dot11RadioEnvironmentMeasurementPeriod` not equal to 0 shall start a STA measurement period at the same time as a radio environment measurement period.

32. Next Generation V2X (NGV) PHY specification

32.1 Introduction

32.1.1 Introduction to NGV PHY

Clause 32 specifies the PHY entity for an NGV orthogonal frequency division multiplexing (OFDM) system. In addition to the requirements in Clause 32, an NGV STA shall be capable of transmitting and receiving 10 MHz PPDU specified in Clause 17.

The NGV PHY is mainly based on the VHT PHY defined in Clause 21, which in turn is based on the HT PHY defined in Clause 19, which in turn is further based on the OFDM PHY defined in Clause 17. The NGV PHY preamble structure, BPSK-dual carrier modulation (DCM), and midamble structure are based on the HE PHY defined in Clause 27.

The NGV PHY provides support for both 10 MHz and contiguous 20 MHz channel widths. The NGV PHY data subcarrier frequency spacing is a half of VHT PHY and HT PHY subcarrier frequency spacing defined in Clause 21 and Clause 19, respectively.

The NGV PHY data subcarriers are modulated using binary phase shift keying (BPSK), BPSK-DCM, quadrature phase shift keying (QPSK), 16-quadrature amplitude modulation (16-QAM), 64-QAM, and 256-QAM. The NGV PHY preamble is encoded by a convolutional encoder and the NGV PHY data payload is encoded by an LDPC encoder. STBC is not employed.

The NGV PHY supports the regulatory requirements in 32.3.15.

An NGV STA shall support the following features:

- Single spatial stream
- NGV-MCS 0 to 9 and NGV-MCS 15
- Three LTF formats: NGV-LTF-1x, NGV-LTF-2x, and NGV-LTF-2x-Repeat
- LDPC coding (transmit and receive)
- Midamble periodicity of 4, 8, 16 OFDM symbols
- 10 MHz NGV PPDU
- NON_NGV_10 PPDU and repetitive NON_NGV_10 PPDU
- Transmit spectrum mask C2 when transmitting a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU, if 20 MHz NGV PPDU is supported
- Transmit spectrum mask A, B, C, or D when transmitting a 10 MHz NGV PPDU, if the STA belongs to transmit power class A, B, C, or D, respectively

An NGV STA may support the following features:

- Transmission and reception of SU MIMO with two spatial streams
- NGV ranging NDP for NGV ranging as described in 31.4
- 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU

32.1.2 Scope

The services provided to the MAC by the NGV PHY consist of the following protocol functions:

- a) A function that maps the PSDU received from the MAC into a PPDU for transmission to one or more receiving STAs.

- b) A function that defines the characteristics and method of transmitting and receiving data through a wireless medium between two or more STAs. Depending on the PPDU format, these STAs support a mixture of NGV and Clause 17 PHYs.

32.1.3 NGV PHY functions

32.1.3.1 General

The NGV PHY contains two functional entities: the PHY function, and the physical layer management function (via the PLME). These functions are described in detail in 32.3 and 32.4, respectively. The NGV PHY service is provided to the MAC through the PHY service primitives defined in Clause 8. The NGV PHY service interface is described in 32.2.

32.1.3.2 PHY layer management entity (PLME)

The PLME performs management of the local PHY functions in conjunction with the MLME.

32.1.3.3 Service specification method

The models represented by figures and state diagrams are intended to be illustrations of the functions provided. It is important to distinguish between a model and a real implementation. The models are optimized for simplicity and clarity of presentation.

The service of a layer is the set of capabilities that it offers to a user in the next higher layer. Abstract services are specified here by describing the service primitives and parameters that characterize each service. This definition is independent of any particular implementation.

32.1.4 PPDU Formats

The structure of the PPDU transmitted by an NGV STA is determined by the TXVECTOR parameters as defined in Table 32-1.

In an NGV STA the FORMAT parameter determines the overall structure of the PPDU and can take one of the following values:

- Non-NGV format of 10 MHz bandwidth (NON_NGV_10), for the PPDU format as specified in Clause 17 for 10 MHz channel spacing including non-NGV duplicate format.
- NGV format (NGV), for the PPDU format as specified in Clause 32.

NOTE—Required support for these formats is defined in 17.1 and 32.1.1.

32.2 NGV PHY service interface

32.2.1 Introduction

The PHY provides an interface to the MAC through an extension of the generic PHY service interface defined in 8.3.4. The interface includes TXVECTOR, RXVECTOR, and PHYCONFIG_VECTOR.

The MAC uses the TXVECTOR to supply the PHY with per-PPDU transmit parameters. The PHY uses the RXVECTOR to inform the MAC of the received PPDU parameters. The MAC uses the PHYCONFIG_VECTOR to configure the PHY for operation that is independent of frame transmission or reception.

32.2.2 TXVECTOR and RXVECTOR

The parameters in Table 32-1 are defined as part of the TXVECTOR parameter list in the PHY-TXSTART.request primitive and/or as part of the RXVECTOR parameter list in the PHY-RXSTART.indication primitives.

Table 32-1—TXVECTOR and RXVECTOR parameters

Parameter	Condition	Value	TXVECTOR	RXVECTOR
FORMAT		Determines the format of the PPDU. Enumerated type: NON_NGV_10 indicates the PPDU format for 10 MHz channel spacing as defined in Clause 17 or non-NGV duplicate PPDU format as defined in 32.3.9.10. NGV indicates NGV PPDU format as defined in Clause 32.	Y	Y
L_LENGTH	FORMAT is NGV	Not present. NOTE—The value of the Length field of the L-SIG in an NGV PPDU is calculated as in Equation (32-8).	N	N
	Otherwise	Indicates the length of the PSDU in octets in the range 1 to 4095. The value is used by the PHY to determine the number of octet transfers that occur between the MAC and the PHY.	Y	Y
L_DATARATE	FORMAT is NGV	Not present. NOTE—The RATE field in the L-SIG field in an NGV PPDU is set to the value representing 3 Mb/s in the 10 MHz channel spacing column of Table 17-6.	N	N
	Otherwise	Indicates the rate, in Mb/s, used to transmit the PSDU. See corresponding entry in Table 17-1 and Table 17-2.	Y	Y
SERVICE	FORMAT is NGV	Not present.	N	N
	Otherwise	Scrambler initialization, null.	Y	N
TXPWR_LEVEL_INDEX	FORMAT is NGV	The allowed values for the TXPWR_LEVEL_INDEX parameter are in the range 1 to $N/2$, where N is the number of octets in dot11TxPowerLevelExtended. This parameter is used to indicate the available transmit output power levels defined in dot11TxPowerLevelExtended that shall be used for the current transmission.	Y	N
	Otherwise	1–8 Refer to 17.2.2.5 for the definition of TXPWR_LEVEL_INDEX.	Y	N

Table 32-1—TXVECTOR and RXVECTOR parameters (*continued*)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
RSSI	FORMAT is NGV	The allowed values for the RSSI parameter are in the range 0 to 255. This parameter is a measure by the PHY of the power observed at the antenna connector used to receive the current PPDU measured during the reception of the NGV-LTF field. RSSI is intended to be used in a relative manner, and it shall be a monotonically increasing function of the received power.	N	Y
	Otherwise	0–RSSI maximum. See 17.2.3.3 for the definition of RSSI.	N	Y
NGV-MCS	FORMAT is NGV	Indicates the modulation and coding scheme used in the transmission of the PPDU. Integer in the range: 0 to 8 and 15 for 10 MHz PPDU. 0 to 9 and 15 for 20 MHz PPDU. NOTE—Set to 0 for NGV ranging NDP. See 32.3.16	Y	Y
	Otherwise	Not present.	N	N
CH_BANDWIDTH	FORMAT is NGV	Indicates the channel width of the transmitted PPDU. Enumerated type: CBW10 for 10 MHz. CBW20 for 20 MHz.	Y	Y
	Otherwise	Not present.	N	N
N_TX	FORMAT is NGV	Indicates the number of transmit chains.	Y	N
	Otherwise	Not present.	N	N
MIDAMBLE_PERIODICITY	FORMAT is NGV	Indicates the midamble periodicity in the Data field. Integer values: Set to 0 to indicate the midamble periodicity of 4 OFDM symbols. Set to 1 to indicate the midamble periodicity of 8 OFDM symbols. Set to 2 to indicate the midamble periodicity of 16 OFDM symbols. Other values are reserved.	Y	Y
	Otherwise	Not present.	N	N

Table 32-1—TXVECTOR and RXVECTOR parameters (*continued*)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
NGV_LTF_TYPE	FORMAT is NGV	Indicates the type of NGV-LTF. Set to 0 to indicate NGV-LTF-2x is used in the transmitted PPDU. Set to 1 to indicate NGV-LTF-1x is used in the transmitted PPDU. NOTE—When NGV-MCS 15 is indicated in 10 MHz, Repeated NGV-LTF-2x is used in the transmitted PPDU. See 32.3.8.10.	Y	Y
	Otherwise	Not present.	N	N
N_PPDU_REP	FORMAT is NGV	Not present.	N	N
	FORMAT is NON_NGV_10	Indicates the number of repetitive transmissions of NON_NGV_10 PPDU after the initial transmission. The allowed values are in the range 0 to 3.	Y	Y
NUM_SS	FORMAT is NGV	Indicates the number of spatial streams. The allowed value: Set to 0 to indicate the number of spatial streams is one. Set to 1 to indicate the number of spatial streams is two	Y	Y
	Otherwise	Not present.	N	N
NON_NGV_MODULATION	FORMAT is NON_NGV_10	In the TXVECTOR, indicates the format of the transmitted non-NGV PPDU. In the RXVECTOR, indicates the estimated format of the received non-NGV PPDU. Enumerated type: OFDM indicates Clause 17 format. NON_NGV_10_DUP_OFDM indicates non-NGV duplicate PPDU format.	Y	Y
	Otherwise	Not present.	N	N
APEP_LENGTH	FORMAT is NGV	If 0, indicates an NGV ranging NDP. Otherwise, indicates the number of octets in the range 1 to aPSDUMaxLength in the A-MPDU pre-EOF padding (see Table 32-20) that is carried in the PSDU.	Y	N
	Otherwise	Not present.	N	N

Table 32-1—TXVECTOR and RXVECTOR parameters (*continued*)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
PSDU_LENGTH	FORMAT is NGV	Indicates the number of octets in the range 0 to aPSDUMaxLength octets (see Table 32-20) in the NGV PSDU. The value of 0 indicates an NGV ranging NDP.	N	Y
	Otherwise	Not present.	N	N
DYN_BANDWIDTH_IN_NON_NGV	FORMAT is NON_NGV_10	In the TXVECTOR, if present, indicates whether the transmitter is capable of Static or Dynamic bandwidth operation. In the RXVECTOR, if valid, indicates whether the transmitter is capable of Static or Dynamic bandwidth operation. Enumerated type: Static if the transmitter is capable of Static bandwidth operation. Dynamic if the transmitter is capable of Dynamic bandwidth operation. NOTE—In the RXVECTOR, the validity of this parameter is determined by the MAC based on the contents of the received MPDU.	O	Y
	Otherwise	Not present.	N	N
CH_BANDWIDTH_IN_NON_NGV	FORMAT is NON_NGV_10	In the TXVECTOR, if present, indicates the channel width of the transmitted PPDU, which is signaled via the scrambling sequence. In the RXVECTOR, if valid, indicates the channel width of the received PPDU, which is signaled via the scrambling sequence. Enumerated type: CBW10, CBW20. NOTE—In the RXVECTOR, the validity of this parameter is determined by the MAC based on the contents of the currently received MPDU (e.g., RTS) or the previous MPDU in an exchange (e.g., RTS preceding a CTS).	O	Y
	Otherwise	Not present.	N	N

Table 32-1—TXVECTOR and RXVECTOR parameters (*continued*)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
TIME_OF_DEPARTURE_REQUESTED	Enumerated type: true indicates that the MAC entity requests that the PHY entity measures and reports time of departure parameters corresponding to the time when the first frame energy is sent by the transmitting port. false indicates that the MAC entity requests that the PHY entity neither measures nor reports time of departure parameters.		O	N
LTF_REP	FORMAT is NGV and APEP_LENGTH is 0	Indicates the number of repetitions of the NGV-LTF symbols. Set to the number of repetitions.	O	N
	Otherwise	Not present.	N	N
NOTE—In the “TXVECTOR” and “RXVECTOR” columns, the following apply: Y = Present; N = Not present; O = Optional.				

32.2.3 PHY CONFIG_VECTOR

The PHYCONFIG_VECTOR carried in a PHY-CONFIG.request primitive for the NGV PHY contains an OPERATING_CHANNEL parameter, which identifies the operating or primary channel. The PHY shall set dot11CurrentPrimaryChannel to the value of parameter OPERATING_CHANNEL.

The PHYCONFIG_VECTOR carried in a PHY-CONFIG.request primitive for the NGV PHY contains a SECONDARY_CHANNEL_OFFSET parameter, which takes one of the following values:

- SECONDARY_CHANNEL_NONE if no secondary channel is present; in this case the PHY shall set dot11CurrentSecondaryChannel to 0.
- SECONDARY_CHANNEL_ABOVE if the channel number of the secondary channel is greater than the channel number of the primary channel; in this case the PHY shall set dot11CurrentSecondaryChannel to dot11CurrentPrimaryChannel + 2.
- SECONDARY_CHANNEL_BELOW if the channel number of the secondary channel is smaller than the channel number of the primary channel; in this case the PHY shall set dot11CurrentSecondaryChannel to dot11CurrentPrimaryChannel – 2.

32.2.4 Effects of CH_BANDWIDTH parameter on PPDU format

Table 32-2 shows the valid combinations of the FORMAT, NON_NGV_MODULATION, and CH_BANDWIDTH parameters and the corresponding PPDU format.

Table 32-2— Interpretation of FORMAT, NON_NGV_MODULATION, and CH_BANDWIDTH parameters

FORMAT	NON_NGV_MODULATION	CH_BANDWIDTH	PPDU format
NGV	N/A	CBW10	The STA transmits an NGV PPDU of 10 MHz bandwidth.
NGV	N/A	CBW20	The STA transmits an NGV PPDU of 20 MHz bandwidth.
NON_NGV_10	OFDM	N/A	The STA transmits a non-NGV PPDU of 10 MHz bandwidth.
NON_NGV_10	NON_NGV_10_DUP_OFDM	N/A	The STA transmits a NON_NGV_10 PPDU with NON_NGV_MODULATION set to NON_NGV_10_DUP_OFDM using two adjacent 10 MHz channels as defined in 32.3.9.10. The 10 MHz channel higher in frequency is rotated +90° relative to the 10 MHz channel lower in frequency as defined in Equation (32-5).

32.2.5 Support for NON_NGV format

32.2.5.1 General

An NGV STA logically contains Clause 17 and Clause 32 PHYs. The MAC interfaces to the PHYs via the Clause 17 and Clause 32 PHY service interfaces as shown in Figure 32-1, Figure 32-2, and Figure 32-3.

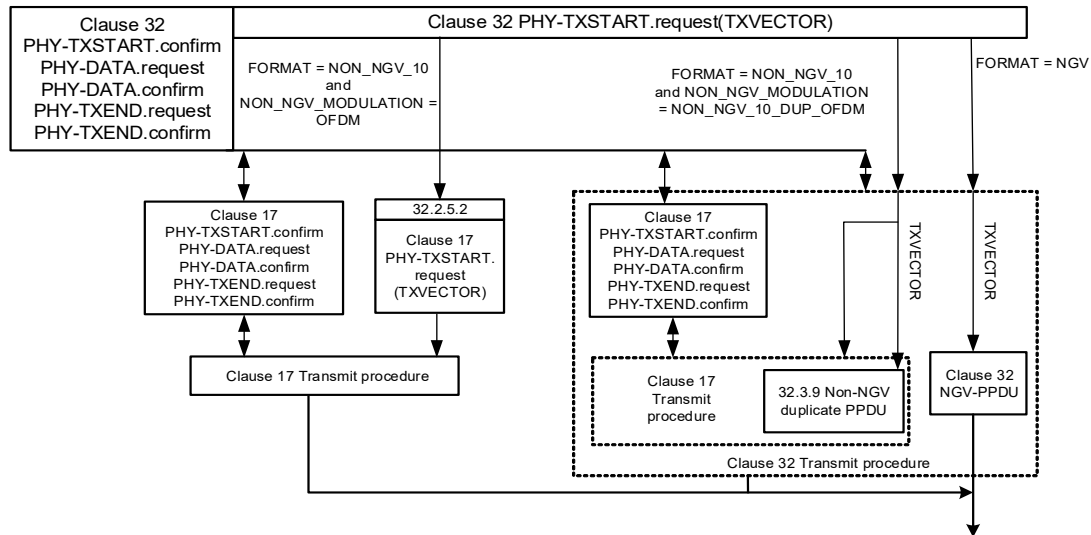


Figure 32-1—PHY interaction on transmit for various PDU formats

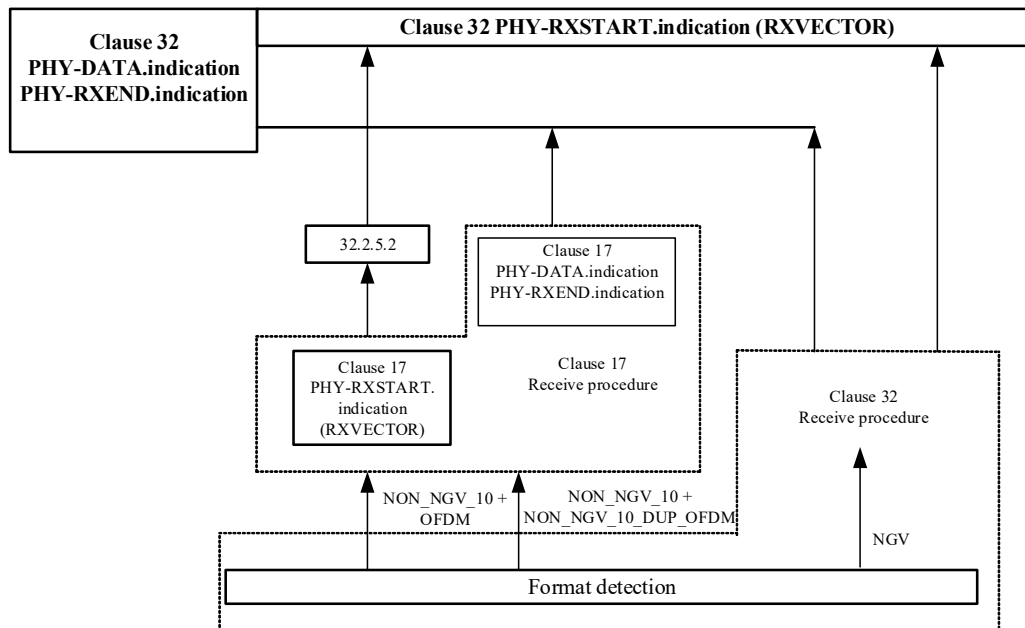


Figure 32-2—PHY interaction on receive for various PDU formats

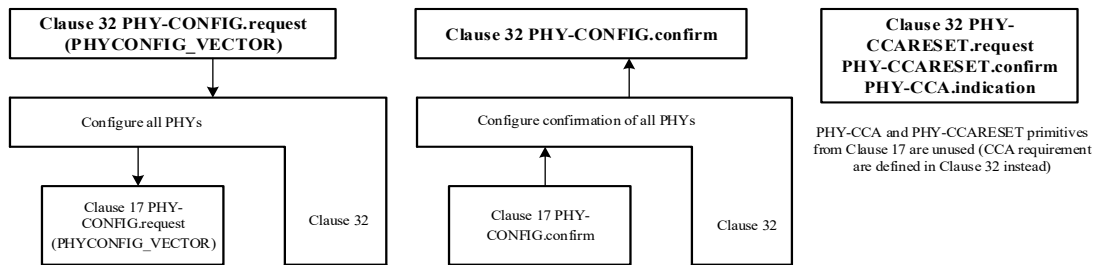


Figure 32-3—PHY-CONFIG and CCA interaction with Clause 17 and Clause 32

32.2.5.2 Support for NON_NGV format

When a PHY-TXSTART.request(TXVECTOR) primitive with the FORMAT parameter equal to NON_NGV_10 is issued, the behavior of the NGV PHY is defined in Clause 17 with additional requirements described in the following subclauses:

- 32.3.11.1 instead of 17.3.9.3
- 32.3.11.4.2 instead of 17.3.9.7.2

where the Clause 32 TXVECTOR parameters in Table 32-1 are mapped to Clause 17 TXVECTOR parameters in Table 17-1 according to Table 32-3.

When the NGV PHY receives a Clause 32 PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive, the NGV PHY shall, for the purpose of OFDM PPDU transmission and reception, behave as if it were a Clause 17 PHY that had received a PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive but with the SECONDARY_CHANNEL_OFFSET parameter discarded from PHYCONFIG_VECTOR.

As defined in 32.3.14, once a PPDU is received and detected as a non-NGV PPDU, the behavior of the NGV PHY is defined in Clause 17. The RXVECTOR parameters from the Clause 17 PHY-RXSTART.indication primitive are mapped to the Clause 32 RXVECTOR parameters as defined in Table 32-3. NGV PHY parameters not listed in the table are not present.

Table 32-3—Mapping of the NGV PHY parameters for non-NGV operation

NGV parameters	5.9 GHz operation defined by Clause 17	Parameter list
L_LENGTH	L_LENGTH	TXVECTOR/RXVECTOR
L_DATARATE	L_DATARATE	TXVECTOR/RXVECTOR
TXPWR_LEVEL_INDEX	TXPWR_LEVEL_INDEX	TXVECTOR
RSSI	RSSI	RXVECTOR
SERVICE	SERVICE	TXVECTOR/RXVECTOR
OPERATING_CHANNEL	OPERATING_CHANNEL	PHYCONFIG_VECTOR
SECONDARY_CHANNEL_OFFSET	Discarded	PHYCONFIG_VECTOR

32.2.5.3 Repetition transmission of NON_NGV_10 PPDU

NGV STA shall support NON_NGV_10 PPDU repetition transmission mode. When a PHY-TXSTART.request(TXVECTOR) primitive with the FORMAT parameter equal to NON_NGV_10 and N_PPDU_REP being nonzero, the NON_NGV_10 repetition transmission mode shall be used by the PHY layer.

In this mode, the NON_NGV_10 PPDU is transmitted $1 + N_PPDU_REP$ times in sequence, where N_PPDU_REP is the number of repetitions following the first transmission, and N_PPDU_REP is indicated through TXVECTOR.

The time separation between every two repeated transmissions is indicated by parameter *aSIFSTime* (see Table 32-20), as illustrated in Figure 32-4.

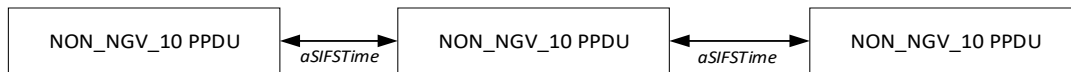


Figure 32-4—Example of NON_NGV_10 repetition transmission with $N_PPDU_REP = 2$

NOTE—All the following N_PPDU_REP repetition PDUs use the same scrambler as the first NON_NGV_10 PPDU.

32.2.5.4 TXSTATUS parameters

The parameters listed in Table 32-4 are defined as part of the TXSTATUS parameter list in the PHY-TXSTART.confirm(TXSTATUS) service primitive.

Table 32-4—TXSTATUS parameters

Parameter	Value
TIME_OF_DEPARTURE	0 to $2^{32} - 1$. The locally measured time when the first frame energy is sent by the transmitting port, in units equal to $1/\text{TIME_OF_DEPARTURE_ClockRate}$. This parameter is present only if TIME_OF_DEPARTURE_REQUESTED is true in the corresponding request.
TIME_OF_DEPARTURE_ClockRate	0 to $2^{16} - 1$. The clock rate, in units of MHz, is used to generate the TIME_OF_DEPARTURE value. This parameter is present only if TIME_OF_DEPARTURE_REQUESTED is true in the corresponding request.
TX_START_OF_FRAME_OFFSET	0 to $2^{32} - 1$. An estimate of the offset (in 10 nanosecond units) from the point in time at which the start of the preamble corresponding to the frame was transmitted at the transmit antenna connector to the point in time at which this primitive is issued to the MAC.

32.3 NGV PHY

32.3.1 Introduction

This subclause provides the procedure by which PSDUs are converted to and from transmissions on the wireless medium.

During transmission, a single PSDU is processed and appended to the NGV PHY preamble including L-STF, L-LTF, L-SIG, RL-SIG, NGV-SIG, RNGV-SIG, NGV-STF, and NGV-LTF fields to create the NGV PPDU. At the receivers, the PHY preamble is processed to aid in the detection, demodulation, and delivery of the PSDU.

32.3.2 NGV PPDU format

A single PPDU format is defined for this PHY: the NGV PPDU format. The format of NGV PPDU is defined in Figure 32-5.

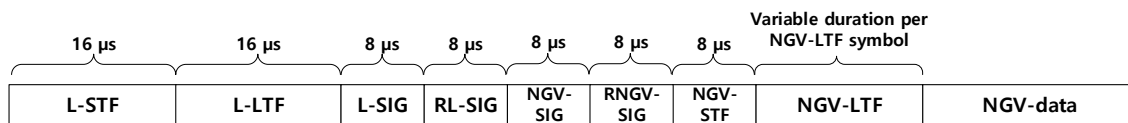


Figure 32-5—NGV PPDU format

The fields of the NGV PPDU format are summarized in Table 32-5.

Table 32-5—Fields of the NGV PPDU

Field	Description
L-STF	Non-HT Short Training field
L-LTF	Non-HT Long Training field
L-SIG	Non-HT SIGNAL field
RL-SIG	Repeated Non-HT SIGNAL field
NGV-SIG	NGV Signal field
RNGV-SIG	Repeated NGV Signal field
NGV-STF	NGV Short Training field
NGV-LTF	NGV Long Training field
Data	The Data field carrying the PSDU(s)

The RL-SIG, NGV-SIG, RNGV-SIG, NGV-STF, and NGV-LTF fields are present only in NGV PPDU formats. The RNGV-SIG field is equal to NGV-SIG field. The fields present in a non-NGV PPDU are the L-STF, L-LTF, L-SIG, and Data fields. The number of symbols in the NGV-LTF field, $N_{\text{NGV-LTF}}$, can be 1, 2, or 4 and is determined by the total number of spatial streams being transmitted in the NGV PPDU and whether the NGV-LTF is repeated in NGV ranging NDP (see Table 32-11).

32.3.3 Transmitter block diagram

The generation of each field in an NGV PPDU uses many of the following blocks:

- a) PHY padding
- b) Scrambler
- c) FEC (BCC or LDPC) encoders
- d) Stream parser
- e) BCC interleaver
- f) Constellation mapper
- g) Pilot insertion
- h) Replication over two 10 MHz channels (for BW = 20 MHz)
- i) CSD per spatial stream insertion
- j) Spatial mapper
- k) IDFT
- l) CSD per chain insertion
- m) GI insertion
- n) Windowing

Figure 32-6 and Figure 32-7 show example transmitter block diagrams. The actual structure of the transmitter is implementation dependent.

In particular, Figure 32-6 shows the transmit process for the L-SIG, RL-SIG, NGV-SIG, and RNGV-SIG fields of an NGV PPDU. The BCC encoder and interleaver are not used when generating the L-STF and L-LTF fields. The same BCC interleaver is applied to both the NGV-SIG field and the RNGV-SIG field such that the baseband signals of the NGV-SIG field symbol and the RNGV-SIG field symbol are exactly the same.

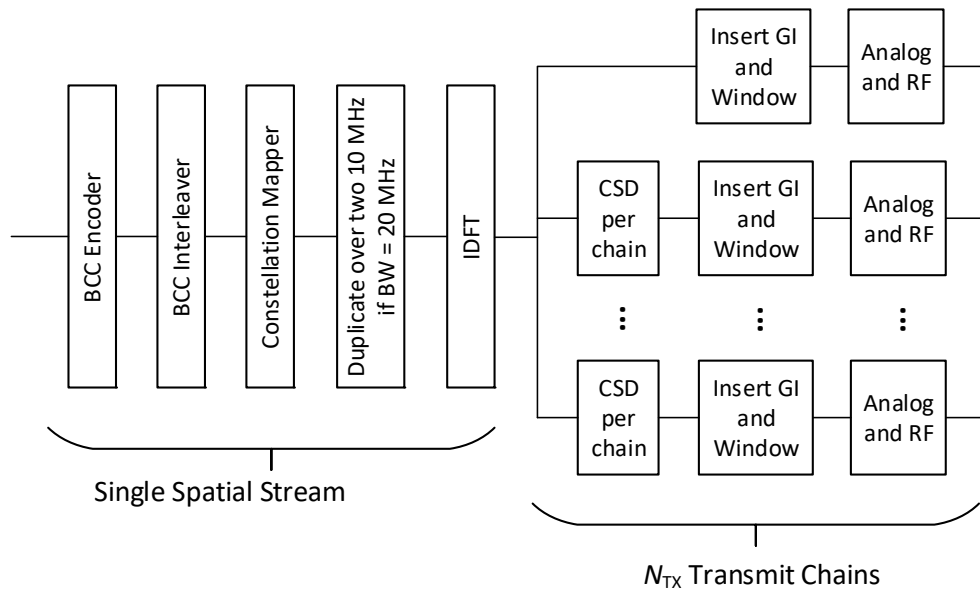
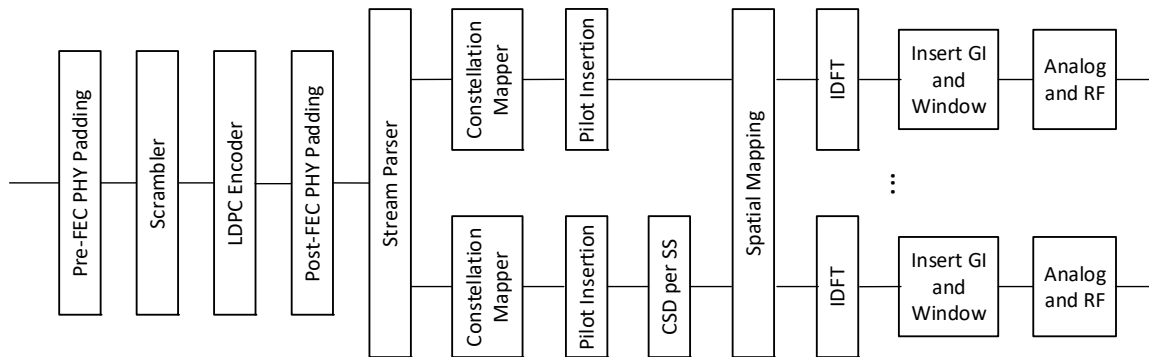


Figure 32-6—Transmitter block diagram for the L-SIG, RL-SIG, NGV-SIG, and RNGV-SIG fields for an NGV PPDU

Figure 32-7 shows the transmitter blocks used to generate the Data field of an NGV transmission. The BPSK-DCM tone mapping, which is part of the constellation mapper block, is applied only if BPSK-DCM is indicated by NGV-MCS field value 15 in NGV-SIG field.



NOTE—Transmission with two spatial streams is optional.

Figure 32-7—Transmitter block diagram for the Data field of an NGV transmission

32.3.4 Overview of the PPDU encoding process

32.3.4.1 General

This subclause provides an overview of the NGV PPDU encoding process.

32.3.4.2 Construction of L-STF

Construct the L-STF field as defined in 32.3.8.3 with the following highlights:

- a) Determine the CH_BANDWIDTH from the TXVECTOR.
- b) Sequence generation: Generate the L-STF sequence over the CH_BANDWIDTH as described in 32.3.8.3.
- c) Phase rotation: Apply appropriate phase rotation for each 10 MHz subchannel as described in 32.3.7.3 and 32.3.7.4.
- d) IDFT: Compute the inverse discrete Fourier transform.
- e) CSD: Apply CSD for each transmit chain as described in 32.3.8.2.1.
- f) Insert GI and apply windowing: Prepend a GI and apply windowing as described in 32.3.7.3.
- g) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 32.3.7.3 and 32.3.8 for details.

32.3.4.3 Construction of the L-LTF

Construct the L-LTF field as defined in 32.3.8.4 with the following highlights:

- a) Determine the CH_BANDWIDTH from the TXVECTOR.
- b) Sequence generation: Generate the L-LTF sequence over the CH_BANDWIDTH as described in 32.3.8.4.
- c) Phase rotation: Apply appropriate phase rotation for each 10 MHz subchannel as described in 32.3.7.3 and 32.3.7.4.
- d) IDFT: Compute the inverse discrete Fourier transform.
- e) CSD: Apply CSD for each transmit chain as described in 32.3.8.2.1.
- f) Insert GI and apply windowing: Prepend a GI and apply windowing as described in 32.3.7.3.
- g) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 32.3.7.3 and 32.3.8 for details.

32.3.4.4 Construction of the L-SIG and RL-SIG

Construct the L-SIG field as the SIGNAL field defined in 32.3.8.5 with the following highlights:

- a) In an NGV PPDU set the RATE subfield in the SIGNAL field to 3 Mb/s in the 10 MHz channel spacing column of Table 17-6. Set the Length, Parity, and Tail bits in the SIGNAL field as described in 32.3.8.5.
- b) BCC encoder: Encode the SIGNAL field by a convolutional encoder at the rate of $R=1/2$ as described in 17.3.5.6.
- c) BCC interleaver: Interleave as described in 17.3.5.7.
- d) Constellation Mapper: BPSK modulate as described in 17.3.5.8.
- e) Pilot insertion: Insert pilots as described in 17.3.5.9.
- f) Duplication and phase rotation: Duplicate the L-SIG field over each 10 MHz of the CH_BANDWIDTH. Apply appropriate phase rotation for each 10 MHz subchannel as described in 32.3.7.3 and 32.3.7.4.
- g) IDFT: Compute the inverse discrete Fourier transform.
- h) CSD: Apply CSD for each transmit chain as described in 32.3.8.2.1.
- i) Insert GI and apply windowing: Prepend a GI and apply windowing as described in 32.3.7.3.

- j) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 32.3.7.3 and 32.3.8 for details.

Construct the RL-SIG field as the repeat SIGNAL field defined in 32.3.8.6 with the same process as for the L-SIG field.

32.3.4.5 Construction of the NGV-SIG and RNGV-SIG

The NGV-SIG field consists of one symbol as defined in 32.3.8.7 and is constructed as follows:

- a) Obtain the NGV-SIG field values from the TXVECTOR entries of NGV-MCS, CH_BANDWIDTH, MIDAMBLE_PERIODICITY, NGV_LTF_TYPE, NUM_SS, and LTF_REP. Add the reserved bits, append the calculated CRC, then append the tail bits as shown in 32.3.8.7. This results in 24 uncoded bits.
- b) BCC encoder: Encode the data by a convolutional encoder at the rate of $R=1/2$ as described in 17.3.5.6.
- c) BCC interleaver: Interleave as described in 17.3.5.7.
- d) Constellation Mapper: BPSK modulate the 48 interleaved bits as described in 17.3.5.8 to form the NGV-SIG symbol.
- e) Pilot insertion: Insert pilots as described in 17.3.5.9.
- f) Duplication and phase rotation: Duplicate NGV-SIG over each 10 MHz of the CH_BANDWIDTH. Apply appropriate phase rotation for each 10 MHz subchannel as described in 32.3.7.3 and 32.3.7.4.
- g) IDFT: Compute the inverse discrete Fourier transform.
- h) CSD: Apply CSD for each transmit chain as described in 32.3.8.2.1.
- i) Insert GI and apply windowing: Prepend a GI and apply windowing as described in 32.3.7.3.
- j) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 32.3.7.3 and 32.3.8 for details.

Construct the RNGV-SIG field as the repeat NGV-SIG field defined in 32.3.8.8 with the same process as for the NGV-SIG field.

32.3.4.6 Construction of NGV-STF

The NGV-STF field is defined in 32.3.8.9 and is constructed as follows:

- a) Sequence generation: Generate the NGV-STF in the frequency domain over the bandwidth indicated by CH_BANDWIDTH as described in 32.3.8.9.
- b) Phase rotation: Apply appropriate phase rotation for each 10 MHz subchannel as described in 32.3.7.3 and 32.3.7.4.
- c) CSD: Apply CSD for each spatial stream as described in 32.3.8.2.2.
- d) Spatial mapping: Apply the Q matrix as described in 32.3.9.8.1.
- e) IDFT: Compute the inverse discrete Fourier transform.
- f) Insert GI and apply windowing: Prepend a GI and apply windowing as described in 32.3.7.3.
- g) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 32.3.7.3 and 32.3.8 for details.

32.3.4.7 Construction of NGV-LTF

The NGV-LTF field is defined in 32.3.8.10 and constructed as follows:

- a) Sequence generation: Generate the NGV-LTF sequence in the frequency domain over the CH_BANDWIDTH as described in 32.3.8.10.
- b) Phase rotation: Apply appropriate phase rotation for each 10 MHz subchannel as described in 32.3.7.3 and 32.3.7.4.
- c) $A_{NGV-LTF}$ matrix mapping: Apply the $P_{NGV-LTF}$ matrix to the data tones of the NGV-LTF sequence and apply the $R_{NGV-LTF}$ matrix to the pilot tones as described in 32.3.8.10.
- d) CSD: Apply CSD for each spatial stream as described in 32.3.8.2.2.
- e) Spatial mapping: Apply the Q matrix as described in 32.3.9.8.1.
- f) IDFT: Compute the inverse discrete Fourier transform.
- g) Insert GI and apply windowing: Prepend a GI and apply windowing as described in 32.3.7.3.
- h) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 32.3.7.3 and 32.3.8 for details.

32.3.4.8 Construction of the Data field in an NGV PPDU

The construction of the Data field in an NGV PPDU proceeds as follows:

- a) Construct the SERVICE field as described in 32.3.9.2 and append the PSDU to the SERVICE field.
- b) PHY padding: Append the PHY pad bits to the PSDU. There are no tail bits.
- c) Scrambler: Scramble the PHY padded data.
- d) LDPC encoder: The scrambled bits are encoded using the LDPC code with the APEP_LENGTH in the TXVECTOR as described in 32.3.9.4.
- e) Stream parser: The output of the LDPC encoder is rearranged into blocks as described in 32.3.9.5.
- f) Phase rotation: Apply appropriate phase rotation for each 10 MHz subchannel as described in 32.3.7.3 and 32.3.7.4.
- g) CSD: Apply CSD for each spatial stream as described in 32.3.8.2.2.
- h) Spatial mapping: Apply the Q matrix as described in 32.3.9.8.1.
- i) IDFT: Compute the inverse discrete Fourier transform.
- j) Insert GI and apply windowing: Prepend a GI and apply windowing as described in 32.3.7.3.
- k) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 32.3.7.3 and 32.3.8 for details.

32.3.5 NGV modulation and coding schemes

The NGV-MCS is a value that determines the modulation and coding used in the Data field of the PPDU. It is a compact representation that is carried in the NGV-SIG field and RNGV-SIG field for NGV PPDU. Rate dependent parameters for the full set of NGV-MCSs are shown in Table 32-21 to Table 32-24. These tables give rate-dependent parameters for NGV-MCSs with indices n , where $n = 0, \dots, 15$, with number of spatial streams from 1 to 2 and bandwidth options of 10 MHz and 20 MHz. Equal modulation is applied to all streams.

The use of BPSK-DCM on the Data field of an NGV PPDU is indicated as NGV-MCS 15 in NGV-SIG field.

32.3.6 Timing related parameters

Table 32-6 defines the timing related parameters.

Table 32-6—Timing related constants

Parameter	CBW10	CBW20	Description
N_{SD}	52	108	Number of data subcarriers
N_{SP}	4	6	Number of pilot subcarriers
N_{ST}	56	114	Total number of subcarriers See NOTE
N_{SR}	28	58	Highest data subcarrier index
Δ_F	156.25 kHz		Subcarrier frequency spacing
T_{DFT}	6.4 μ s		IDFT/DFT period
T_{GI}	1.6 μ s = $T_{DFT}/4$		Guard interval duration
T_{GI2}	3.2 μ s		Double guard interval
T_{SYM}	8 μ s = $T_{DFT} + T_{GI} = 1.25 \times T_{DFT}$		Symbol interval
T_{L-STF}	16 μ s = $10 \times T_{DFT}/4$		Non-HT Short Training field duration
T_{L-LTF}	16 μ s = $2 \times T_{DFT} + T_{GI2}$		Non-HT Long Training field duration
T_{L-SIG}	8 μ s		Non-HT SIGNAL field duration
T_{RL-SIG}	8 μ s		Repeated Non-HT SIGNAL field duration
$T_{NGV-SIG}$	8 μ s		NGV SIGNAL field duration
$T_{RNGV-SIG}$	8 μ s		Repeated NGV SIGNAL field duration
$T_{NGV-STF}$	8 μ s		NGV Short Training field duration
$T_{NGV-LTF-2X}$	8 μ s		Duration of each NGV-LTF-2x symbol
$T_{NGV-LTF-1X}$	4.8 μ s		Duration of each NGV-LTF-1x symbol
$T_{NGV-LTF-2X-Repeat}$	14.4 μ s		Duration of each repeated NGV-LTF-2x symbol
$T_{NGV-LTF}$	$T_{NGV-LTF-2X}$, $T_{NGV-LTF-1X}$ or $T_{NGV-LTF-2X-Repeat}$ depending on the LTF format used		Duration of each OFDM symbol in NGV-LTF field
$N_{service}$	16		Number of bits in the SERVICE field
N_{tail}	6		Number of tail bits per BCC encoder
NOTE — $N_{ST} = N_{SD} + N_{SP}$			

Table 32-7 defines parameters used frequently in Clause 32.

Table 32-7—Frequently used parameters

Symbol	Explanation
N_{CBPS}	Number of coded bits per symbol
N_{CBPSS}	Number of coded bits per symbol per spatial stream
N_{DBPS}	Number of data bits per symbol
N_{BPSCS}	Number of coded bits per subcarrier per spatial stream
N_{RX}	Number of receive chains
N_{SS}	Number of spatial streams
N_{TX}	Number of transmit chains
$N_{\text{NGV-LTF}}$	Number of NGV-LTF symbols
R	R is the coding rate

32.3.7 Mathematical description of signals

32.3.7.1 Notation

For a description of the conventions used for the mathematical description of the signals, see 17.3.2.5, and 21.3.7.1.

32.3.7.2 Subcarrier indices in use

For a 10 MHz non-NGV PPDU, the signal is transmitted on subcarriers -26 to -1 and 1 to 26 , with 0 being the center (DC) subcarrier. See 17.3.2.6.

For a 10 MHz NGV PPDU transmission, the 10 MHz is divided into 64 subcarriers. The signal is transmitted on subcarriers -28 to -1 and 1 to 28 , with 0 being the center (DC) subcarrier.

For a 20 MHz NGV PPDU transmission, the 20 MHz is divided into 128 subcarriers. The signal is transmitted on subcarriers -58 to -2 and 2 to 58 .

32.3.7.3 Transmitted signal

The transmitted signal is described in complex baseband signal notation. The actual transmitted signal is related to the complex baseband signal by the relation shown in Equation (32-1).

$$r_{RF}^{i_{TX}} = \text{Re}\{r_{PPDU}^{i_{TX}}(t) \exp(j2\pi f_c t)\}, \quad i_{TX} = 1, \dots, N_{TX} \quad (32-1)$$

where

$r_{PPDU}^{i_{TX}}(t)$ represents the complex baseband signal of transmit chain i_{TX}

f_c represents the center frequency of the PPDU

The transmitted RF signal is derived by upconverting the complex baseband signal, which consists of several fields. The timing boundaries for the various fields are shown in Figure 32-8, where $N_{\text{NGV-LTF}}$ is the number of NGV-LTF symbols as defined in Table 32-11.

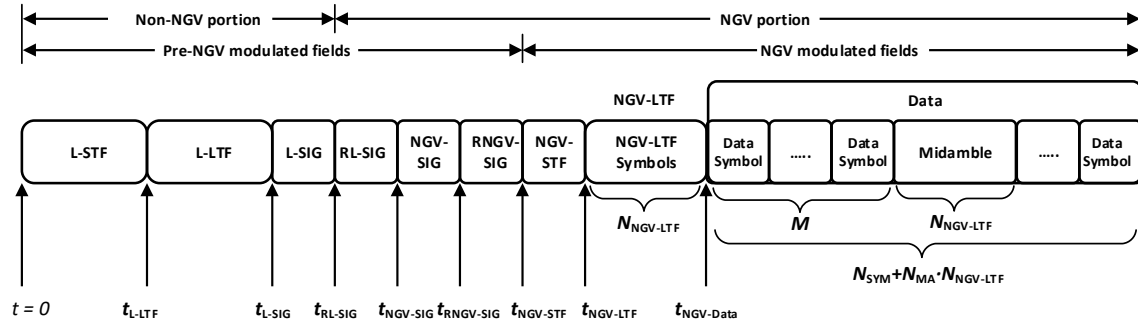


Figure 32-8—Timing boundaries for NGV PPDU fields

The time offset, t_{Field} , determines the starting time of the corresponding field relative to the start of L-STF ($t = 0$).

The signal transmitted on transmit chain i_{TX} shall be as shown in Equation (32-2).

$$\begin{aligned}
 r_{\text{PPDU}}^{i_{\text{TX}}}(t) = & r_{\text{L-STF}}^{i_{\text{TX}}}(t) + r_{\text{L-LTF}}^{i_{\text{TX}}}(t - t_{\text{L-LTF}}) \\
 & + r_{\text{L-SIG}}^{i_{\text{TX}}}(t - t_{\text{L-SIG}}) + r_{\text{RL-SIG}}^{i_{\text{TX}}}(t - t_{\text{RL-SIG}}) + r_{\text{NGV-SIG}}^{i_{\text{TX}}}(t - t_{\text{NGV-SIG}}) \\
 & + r_{\text{RNGV-SIG}}^{i_{\text{TX}}}(t - t_{\text{RNGV-SIG}}) + r_{\text{NGV-STF}}^{i_{\text{TX}}}(t - t_{\text{NGV-STF}}) + r_{\text{NGV-LTF}}^{i_{\text{TX}}}(t - t_{\text{NGV-LTF}}) \\
 & + r_{\text{NGV-Data}}^{i_{\text{TX}}}(t - t_{\text{NGV-Data}})
 \end{aligned} \tag{32-2}$$

where

$$\begin{aligned}
 1 & \leq i_{\text{TX}} \leq N_{\text{TX}} \\
 t_{\text{L-LTF}} & = T_{\text{L-STF}} \\
 t_{\text{L-SIG}} & = t_{\text{L-LTF}} + T_{\text{L-LTF}} \\
 t_{\text{RL-SIG}} & = t_{\text{L-SIG}} + T_{\text{L-SIG}} \\
 t_{\text{NGV-SIG}} & = t_{\text{RL-SIG}} + T_{\text{RL-SIG}} \\
 t_{\text{RNGV-SIG}} & = t_{\text{NGV-SIG}} + T_{\text{RNGV-SIG}} \\
 t_{\text{NGV-STF}} & = t_{\text{RNGV-SIG}} + T_{\text{RNGV-SIG}} \\
 t_{\text{NGV-LTF}} & = t_{\text{NGV-STF}} + T_{\text{NGV-STF}} \\
 t_{\text{NGV-Data}} & = t_{\text{NGV-LTF}} + N_{\text{NGV-LTF}} \cdot T_{\text{NGV-LTF}}
 \end{aligned}$$

Each field, $r_{\text{Field}}^{i_{\text{TX}}}(t)$, is defined as the summation of one or more subfields, where each subfield is defined to be an inverse discrete Fourier transform as specified in Equation (32-3).

$$r_{Subfield}^{i_{TX}}(t) = \frac{1}{\sqrt{N_{Field}^{Tone} \times N_{Norm}}} w_{T_{Subfield}}(t) \eta_{Field} \sum_{k=-N_{SR}}^{N_{SR}} \sum_{m=1}^{N_{SS}} [Q_k]_{i_{TX},m} \Upsilon_{k,BW} X_k^m \exp(j2\pi k \Delta_F (t - T_{GI,Field} - T_{CS,NGV}(m))) \quad (32-3)$$

This general representation holds for all subfields. The total power of the time domain NGV modulated field signals summed over all transmit chains should not exceed the total power of the time domain pre-NGV modulated field signals summed over all transmit chains. For notational simplicity, the parameter BW is omitted from some bandwidth dependent terms.

In Equation (32-3), the following notations are used:

N_{Field}^{Tone}	Table 32-8 summarizes the various values of N_{Field}^{Tone} as a function of bandwidth.
N_{Norm}	For pre-NGV modulated fields, $N_{Norm} = N_{TX}$. For NGV modulated fields $N_{Norm} = N_{SS}$, where N_{SS} is given in Table 32-7.
$w_{T_{Subfield}}(t)$	is a windowing function. An example function, $w_{T_{Subfield}}(t)$, is given in 17.3.2.5. $T_{Subfield}$ is T_{L-STF} for L-STF, T_{L-LTF} for L-LTF, T_{L-SIG} for L-SIG, T_{RL-SIG} for RL-SIG, $T_{NGV-SIG}$ for NGV-SIG, $T_{RNGV-SIG}$ for RNGV-SIG, $T_{NGV-STF}$ for NGV-STF, $T_{NGV-LTF}$ for NGV-LTF symbol and midamble symbol, and T_{SYM} for Data symbol.
Q_k	is the spatial mapping matrix for the subcarrier k . For pre-NGV modulated fields, Q_k is a column vector with N_{TX} elements with element i_{TX} being $e^{-j2\pi k \Delta_F T_{CS}^{i_{TX}}}$, where $T_{CS}^{i_{TX}}$ represents the cyclic shift for transmitter chain i_{TX} whose values are given in Table 21-10. For NGV modulated fields, Q_k is a matrix with N_{TX} rows and N_{SS} columns as further described in 32.3.9.8.1.
$\Upsilon_{k,BW}$	is defined in 32.3.7.4.
Δ_F	is the subcarrier frequency spacing given in Table 32-6.
X_k^m	is the frequency domain symbol in subcarrier k of spatial stream m . Some of the X_k^m within $-N_{SR} \leq k \leq N_{SR}$ have a value of 0. Examples of such cases include the DC tones, guard tones on each side of the transmit spectrum, as well as the unmodulated tones of L-STF and NGV-STF fields.

NOTE—The multiplication matrices $A_{NGV-LTF}^k$ and $P_{NGV-LTF}$ are included in the calculation of X_k^m for the NGV-LTF field.

$T_{GI,Field}$	is the guard interval duration used for each OFDM symbol in the field. For L-STF and NGV-STF, $T_{GI,Field} = T_{GI}$ but it can be omitted from Equation (32-3) due to the periodic property of L-STF and NGV-STF over every 1.6 μ s. For the L-SIG, RL-SIG, NGV-SIG, RNGV-SIG, NGV-LTF, and Data fields, $T_{GI,Field}$ is defined in the “Guard interval duration” column of Table 32-8.
$T_{CS,NGV}(m)$	For pre-NGV modulated fields, $T_{CS,NGV}(m) = 0$. For NGV modulated fields, $T_{CS,NGV}(m)$ represents the cyclic shift per spatial stream, whose value is given in 21.3.8.3.2, where the row for $N_{STS,total} = 1$ corresponds to $N_{SS} = 1$ and the row for $N_{STS,total} = 2$ corresponds to $N_{SS} = 2$.

η_{Field} is the power scale factor of a given field within an OFDM symbol for NGV PPDU. For the L-STF and L-LTF fields, η_{Field} is $\sqrt{2}$ when NGV-MCS 0 or NGV-MCS 15 and one spatial stream is used for data field in 10 MHz NGV PPDU; otherwise it is 1. For other fields of NGV PPDU, $\eta_{Field} = 1$.

Table 32-8—Tone scaling factor and guard interval duration values for PHY fields

Field	N_{Field}^{Tone}		Guard interval duration
	10 MHz	20 MHz	
L-STF	12	24	—
L-LTF	52	104	T_{GI2}
L-SIG	52	104	T_{GI}
RL-SIG	52	104	T_{GI}
NGV-SIG	52	104	T_{GI}
RNGV-SIG	52	104	T_{GI}
NGV-STF	12	24	—
NGV-LTF-1x	28	58	T_{GI}
NGV-LTF-2x	56	114	T_{GI}
NGV-LTF-2x-Repeat	56	114	T_{GI}
Data	56	114	T_{GI}
NON_NGV_10_DUP_OFDM-Data	—	104	T_{GI}

32.3.7.4 Definition of tone rotation

The function $\Upsilon_{k, BW}$ is used to represent a rotation of the tones. The value of BW in $\Upsilon_{k, BW}$ is determined by the TXVECTOR parameter CH_BANDWIDTH as defined in Table 32-9.

Table 32-9—CH_BANDWIDTH

CH_BANDWIDTH	$\Upsilon_{k, BW}$
CBW10	$\Upsilon_{k, 10}$
CBW20	$\Upsilon_{k, 20}$

For a 10 MHz NGV PPDU transmission:

$$\Upsilon_{k, 10} = 1 \quad (32-4)$$

For a 20 MHz NGV PPDU transmission:

$$\gamma_{k,20} = \begin{cases} 1, & k < 0 \\ j, & k \geq 0 \end{cases} \quad (32-5)$$

32.3.8 NGV preamble

32.3.8.1 Introduction

An NGV preamble is defined to carry the required information to operate in a system with multiple transmit and multiple receive antennas. To maintain compatibility with non-NGV STAs, specific non-NGV fields are defined that can be received by non-NGV STAs compliant with Clause 17. The non-NGV fields are followed by NGV fields specific to NGV STAs.

32.3.8.2 Cyclic shift

32.3.8.2.1 Cyclic shift for pre-NGV modulated fields

The cyclic shift value $N_{CS}^{i_{TX}}$ for the L-STF, L-LTF, L-SIG, RL-SIG, NGV-SIG, and RNGV-SIG fields of the PPDU for transmit chain i_{TX} out of a total of N_{TX} are defined in Table 21-10.

32.3.8.2.2 Cyclic shift for NGV modulated fields

The cyclic shift values defined in this subclause apply to the NGV-STF, NGV-LTF, and Data fields of the NGV PPDU.

Throughout the NGV modulated fields of the preamble, cyclic shifts are applied to prevent unintended beamforming when correlated signals are transmitted in multiple spatial streams. The same cyclic shift is also applied to these streams during the transmission of the Data field of the NGV PPDU. The cyclic shift value for the NGV modulated fields for spatial stream m out of N_{SS} , the number of spatial streams, is shown in Table 21-11 where the values in the row for $N_{STS} = m$ is used.

32.3.8.3 L-STF definition

The frequency domain sequence of the L-STF field for a 10 MHz or 20 MHz transmission is defined by Equation (19-8) and Equation (19-9), respectively, in 19.3.9.3.3.

The time domain representation of the signal on transmit chain i_{TX} shall be as specified in Equation (32-6).

$$r_{L-STF}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \times N_{L-STF}^{Tone}}} w_{\Gamma_{L-STF}}(t) \eta_{L-STF} \sum_{i_{BW}=0}^{N_{10\text{ MHz}}-1} \sum_{k=-26}^{26} (\gamma_{(k-K_{shift}(i_{BW})), BW} S_{k, 10} \times \exp(j2\pi(k-K_{shift}(i_{BW}))\Delta_F(t-T_{CS}^{i_{TX}}))) \quad (32-6)$$

where

η_{L-STF} is the modulation dependent power scaling factor for the L-STF field with the following value:

$$\eta_{L-STF} = \begin{cases} \sqrt{2}, & \text{when NGV-MCS 0 or NGV-MCS 15 and one spatial stream is used on Data field for 10 MHz transmission} \\ 1, & \text{otherwise} \end{cases}$$

$$N_{10\text{ MHz}} = \begin{cases} 1, & \text{If CH_BANDWIDTH is CBW10} \\ 2, & \text{If CH_BANDWIDTH is CBW20} \end{cases}$$

$$K_{shift}(i) = (N_{10\text{ MHz}} - 1 - 2i) \times 32$$

$T_{CS}^{(i_{TX})}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 32.3.8.2.1

$\gamma_{k, BW}$ is defined in Equation (32-4) and Equation (32-5)

N_{L-STF}^{Tone} has the value given in Table 32-8

$S_{k, 10}$ is defined as $S_{-26, 26}$ in Equation (19-8)

32.3.8.4 L-LTF definition

For a 10 MHz or 20 MHz transmission, the L-LTF pattern in the NGV preamble is defined by Equation (19-11) and Equation (19-12) in 19.3.9.3.4, respectively.

The time domain representation of the signal on transmit chain i_{TX} shall be as defined in Equation (32-7).

$$r_{L-LTF}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \times N_{L-LTF}^{Tone}}} w_{\Gamma_{L-LTF}}(t) \eta_{L-LTF} \sum_{i_{BW}=0}^{N_{10\text{ MHz}}-1} \sum_{k=-26}^{26} (\gamma_{(k-K_{shift}(i_{BW})), BW} L_{k, 10} \times \exp(j2\pi(k-K_{shift}(i_{BW}))\Delta_F(t-T_{CS}^{i_{TX}}))) \quad (32-7)$$

where

η_{L-LTF} is a modulation dependent scaling factor for the L-LTF field with the same value as η_{L-STF}

$T_{CS}^{(i_{TX})}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 32.3.8.2.1

$\gamma_{k, BW}$ is defined in Equation (32-4) and Equation (32-5)

N_{L-LTF}^{Tone} has the value given in Table 32-8

$N_{10\text{ MHz}}$ is defined in 32.3.8.3

$$K_{shift}(i) = (N_{10\text{ MHz}} - 1 - 2i) \times 32$$

$L_{k,10}$ is defined as $L_{-26,26}$ in Equation (19-11)

32.3.8.5 L-SIG definition

The L-SIG field is used to communicate rate and length information. The structure of the L-SIG field is defined in Figure 17-5.

In an NGV PPDU, the RATE field shall be set to the value representing 3 Mb/s in the 10 MHz channel spacing column of Table 17-6. In a non-NGV duplicate PPDU, the RATE field is defined in 17.3.4.2 using the L_DATARATE parameter in the TXVECTOR.

The Length field shall be set to the value given by Equation (32-8).

$$\text{Length} = \frac{\text{TXTIME} - 40}{8} \times 3 - 3 \quad (32-8)$$

where

TXTIME (in μs) is defined in 32.4.3.

The LSB of the binary expression of the value of the Length field shall be mapped to B5. In a non-NGV duplicate PPDU, the LENGTH field is defined in 17.3.4.3 using the L_LENGTH parameter in the TXVECTOR.

The Reserved (R) field shall be set to 0.

The Parity (P) field has the even parity of bits 0–16.

The SIGNAL TAIL field shall be set to 0.

The L-SIG field shall be encoded, interleaved, and mapped following the steps described in 17.3.5.6, 17.3.5.7, and 17.3.5.8. The stream of 48 complex numbers generated by these steps is denoted by d_k , $k = 0, \dots, 47$ and are mapped to subcarriers $[-26, 26]$. Pilots shall be inserted as described in 17.3.5.9.

The time domain waveform of the L-SIG field shall be as given by Equation (32-9).

$$r_{L-SIG}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \times N_{L-SIG}^{Tone}}} w_{T_{L-SIG}}(t) \sum_{i_{BW}=0}^{N_{10\text{ MHz}}-1} \sum_{k=-26}^{26} \gamma_{(k-K_{shift}(i_{BW})), BW}(D_{k,10} + p_0 P_k) \times \exp(j2\pi(k - K_{shift}(i_{BW}))\Delta_F(t - T_{GI} - T_{CS}^{i_{TX}})) \quad (32-9)$$

where

$N_{10\text{ MHz}}$ is defined in 32.3.8.3

$$K_{shift}(i) = (N_{10\text{ MHz}} - 1 - 2i) \times 32$$

$$D_{k,10} = \begin{cases} 0, & k = 0, \pm 7, \pm 21 \\ d_{M_{10}^r(k)}, & \text{otherwise} \end{cases}$$

$$M_{10}^r(k) = \begin{cases} k + 26, & -26 \leq k \leq -22 \\ k + 25, & -20 \leq k \leq -8 \\ k + 24, & -6 \leq k \leq -1 \\ k + 23, & 1 \leq k \leq 6 \\ k + 22, & 8 \leq k \leq 20 \\ k + 21, & 22 \leq k \leq 26 \end{cases} \quad (32-10)$$

- P_k is defined in 17.3.5.10
- p_0 is the first pilot value in the sequence defined in 17.3.5.10
- N_{L-SIG}^{Tone} has the value given in Table 32-8
- $\gamma_{k,BW}$ is defined in Equation (32-4) and Equation (32-5)
- $T_{CS}^{i_{TX}}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 32.3.8.2.1

NOTE— $M_{10}^r(k)$ is a “reverse” function of the $M(k)$ defined in 17.3.5.10.

32.3.8.6 RL-SIG definition

The RL-SIG field is a repeat of the L-SIG field and is used to differentiate an NGV PPDU from a non-NGV PPDU. RL-SIG field shall be modulated the same as L-SIG field.

The time domain waveform of the RL-SIG field shall be as given by Equation (32-11).

$$r_{RL-SIG}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \times N_{RL-SIG}^{Tone}}} w_{T_{L-SIG}}(t) \sum_{i_{BW}=0}^{N_{10\text{ MHz}}-1} \sum_{k=-26}^{26} \gamma_{(k-K_{shift}(i_{BW})), BW}(D_{k,10} + p_1 P_k) \times \exp(j2\pi(k - K_{shift}(i_{BW}))\Delta_F(t - T_{GI} - T_{CS}^{i_{TX}})) \quad (32-11)$$

where

- N_{RL-SIG}^{Tone} has the value given in Table 32-8.
- p_1 is the second pilot value in the sequence defined in 17.3.5.10

32.3.8.7 NGV-SIG definition

The NGV-SIG field carries information required to interpret NGV PPDU. The NGV-SIG field contains the fields listed in Table 32-10.

The NGV-SIG field is composed of one OFDM symbol, containing 24 data bits, as shown in Table 32-10. NGV-SIG is transmitted before RNGV-SIG. The NGV-SIG symbol shall be BCC encoded at rate, $R = 1/2$, be interleaved, be mapped to a BPSK constellation, and have pilots inserted following the steps described in 17.3.5.6, 17.3.5.7, 17.3.5.8, and 17.3.5.9. The stream of 48 complex numbers generated by these steps (before pilot insertion) is denoted by d_k , $k = 0, \dots, 47$.

Table 32-10—Fields in the NGV-SIG field

Bit	Field	Number of bits	Description
B0–B1	PHY Version	2	Set to 0 for NGV PHY. Values 1–3 are reserved.
B2	Bandwidth	1	Set to 0 for 10 MHz, and set to 1 for 20 MHz.
B3–B6	NGV-MCS	4	For 10 MHz PPDU, set to n for NGV-MCS n , where $n = 0, 1, 2, \dots, 8$, and 15. Value 9 is not valid and values 10–14 are reserved. For 20 MHz PPDU, set to n for NGV-MCS n , where $n = 0, 1, 2, \dots, 9$, and 15. MCS 10–14 are reserved.
B7	N_{SS}	1	Set to 0 for one spatial stream, and set to 1 for two spatial streams.
B8–B9	Midamble Periodicity	2	Set to 0 for 4 symbols. Set to 1 for 8 symbols. Set to 2 for 16 symbols. Value 3 is reserved.
B10	LTF Format	1	Set to 0 for NGV-LTF-2x. Set to 1 for NGV-LTF-1x. NOTE—If NGV-LTF-2x-Repeat is used (see 32.3.8.10) with B2 set to 0, B3–B6 set to 15 and B7 set to 0, this field is reserved and set to 1.
B11	LDPC Extra OFDM Symbol	1	Set to 1 if the LDPC PPDU encoding process results in an extra OFDM symbol as described in 21.3.10.5.4. Set to 0 otherwise.
B12	LTF Repetition	1	Set to 1 for NGV ranging NDP with NGV-LTF repetition. Set to 0 otherwise.
B13	Reserved	1	Reserved and set to 1.
B14–B17	CRC	4	CRC is calculated as in 27.3.11.7.3 with $L = 13$.
B18–B23	Tail	6	Used to terminate the trellis of the convolutional decoder. Set to 0.

The time domain waveform for the NGV-SIG field in an NGV PPDU shall be as specified in Equation (32-12).

$$r_{NGV-SIG}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \times N_{NGV-SIG}^{Tone}}} w_{T_{NGV-SIG}}(t) \sum_{i_{BW}=0}^{N_{10MHz}-1} \sum_{k=-26}^{26} \gamma_{(k-K_{shift}(i_{BW})), BW}(D_{k, 10} + p_2 P_k) \times \exp(j2\pi(k - K_{shift}(i_{BW}))\Delta_F(t - T_{GI} - T_{CS}^{i_{TX}})) \quad (32-12)$$

where

$N_{10\text{ MHz}}$ and $K_{shift}(i)$ are defined in 32.3.8.5

$$D_{k,10} = \begin{cases} 0, & k = 0, \pm 7, \pm 21 \\ d_{M'_{10}(k)}, & \text{otherwise} \end{cases}$$

$M'_{10}(k)$ is defined in Equation (32-10)

P_k and p_2 are defined in 17.3.5.10

$N_{NGV-SIG}^{Tone}$ has the value given in Table 32-8

$\Upsilon_{k,BW}$ is defined in Equation (32-4) and Equation (32-5)

$T_{CS}^{i_{TX}}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 32.3.8.2.1

NOTE—In NGV PPDU, the NGV-SIG is transmitted with the same number of subcarriers and the same cyclic shifts as the preceding non-NGV portion of the preamble.

32.3.8.8 RNGV-SIG definition

The RNGV-SIG field is a repeat of the NGV-SIG field and is configured identically to the NGV-SIG.

The time domain waveform of the RNGV-SIG field shall be as given by Equation (32-13).

$$r_{RNGV-SIG}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \times N_{RNGV-SIG}^{Tone}}} w_{T_{RNGV-SIG}}(t) \sum_{i_{BW}=0}^{N_{10\text{ MHz}}-1} \sum_{k=-26}^{26} \Upsilon_{(k-K_{shift}(i_{BW})), BW}(D_{k,10} + p_3 P_k) \quad (32-13) \\ \times \exp(j2\pi(k - K_{shift}(i_{BW}))\Delta_F(t - T_{GI} - T_{CS}^{i_{TX}}))$$

where

$N_{RNGV-SIG}^{Tone}$ has the value given in Table 32-8

p_3 is the fourth pilot value in the sequence defined in 17.3.5.10

32.3.8.9 NGV-STF definition

The main purpose of the NGV-STF field is to improve automatic gain control estimation in a MIMO transmission. The duration of the NGV-STF field is 8 μ s. The frequency domain sequence used to construct the NGV-STF field in a 10 MHz transmission is identical to the VHT-STF field for 20 MHz transmission. In a 20 MHz transmission, the NGV-STF field is constructed by using the frequency domain sequence for the VHT-STF field of 40 MHz transmission and applying appropriate phase rotations per 10 MHz subchannel.

For a 10 MHz transmission, the frequency domain sequence is given by Equation (32-14).

$$NGVS_{-28,28} = VHTS_{-28,28} \quad (32-14)$$

where $VHTS_{-28,28}$ is defined in Equation (21-29).

For a 20 MHz transmission, the frequency domain sequence is given by Equation (32-15).

$$NGVS_{-58,58} = VHTS_{-58,58} \quad (32-15)$$

where $VHTS_{-58,58}$ is defined in Equation (21-30).

NOTE—Equation (32-14) and Equation (32-15) do not show the phase rotation per 10 MHz subchannel.

The time domain representation of the signal on transmit chain i_{TX} shall be as specified in Equation (32-16).

$$r_{NGV-STF}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{SS} \times N_{NGV-STF}^{Tone}}} w_{T_{NGV-STF}}(t) \sum_{k=-N_{SR}}^{N_{SR}} \sum_{m=1}^{N_{SS}} [Q_k]_{i_{TX},m} \gamma_{k,BW} NGVS_k \quad (32-16)$$

$$\times \exp(j2\pi k \Delta_F(t - T_{CS,NGV}(m)))$$

where

$N_{NGV-STF}^{Tone}$ has the value given in Table 32-8

$T_{CS,NGV}(m)$ is given in 32.3.8.2.2

Q_k is defined in 32.3.7.3

$\gamma_{k,BW}$ is defined in defined in Equation (32-4) and Equation (32-5)

32.3.8.10 NGV-LTF definition

The NGV-LTF field provides means for the receiver to estimate the MIMO channel between the set of constellation mapper outputs and the receive chains. The transmitter provides training for N_{SS} spatial streams (spatial mapper inputs) used for the transmission of the PSDU(s). For each tone, the MIMO channel that can be estimated is an $N_{RX} \times N_{SS}$ matrix. An NGV transmission has a preamble that contains NGV-LTF symbols, where the data tones of each NGV-LTF symbol are multiplied by entries belonging to a matrix $P_{NGV-LTF}$, to enable channel estimation at the receiver. The pilot tones of each NGV-LTF symbol are multiplied by the entries of a matrix $R_{NGV-LTF}$ defined in the following text. The multiplication of the pilot tones in the NGV-LTF symbol by the $R_{NGV-LTF}$ matrix instead of the $P_{NGV-LTF}$ matrix allows receivers to track phase and frequency offset during MIMO channel estimation using the NGV-LTF. The number of NGV-LTF symbols, $N_{NGV-LTF}$, is a function of the number of spatial streams, N_{SS} , as shown in Table 32-11 and LTF_REP as shown in 32.3.16. As a result the NGV-LTF field consists of one, two, or four symbols.

Table 32-11—Number of NGV-LTFs required for different numbers of spatial streams

N_{SS}	$N_{NGV-LTF}$
1	$1 \times (\text{LTF_REP} + 1)$
2	$2 \times (\text{LTF_REP} + 1)$

An NGV PPDU supports three NGV-LTF formats: NGV-LTF-1x, NGV-LTF-2x, and NGV-LTF-2x-Repeat. NGV-LTF-2x is the default LTF format; NGV-LTF-1x is used for high efficiency transmission of one spatial stream and NGV-LTF-2x-Repeat is used for extended range transmissions. A transmitter shall use the NGV-LTF-2x-Repeat format if and only if the NGV Data field of a 10 MHz NGV PPDU is modulated with BPSK-DCM and one spatial stream; the transmitter shall use the NGV-LTF-2x if the NGV Data field is modulated with two spatial streams; the transmitter shall use the NGV-LTF-2x for NGV ranging NDP; otherwise, the transmitter can choose either NGV-LTF-2x or NGV-LTF-1x. The NGV-LTF format of NGV-LTF-1x or NGV-LTF-2x is signaled by B10 in NGV-SIG field. The NGV-LTF-2x-Repeat format is signaled by the combination of B2 set to 0, B3–B6 set to 15 and B7 set to 0 in NGV-SIG field, and B10 is Reserved and set to 1. The duration of each NGV-LTF symbol is $T_{NGV-LTF}$, defined in Equation (32-17).

$$T_{NGV-LTF} = \begin{cases} T_{NGV-LTF-1x}, & \text{if NGV-LTF-1x} \\ T_{NGV-LTF-2x}, & \text{if NGV-LTF-2x} \\ T_{NGV-LTF-2x-Repeat}, & \text{if NGV-LTF-2x-Repeat} \end{cases} \quad (32-17)$$

Let LTF_{left} and LTF_{right} be the sequences defined in Equation (32-18) and Equation (32-19), respectively.

$$LTF_{left} = \{1, 1, -1, -1, 1, 1, -1, 1, -1, 1, 1, 1, 1, 1, -1, -1, 1, 1, -1, 1, -1, 1, 1, 1\} \quad (32-18)$$

$$LTF_{right} = \{1, -1, -1, 1, 1, -1, 1, -1, 1, -1, -1, -1, -1, -1, 1, 1, -1, -1, 1, -1, 1, -1, 1, 1\} \quad (32-19)$$

NOTE 1— LTF_{left} is identical to the leftmost 26 elements of Equation (17-8), and LTF_{right} is identical to the rightmost 26 elements of Equation (17-8).

In a 10 MHz transmission, the NGV-LTF-1x sequence transmitted is given by Equation (32-20).

$$NGV-LTF_{-28,28} = \{1, 0, 1, 0, -1, 0, 1, 0, -1, 0, -1, 0, 1, 0, 1, 0, -1, 0, 1, 0, 1, 0, 1, 0, 0, 0, -1, 0, 1, 0, -1, 0, -1, 0, -1, 0, -1, 0, 1, 0, -1, 0, -1, 0, -1, 0, 1, 0, 1, 0, -1\} \quad (32-20)$$

In a 10 MHz transmission, the NGV-LTF-2x sequence transmitted is given by Equation (32-21).

$$\begin{aligned} NGV-LTF_{-28,28} &= \{1, 1, LTF_{left}, 0, LTF_{right}, -1, -1\} \\ &= VHT-LTF_{-28,28} \end{aligned} \quad (32-21)$$

where

$VHT-LTF_{-28,28}$ is defined in Equation (21-36).

In a 20 MHz transmission, the NGV-LTF-1x sequence transmitted is given by Equation (32-22).

$$\begin{aligned} NGV-LTF_{-58,58} &= \{1, 0, -1, 0, 1, 0, -1, 0, -1, 0, 1, 0, 1, 0, -1, 0, 1, 0, 1, 0, 1, 0, 1, 0, -1, 0, 1, 0, 1, 0, \\ &1, 0, 1, 0, 1, 0, -1, 0, 1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, 1, 0, -1, 0, -1, 0, -1, 0, \\ &1, 0, 1, 0, -1, 0, 1, 0, 0, 0, -1, 0, 1, 0, 1, 0, -1, 0, 1, 0, -1, 0, -1, 0, 1, 0, 1, 0, 1, 0, \\ &-1, 0, 1, 0, 1, 0, 1, 0, 1, 0, -1, 0, 1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, 1, 0, -1, \\ &0, -1, 0, -1, 0, 1, 0, 1\} \end{aligned} \quad (32-22)$$

In a 20 MHz transmission, the NGV-LTF-2x sequence transmitted is given by Equation (32-23).

$$\begin{aligned} NGV-LTF_{-58,58} &= \{LTF_{left}, 1, LTF_{right}, -1, -1, -1, 1, 0, 0, 0, -1, 1, 1, -1, LTF_{left}, 1, LTF_{right}\} \quad (32-23) \\ &= VHT-LTF_{-58,58} \end{aligned}$$

where

$VHT-LTF_{-58,58}$ is defined in Equation (21-37).

NOTE 2—Equation (32-20), Equation (32-21), Equation (32-22), and Equation (32-23) do not show the phase rotation per 10 MHz subchannel.

The generation of the time domain NGV-LTF symbols is shown in Figure 32-9.

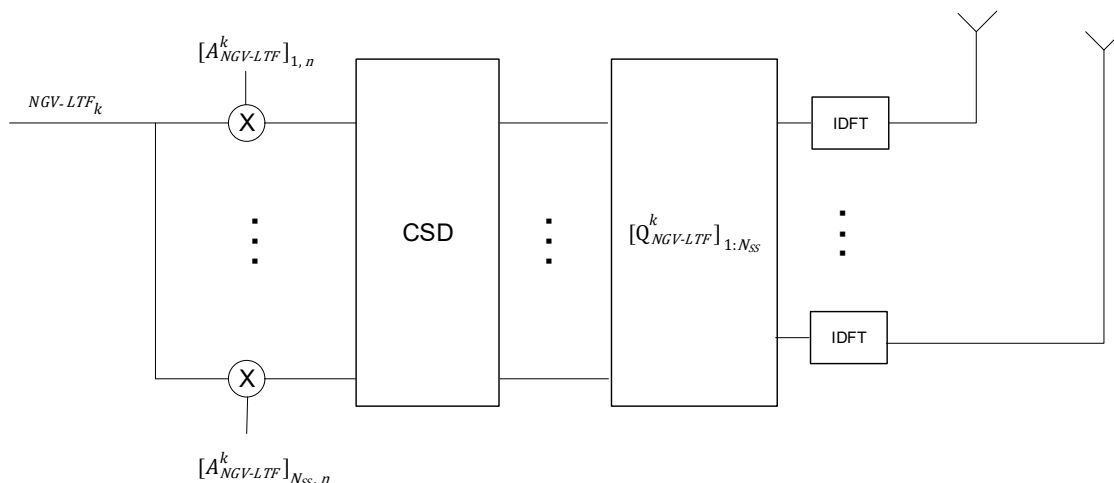


Figure 32-9—Generation of NGV-LTF symbols

The generation of a time domain symbol of NGV-LTF-1x is equivalent to modulating every other tone in an OFDM symbol of 6.4 μ s excluding GI, and then only transmit the first half of the OFDM symbol in the time domain, as shown in Figure 32-10.

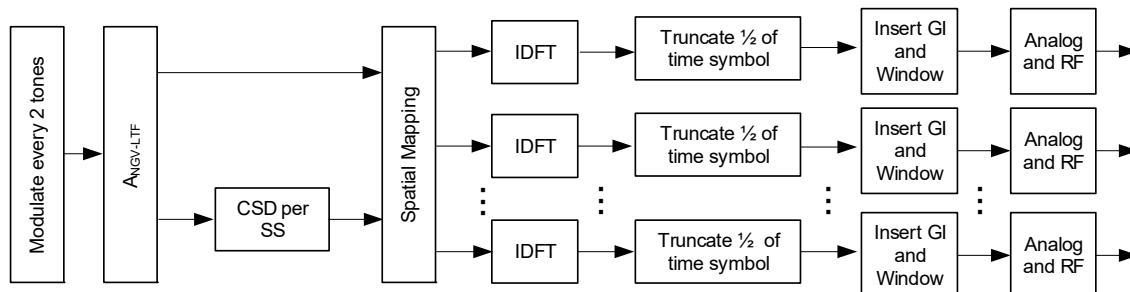


Figure 32-10—Generation of NGV-LTF-1x symbols

NGV-LTF-2x-Repeat is constructed by repeating the time domain symbol of NGV-LTF-2x excluding GI and preappend one cyclic prefix of duration 1.6 μ s. The generation of a time domain symbol of

NGV-LTF-2x-Repeat is equivalent to modulating every tone in an OFDM symbol of 6.4 μ s excluding GI, and then the OFDM symbol is repeated in the time domain, as shown in Figure 32-11.

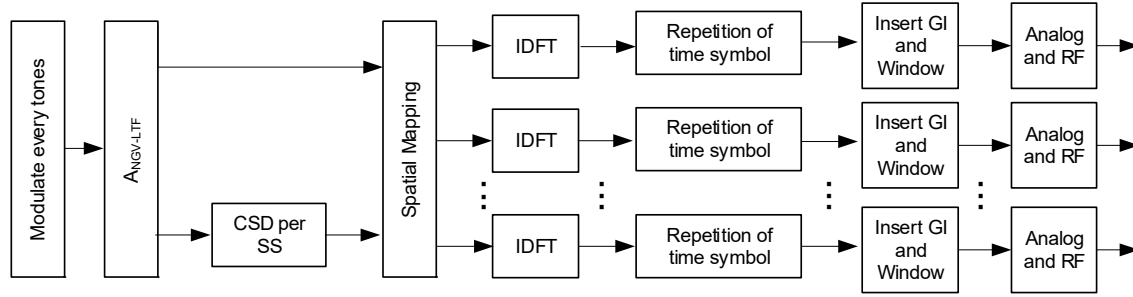


Figure 32-11—Generation of NGV-LTF-2x-Repeat symbols

$A_{NGV-LTF}^k$ is given by Equation (32-24).

$$A_{NGV-LTF}^k = \begin{cases} R_{NGV-LTF}, & \text{if } k \in K_{pilot} \\ P_{NGV-LTF}, & \text{otherwise} \end{cases} \quad (32-24)$$

where

K_{pilot} is the set of subcarrier indices for the pilot tones:

For a 10 MHz transmission, $K_{pilot} = \{\pm 8, \pm 22\}$

For a 20 MHz transmission, $K_{pilot} = \{\pm 12, \pm 26, \pm 54\}$

$R_{NGV-LTF}$ is an $N_{SS} \times N_{NGV-LTF}$ matrix whose elements are defined in Equation (32-25) and Equation (32-26)

$$[R_{NGV-LTF}]_{m,n} = [P_{NGV-LTF}]_{1,n}, \quad 1 \leq n \leq N_{NGV-LTF}, \quad 1 \leq m \leq N_{SS} \quad (32-25)$$

$$P_{NGV-LTF} = [P_{2 \times 2} P_{2 \times 2}] \quad (32-26)$$

where

$P_{2 \times 2} = P_{4 \times 4 \{1:2\} \{1:2\}}$ is a submatrix of $P_{4 \times 4}$ with elements of the first two rows and two columns of $P_{4 \times 4}$, which is defined in Equation (19-27)

The time domain representation of the waveform transmitted on transmit chain i_{TX} shall be as described by Equation (32-27).

$$\begin{aligned}
 r_{NGV-LTF}^{(i_{TX})}(t) &= \frac{1}{\sqrt{N_{SS} \times N_{NGV-LTF}^{Tone}}} \sum_{n=0}^{N_{NGV-LTF}-1} w_{T_{NGV-LTF}}(t - nT_{NGV-LTF}) \\
 &\times \sum_{k=-N_{SR}}^{N_{SR}} \sum_{m=1}^{N_{SS}} [Q_k]_{i_{TX},m} \gamma_{k,BW} [A_{NGV-LTF}^k]_{m,(n+1)} NGV-LTF_k \\
 &\times \exp(j2\pi k \Delta_F(t - nT_{NGV-LTF} - T_{GI} - T_{CS,NGV}(m)))
 \end{aligned} \tag{32-27}$$

where

- $N_{NGV-LTF}^{Tone}$ has the value given in Table 32-8
- $T_{CS,NGV}(m)$ is given in 32.3.8.2.2
- Q_k is defined in 32.3.7.3
- $\gamma_{k,BW}$ is defined in Equation (32-4) and Equation (32-5)
- $A_{NGV-LTF}^k$ is defined in Equation (32-24)

32.3.9 Data field

32.3.9.1 General

The number of OFDM symbols in the Data field is determined by the Length field in L-SIG (see Equation (32-8)), the preamble duration, and the setting of the NGV-LTF and Midamble field in NGV-SIG (see 32.3.8).

For LDPC encoding, the Data field shall consist of the SERVICE field, the PSDU, and the PHY pad bits. The padding flow is as follows. The MAC delivers a PSDU that fills the available octets in the Data field of the PPDU. The PHY determines the number of pad bits to add and appends them to the PSDU. The number of pad bits added always is 0 to 7. The PHY padding bits are calculated using Equation (32-28).

$$N_{PAD} = N_{SYM,init} N_{DBPS} - 8 \times \text{PSDU_LENGTH} - N_{service} \tag{32-28}$$

where

- PSDU_LENGTH is defined in 32.4.3
- $N_{SYM,init}$ is given by Equation (21-62) with $m_{STBC} = 1$

32.3.9.2 SERVICE field

The SERVICE field is as shown in Table 32-12.

Table 32-12—SERVICE field

Bits	Field	Description
B0–B6	Scrambler Initialization	Set to 0
B7–B15	Reserved	Set to 0

32.3.9.3 Scrambler

The SERVICE, PSDU, and PHY pad parts of the Data field shall be scrambled by the scrambler defined in 17.3.5.5. The Clause 17 TXVECTOR parameters CH_BANDWIDTH_IN_NON_HT and DYN_BANDWIDTH_IN_NON_HT are not present; therefore, the initial state of the scrambler is set to a pseudorandom nonzero seed.

32.3.9.4 Coding

The Data field of an NGV PPDU shall be encoded using LDPC coding. The LDPC code and encoding process shall follow as described in 21.3.10.5.4 for a VHT SU PPDU with parameter m_{STBC} set to 1 and the LDPC Extra Symbol field setting in NGV-SIG.

32.3.9.5 Stream parser

After coding and puncturing, the data bit streams at the output of the FEC encoder are processed in groups of N_{CBPS} bits. Each of these groups is rearranged into N_{SS} blocks of N_{CBPSS} bits. This operation is referred to as stream parsing and is described in 21.3.10.6 for a SU transmission.

32.3.9.6 Constellation mapping

The mapping between bits at the output of the stream parser and complex constellation points for BPSK, QPSK, 16-QAM, and 64-QAM follows the rules defined in 17.3.5.8 and 256-QAM follows the rules defined in 21.3.10.9.

The streams of complex numbers out of constellation mapping are denoted as shown in Equation (32-29).

$$d_{k,m,n}, \quad 0 \leq k \leq N_{SD} - 1; \quad 1 \leq m \leq N_{SS}; \quad 0 \leq n \leq N_{SYM} - 1 \quad (32-29)$$

For NGV data portion modulated with NGV-MCS 15, the input stream is broken into groups of N_{CBPS} bits ($B_0, B_1, \dots, B_{N_{CBPS}-1}$). Each bit B_k is BPSK modulated to a sample d_k . This generates the samples for the lower half of the data subcarriers as $d_{k,1,n} = d_k$. For the upper half of the subcarriers, the samples are generated as $d_{k+N_{SD},1,n} = d_{k,1,n} \times e^{j(k+N_{SD})\pi}$, with $k = 0, 1, \dots, N_{SD} - 1$. The N_{SD} here refers to the N_{SD} for NGV-MCS 15, which is half the value of N_{SD} for NGV-MCS 0.

32.3.9.7 Pilot subcarriers

In a 10 MHz transmission, four pilot tones shall be inserted in subcarriers $k = \{-22, -8, 8, 22\}$. The pilot mapping P_n^k for subcarrier k for symbol n shall be as specified in Equation (32-30).

$$\begin{aligned} P_n^{\{-22, -8, 8, 22\}} &= \{ \Psi_{1,n \bmod 4}^{(1)}, \Psi_{1,(n+1) \bmod 4}^{(1)}, \Psi_{1,(n+2) \bmod 4}^{(1)}, \Psi_{1,(n+3) \bmod 4}^{(1)} \} \\ P_n^{k \neq \{-22, -8, 8, 22\}} &= 0 \end{aligned} \quad (32-30)$$

where

$\Psi_{1,m}^{(1)}$ is given by the $N_{STS} = 1$ row of Table 19-19

In a 20 MHz transmission, six pilot tones shall be inserted in subcarriers $k = \{-54, -26, -12, 12, 26, 54\}$. The pilot mapping P_n^k for subcarrier k for symbol n shall be as specified in Equation (32-31).

$$\begin{aligned} P_n^{\{-54, -26, -12, 12, 26, 54\}} &= \{\Psi_{1, n \bmod 6}^{(1)}, \Psi_{1, (n+1) \bmod 6}^{(1)}, \dots, \Psi_{1, (n+5) \bmod 6}^{(1)}\} \\ P_n^{k \neq \{-54, -26, -12, 12, 26, 54\}} &= 0 \end{aligned} \quad (32-31)$$

where

$\Psi_{1, m}^{(1)}$ is given by the $N_{STS} = 1$ row of Table 19-20

The above pilot mapping shall be copied to all spatial streams before the spatial stream cyclic shifts are applied.

The pilot subcarrier locations in the NGV-LTF field for NGV-LTF-1x, NGV-LTF-2x, and NGV-LTF-2x-Repeat are the same as the pilot subcarrier locations in the Data field.

32.3.9.8 OFDM modulation

32.3.9.8.1 Transmission in NGV format

The time domain waveform of the Data field of an NGV PPDU from transmit chain i_{TX} , $1 \leq i_{TX} \leq N_{TX}$ shall be as defined in Equation (32-32) and Equation (32-33).

$$\begin{aligned} r_{NGV\text{-}Data}^{(i_{TX})}(t) &= \sum_{n=0}^{N_{SYM}-1} w_{T_{SYM}} \left(t - nT_{SYM} - \left\lfloor \frac{n}{M} \right\rfloor T_{MA} \right) q_n r_{NGV\text{-}Data, n}^{(i_{TX})} \left(t - \left\lfloor \frac{n}{M} \right\rfloor T_{MA} \right) + \\ &\quad \sum_{i_{MA}=0}^{N_{MA}-1} p_{i_{MA}} r_{NGV\text{-}LTF}^{(i_{TX})}(t - (i_{MA} + 1)M \cdot T_{SYM} - i_{MA}T_{MA}) \end{aligned} \quad (32-32)$$

$$\begin{aligned} r_{NGV\text{-}Data, n}^{(i_{TX})}(t) &= \frac{1}{\sqrt{N_{NGV\text{-}Data}^{Tone} N_{SS}}} \sum_{k=-N_{SR}m}^{N_{SR}} \sum_{m=1}^{N_{SS}} \left(\left[\mathcal{Q}_k \right]_{i_{TX}, m} \Upsilon_{k, BW}(D_{k, m, n, BW} + p_{n+4} P_n^k) \right. \\ &\quad \left. \times \exp(j2\pi k \Delta_F(t - nT_{SYM} - T_{GI, Data} - T_{CS, NGV}(m))) \right) \end{aligned} \quad (32-33)$$

where

$r_{NGV\text{-}Data, n}^{(i_{TX})}(t)$ is the n^{th} OFDM data symbol in the Data field, $0 \leq n < N_{SYM} - 1$

M is the midamble periodicity defined in 32.3.9.9

$r_{NGV\text{-}LTF}^{(i_{TX})}(t)$ is defined in Equation (32-27)

$T_{MA} = T_{NGV\text{-}LTF} N_{NGV\text{-}LTF}$ is the duration of one midamble

p_n is defined in 17.3.5.10

$$q_n = \begin{cases} 1, & \text{for } n < M \\ p_{\left\lfloor \frac{n-M}{M} \right\rfloor}, & \text{for } n \geq M \end{cases}$$

i_{MA} is the index for the midambles, $0 \leq i_{MA} \leq N_{MA} - 1$

P_n^k is defined in 32.3.9.7

$\Upsilon_{k, BW}$ is defined in Equation (32-4) and Equation (32-5)

$D_{k,m,n,BW}$ is the transmitted constellation at subcarrier k , spatial stream m , and Data field OFDM symbol n and is defined in Equation (32-34) and Equation (32-35)

$N_{NGV-Data}^{Tone}$ has the value given in Table 32-8

$T_{CS,NGV}(n)$ is defined in Table 21-11

$T_{GI,Data}$ is the guard interval duration. $T_{GI,Data} = T_{GI}$

In a 10 MHz NGV PPDU transmission:

$$D_{k,m,n,10} = \begin{cases} 0, & k = 0, \pm 8, \pm 22 \\ d_{M_{10}^r(k),m,n}, & \text{otherwise} \end{cases} \quad (32-34)$$

where

$$M_{10}^r(k) = \begin{cases} k+28, & -28 \leq k \leq -23 \\ k+27, & -21 \leq k \leq -9 \\ k+26, & -7 \leq k \leq -1 \\ k+25, & 1 \leq k \leq 7 \\ k+24, & 9 \leq k \leq 21 \\ k+23, & 23 \leq k \leq 28 \end{cases}$$

In a 20 MHz NGV PPDU transmission:

$$D_{k,m,n,20} = \begin{cases} 0, & k = 0, \pm 12, \pm 26, \pm 54 \\ d_{M_{20}^r(k),m,n}, & \text{otherwise} \end{cases} \quad (32-35)$$

where

$$M_{20}^r(k) = \begin{cases} k+58, & -58 \leq k \leq -55 \\ k+57, & -53 \leq k \leq -27 \\ k+56, & -25 \leq k \leq -13 \\ k+55, & -11 \leq k \leq -2 \\ k+52, & 2 \leq k \leq 11 \\ k+51, & 13 \leq k \leq 25 \\ k+50, & 27 \leq k \leq 53 \\ k+49, & 55 \leq k \leq 58 \end{cases}$$

Q_k is a spatial mapping/steering matrix with N_{TX} rows and N_{SS} columns for subcarrier k . Q_k may be frequency dependent. Refer to the examples of Q_k listed in 19.3.11.11.2 for examples of Q_k that could be used for NGV PPDU.

NOTE—Implementations are not restricted to the spatial mapping matrix examples listed in 19.3.11.11.2 and the number of transmit chains N_{TX} could be more than one. The beamforming steering matrices are implementation specific.

32.3.9.9 Midambles

An NGV STA shall include midambles in an NGV PPDU transmission when the number of midamble periods, N_{MA} , is greater than zero in accordance with Equation (32-36). Midambles facilitate updating of the channel estimate during the NGV PPDU reception.

Midambles are present in the Data field of the NGV PPDU every M OFDM symbols, where M is either 4, 8, or 16 as indicated by the Midamble Periodicity field in the NGV-SIG field (see Table 32-10).

The Midamble field shall use the same format as the NGV-LTF field of the same PPDU, which format is signalled in the NGV-SIG field. The generation of midamble is defined in 32.3.8.10.

The midambles and succeeding Data field OFDM symbols shall be randomized to prevent the generation of spectral lines in the NGV PPDU spectrum. The n th midamble and M Data field OFDM symbols following immediately after the n th midamble shall be multiplied by p_n , $n = 0, \dots, N_{MA} - 1$, where p_n is defined in 17.3.5.10. The midamble and the randomization procedure in an NGV PPDU are shown in Figure 32-12.

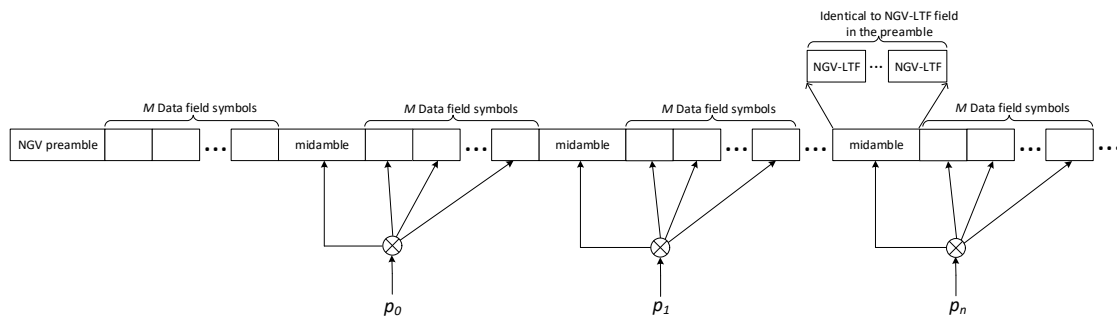


Figure 32-12—NGV PPDU with midamble and randomization procedure

If present, the number of midamble periods, N_{MA} , in a PPDU is calculated using Equation (32-36).

$$N_{MA} = \left\lfloor \frac{N_{SYM} - 1}{M} \right\rfloor \quad (32-36)$$

where N_{SYM} is the total number of data symbols in the Data field, is given in 21.3.10.5.4, and computed using Equation (19-41) in step d) of 19.3.11.7.5.

As shown in Figure 32-12, the first midamble is inserted immediately after the M th OFDM symbol in the Data field, and a midamble is not inserted after the last data OFDM symbol if $\text{mod}(N_{SYM}, M) = 0$.

32.3.9.10 Non-NGV duplicate transmission

When the TXVECTOR parameter FORMAT is NON_NGV_10 and the TXVECTOR parameter NON_NGV_MODULATION is NON_NGV_10_DUP_OFDM, the transmitted PPDU is a non-NGV duplicate. Non-NGV duplicate transmission is used to transmit to STAs that support non-NGV OFDM and may be present in a part of a 20 MHz channel (see Table 32-2). The RL-SIG, NGV-SIG, RNGV-SIG, NGV-STF, and NGV-LTF fields are not transmitted. The L-STF, L-LTF, and L-SIG fields shall be transmitted in the same way as in the NGV transmission, except that the L-SIG field's Rate and Length fields shall be as defined in 17.3.4. The Data field shall be as defined in Equation (32-37).

NOTE—For L-STF and L-LTF fields, the power scale factor is 1 in 20 MHz non-NGV duplicate PPDU.

$$r_{\text{Non-NGV-Data}}^{(i_{TX})}(t) = \frac{1}{\sqrt{N_{TX}N_{\text{NON_NGV_10_DUP_OFDM-Data}}^{\text{Tone}}}} \sum_{n=0}^{N_{SYM}-1} w_{T_{SYM}}(t - nT_{SYM}) \quad (32-37)$$

$$\times \sum_{k=-26}^{26} (D_{k,n} + p_{n+1}P_k)(\exp(j2\pi(k-32)\Delta_F(t - nT_{SYM} - T_{GI} - T_{CS}^{i_{TX}})))$$

$$+ j \times \exp(j2\pi(k+32)\Delta_F(t - nT_{SYM} - T_{GI} - T_{CS}^{i_{TX}})))$$

where

P_k and p_n are defined in 17.3.5.10

$$D_{k,n} = \begin{cases} 0, & k = 0, \pm 7, \pm 21 \\ d_{M_{10}^r(k), 1, n}, & \text{otherwise} \end{cases} \quad (32-38)$$

$M_{10}^r(k)$ is defined in Equation (32-10)

$T_{CS}^{i_{TX}}$ represents the cyclic shift of the transmit chain i_{TX} and is defined in 32.3.8.2.1

$N_{\text{NON_NGV_10_DUP_OFDM-Data}}^{\text{Tone}}$ has the value given in Table 32-8

32.3.10 SU-MIMO

SU-MIMO is defined for an NGV transmitter to transmit two spatial streams to an NGV receiver to improve throughput.

An NGV transmitter equipped with more than one antenna may use the SU-MIMO technique to transmit an NGV PPDU with two spatial streams. The choice of number of spatial streams is described in 31.5.

The steering matrix Q_k , defined in 32.3.7.3, used by the transmitter is implementation specific.

NOTE—The channel in the vehicular environment is fast-varying, so a SU-MIMO transmitter cannot obtain accurate channel information through sounding. Therefore, there is no explicit sounding protocol defined for an NGV transmitter to obtain the channel information prior to the transmission.

32.3.11 Transmit specification

32.3.11.1 Transmit spectrum mask

The transmit spectrum mask by regulatory domain is defined in Annex D and Annex E.

For any STA transmitting a 10 MHz NGV PPDU, the transmit spectrum mask is defined in 17.3.9.3.

For any STA using 20 MHz channel spacing and transmitting a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU, the transmitted spectral density shall have a 0 dBr bandwidth not exceeding 19 MHz and shall not exceed the spectrum mask created using the permitted power spectral density levels listed in Table 32-13.

The transmission of a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU shall meet transmit spectrum mask C2 regardless of the transmit power class to which the STA belongs.

NOTE 1—In the presence of additional regulatory restrictions, the device needs to meet both the regulatory requirements and the mask defined in this subclause.

NOTE 2—For rules regarding TX center frequency leakage levels, see 32.3.11.4.2. The spectral mask requirements in this subclause do not apply to the RF LO.

Table 32-13—Spectrum mask data for 20 MHz channel spacing

STA transmit power class	Permitted power spectral density, dBr				
	± 9.5 MHz offset (±f1)	± 10.0 MHz offset (±f2)	± 10.5 MHz offset (±f3)	± 15 MHz offset (±f4)	± 25 MHz offset (±f5)
Any Class	0	−26	−32	−40	−50

32.3.11.2 Spectral flatness

Spectral flatness measurements shall be conducted using BPSK modulated PPDU. Demodulate the PPDU according to the following (or equivalent) procedure:

- a) Start of PPDU shall be detected.
- b) Transition from L-STF to L-LTF shall be detected and fine timing shall be established.
- c) Coarse and fine frequency offsets shall be estimated.
- d) Symbols in a PPDU shall be derotated according to estimated frequency offset.
- e) For each NGV-LTF symbol, transform the symbol into subcarrier received values, estimate the phase from the pilot subcarriers, and derotate the subcarrier values according to the estimated phase.
- f) For each of the data OFDM symbols: transform the symbol into subcarrier received values.

The spectral flatness test shall be performed over at least 20 PPDU. The PPDU under test shall be at least 16 data OFDM symbols long.

Evaluate spectral flatness using the subcarrier received values or the magnitude of the channel estimation.

Let $E_{i,avg}$ denote the magnitude of the channel estimation on subcarrier i or the average constellation energy of a BPSK modulated subcarrier i in an NGV data symbol.

For the spectral flatness test, the transmitting STA shall be configured to use a spatial mapping matrix Q_k (see 32.3.9.8) with flat frequency response. Each output port under test of the transmitting STA shall be connected through a cable to one input port of the testing instrumentation. The requirements are defined in Table 32-14 and apply to both 10 MHz and 20 MHz transmissions.

Table 32-14—Maximum transmit spectral flatness deviations

Format	Bandwidth of transmission (MHz)	Average subcarrier indices (inclusive)	Tested subcarrier indices (inclusive)	Maximum deviation (dB)
NGV	10	−16 to −1 and +1 to +16	−16 to −1 and +1 to +16	±4
			−28 to −17 and +17 to +28	+4/−6
	20	−42 to −2 and +2 to +42	−42 to −2 and +2 to +42	±4
			−58 to −43 and +43 to +58	+4/−6

32.3.11.3 Transmit center frequency and symbol clock frequency tolerance

The symbol clock frequency and transmit center frequency tolerance shall be ±20 ppm maximum. The transmit center frequency and the symbol clock frequency for all transmit antennas and frequency segments shall be derived from the same reference oscillator.

32.3.11.4 Modulation accuracy

32.3.11.4.1 Introduction to modulation accuracy tests

Transmit modulation accuracy specifications are described in 32.3.11.4.2 and 32.3.11.4.3. The test method is described in 32.3.11.4.4.

32.3.11.4.2 Transmitter center frequency leakage

TX LO leakage shall meet the following requirements for all bandwidths:

- When the RF LO is in the center of the transmitted PPDU BW, the power measured at the center of transmission BW using resolution BW 156.25 kHz shall not exceed the average power per subcarrier of the transmitted PPDU, or equivalently, $(P - 10\log(N_{ST}))$, where P is the transmit power per antenna in dBm, and N_{ST} is defined in Table 32-6.
- When the RF LO is not at the center of the transmitted PPDU BW, the power measured at the location of the RF LO using resolution BW 156.25 kHz shall not exceed the maximum of −32 dB relative to the total transmit power and −20 dBm, or equivalently $\max(P - 32, -20)$, where P is the transmit power per antenna in dBm.

The transmit center frequency leakage is specified per antenna.

32.3.11.4.3 Transmitter constellation error

The relative constellation RMS error, calculated by first averaging over subcarriers, NGV PPDU, and spatial streams (see Equation (19-89)) shall not exceed a data-rate dependent value according to Table 32-15. The number of spatial streams under test shall be equal to the number of utilized transmitting STA antenna (output) ports and also equal to the number of utilized testing instrumentation input ports. In the test, $N_{SS} = N_{STS}$ (no STBC) shall be used and no beamforming steering matrix shall be used. Each output port of the transmitting STA shall be connected through a cable to one input port of the testing instrumentation. The requirements apply to both 10 MHz and 20 MHz transmissions.

32.3.11.4.4 Transmitter modulation accuracy (EVM) test

The transmit modulation accuracy test shall be performed by instrumentation capable of converting the transmitted signals into a stream of complex samples at a sampling rate greater than or equal to the bandwidth of the signal being transmitted.

In this case, transmit modulation accuracy shall meet the required value in Table 32-15 using only the subcarriers within the channel width.

Table 32-15—Allowed relative constellation error versus constellation size and coding rate

Modulation	Coding rate	Relative constellation error (dB)
BPSK-DCM	1/2	–5
BPSK	1/2	–5
QPSK	1/2	–10
QPSK	3/4	–13
16-QAM	1/2	–16
16-QAM	3/4	–19
64-QAM	2/3	–22
64-QAM	3/4	–25
64-QAM	5/6	–27
256-QAM	3/4	–30
256-QAM	5/6	–32

The instrument shall have sufficient accuracy in terms of I/Q arm amplitude and phase balance, DC offsets, phase noise, and analog to digital quantization noise. A possible embodiment of such a setup is converting the signals to a low IF frequency with a microwave synthesizer, sampling the signal with a digital oscilloscope and decomposing it digitally into quadrature components. The sampled signal shall be processed in a manner similar to an actual receiver, according to the following steps, or equivalent procedure:

- Start of PPDU shall be detected.
- Transition from L-STF to L-LTF shall be detected and fine timing shall be established.
- Coarse and fine frequency offsets shall be estimated.
- Symbols in a PPDU shall be derotated according to estimated frequency offset.
- For each NGV-LTF symbol, transform the symbol into subcarrier received values, estimate the phase from the pilot subcarriers, and derotate the subcarrier values according to the estimated phase.
- Estimate the complex channel response coefficient for each of the subcarriers and each of the transmit streams. If midambles are present in the Data field of the PPDU, the channel response coefficients shall be based upon the most recently received midamble symbols.
- For each of the data OFDM symbols: transform the symbol into subcarrier received values, estimate the phase from the pilot subcarriers, derotate the subcarrier values according to the estimated phase, group the results from all of the receiver chains in each subcarrier to a vector, and multiply the vector by a zero-forcing equalization matrix generated from the estimated channel.

- h) For each data-carrying subcarrier in each spatial stream, find the closest constellation point and compute the Euclidean distance from it. If midambles are present in the Data field of the PPDU, the midamble symbols shall not be used to compute the Euclidean distance.
- i) Compute the average across PDUs of the RMS of all errors per PDU as given by Equation (19-89).

The test shall be performed over at least 20 PDUs [N_f as defined in Equation (19-89)]. The PDUs under test shall be at least 16 data OFDM symbols long. Random data shall be used for the symbols.

32.3.11.4.5 Time of departure accuracy

The time of departure accuracy test evaluates `TIME_OF_DEPARTURE` against `aTxPHYTxStartRMS` and `aTxPHYTxStartRMS` against `TIME_OF_DEPARTURE_ACCURACY_TEST_THRESH` as defined in Annex P with the following test parameters:

- `MULTICHANNEL_SAMPLING_RATE` is:

$$10 \times 10^6 \left(1 + \left\lceil \frac{f_H - f_L}{10 \text{ MHz}} \right\rceil \right) \text{ sample/s, for a CH_BANDWIDTH parameter equal to CBW10}$$

$$20 \times 10^6 \left(1 + \left\lceil \frac{f_H - f_L}{10 \text{ MHz}} \right\rceil \right) \text{ sample/s, for a CH_BANDWIDTH parameter equal to CBW20}$$

where

f_H is the nominal center frequency in Hz of the highest channel in the channel set

f_L is the nominal center frequency in Hz of the lowest channel in the channel set. The channel set is the set of channels used for transmitting measurement frames, the channel set comprises channels uniformly spaced across $f_H - f_L \leq 50 \text{ MHz}$.

- `FIRST_TRANSITION_FIELD` is L-STF.
- `SECOND_TRANSITION_FIELD` is L-LTF.
- `TRAINING_FIELD` is the long symbols windowed in a manner that should approximate the windowing described in 17.3.2.5 with $T_{TR} = 100 \text{ ns}$ for 20 MHz channel spacing, $T_{TR} = 200 \text{ ns}$ for 10 MHz channel spacing.
- `TIME_OF_DEPARTURE_ACCURACY_TEST_THRESH` is 80 ns.

NOTE—The indicated windowing applies to the time of departure accuracy test equipment, and not the transmitter or receiver.

32.3.12 Receiver specification

32.3.12.1 General

For receiver minimum input sensitivity, adjacent channel rejection, nonadjacent channel rejection, receiver maximum input level, and CCA sensitivity requirements described in this subclause, the input levels are measured at the antenna connector and are referenced as the average power per receive antenna. The number of spatial streams under test shall be equal to the number of utilized transmitting STA antenna (output) ports and also equal to the number of utilized device under test input ports. Each output port of the transmitting STA shall be connected through a cable to one input port of the device under test.

The requirements on receiver minimum input sensitivity in 32.3.12.2, adjacent channel rejection in 32.3.12.3, and nonadjacent channel rejection in 32.3.12.4 apply to NGV PDUs. NGV-LTF-2x-Repeat format is used if the NGV PDUs are modulated with NGV-MCS 15 and $N_{SS} = 1$ in 10 MHz, otherwise, NGV-LTF-2x format is used.

NOTE—Similar receiver minimum input sensitivity requirements, adjacent channel rejection requirements, and nonadjacent channel rejection requirements are defined for VHT PDUs specified in 21.3.18.

32.3.12.2 Receiver minimum input sensitivity

The packet error ratio (PER) shall be less than 10% for a PSDU with the rate dependent input levels listed in Table 32-16. The PSDU length shall be 2048 octets for BPSK-DCM or 4096 octets for all other modulations.

Table 32-16—Receiver minimum input level sensitivity

Modulation	Rate (R)	Minimum sensitivity (10 MHz PPDU) (dBm)	Minimum sensitivity (20 MHz PPDU) (dBm)
BPSK-DCM	1/2	−88	−82
BPSK	1/2	−85	−82
QPSK	1/2	−82	−79
QPSK	3/4	−80	−77
16-QAM	1/2	−77	−74
16-QAM	3/4	−73	−70
64-QAM	2/3	−69	−66
64-QAM	3/4	−68	−65
64-QAM	5/6	−67	−64
256-QAM	3/4	−62	−59
256-QAM	5/6	N/A	−57

32.3.12.3 Adjacent channel rejection

Adjacent channel rejection for W MHz channels (where W is 10 or 20) shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in 32.3.12.2 and raising the power of the interfering signal of W MHz bandwidth until 10% PER is caused for a PSDU length of 2048 octets for BPSK-DCM or 4096 octets for all other modulations. The difference in power between the signals in the interfering channel and the desired channel is the corresponding adjacent channel rejection. The center frequency of the adjacent channel shall be placed W MHz away from the center frequency of the desired signal.

The interfering signal in the adjacent channel shall be a signal compliant with the NGV PHY, unsynchronized with the signal in the channel under test, and shall have a minimum duty cycle of 50%. The corresponding rejection shall be no less than specified in Table 32-17.

An optional enhanced performance specification is provided for systems requiring improved immunity to out-of-channel interfering emissions. If a STA has dot11ACRType equal to 2, the adjacent channel rejection shall be no less than specified in Table 32-18. The interfering signal in the adjacent channel shall be a signal compliant with the NGV PHY, using class C transmit mask for 10 MHz channel and C2 transmit mask for 20 MHz channel (see D.2.2), unsynchronized with the signal in the channel under test. The corresponding minimum receiver sensitivities for each modulation and coding rate are the same as in Table 32-16.

Table 32-17—Minimum required adjacent and nonadjacent channel rejection level

Modulation	Rate (R)	Adjacent channel rejection (dB) (10 MHz/20 MHz PPDU)	Nonadjacent channel rejection (dB) (10 MHz/20 MHz PPDU)
BPSK-DCM	1/2	19 (10 MHz), 16 (20 MHz)	35 (10 MHz), 32 (20 MHz)
BPSK	1/2	16	32
QPSK	1/2	13	29
QPSK	3/4	11	27
16-QAM	1/2	8	24
16-QAM	3/4	4	20
64-QAM	2/3	0	16
64-QAM	3/4	−1	15
64-QAM	5/6	−2	14
256-QAM	3/4	−7	9
256-QAM	5/6	−9	7
NOTE—The values defined for 256-QAM with 5/6 coding rate only apply to 20 MHz channel.			

Table 32-18—Optional enhanced minimum required adjacent and nonadjacent channel rejection level

Modulation	Rate (R)	Adjacent channel rejection (dB) (10 MHz/20 MHz PPDU)	Nonadjacent channel rejection (dB) (10 MHz/20 MHz PPDU)
BPSK-DCM	1/2	31 (10 MHz), 28 (20 MHz)	45 (10 MHz), 42 (20 MHz)
BPSK	1/2	28	42
QPSK	1/2	25	39
QPSK	3/4	23	37
16-QAM	1/2	20	34
16-QAM	3/4	16	30
64-QAM	2/3	12	26
64-QAM	3/4	11	25
64-QAM	5/6	10	24
256-QAM	3/4	5	19
256-QAM	5/6	3	17
NOTE—The values defined for 256-QAM with 5/6 coding rate only apply to 20 MHz channel.			

32.3.12.4 Nonadjacent channel rejection

Nonadjacent channel rejection for W MHz channels (where W is 10 or 20) shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in Table 32-16, and raising the power of the interfering signal of W MHz bandwidth until a 10% PER occurs for a PSDU length of 2048 octets for BPSK-DCM or 4096 octets for all other modulations. The difference in power between the signals in the interfering channel and the desired channel is the corresponding nonadjacent channel rejection. The nonadjacent channel rejection shall be met with any nonadjacent channels located at least $2 \times W$ MHz away from the center frequency of the desired signal.

The interfering signal in the nonadjacent channel shall be a signal compliant with the NGV PHY, unsynchronized with the signal in the channel under test, and shall have a minimum duty cycle of 50%. The corresponding rejection shall be no less than specified in Table 32-17.

An optional enhanced performance specification is provided for systems requiring improved immunity to out-of-channel interfering emissions. If a STA has `dot11ACRType` equal to 2, the nonadjacent channel rejection shall be no less than specified in Table 32-18. The interfering signal in the nonadjacent channel shall be a signal compliant with the NGV PHY, using class C transmit mask for 10 MHz channel and C2 transmit mask for 20 MHz channel (see D.2.2), unsynchronized with the signal in the channel under test. The corresponding minimum receiver sensitivities for each modulation and coding rate are the same as in Table 32-16.

32.3.12.5 Receiver maximum input level

The receiver shall provide a maximum PER of 10% at a PSDU length of 2048 octets for BPSK-DCM or 4096 octets for all other modulations, for a maximum input level of -30 dBm, measured at each antenna connector for any baseband NGV modulation.

32.3.12.6 CCA sensitivity

32.3.12.6.1 General

The thresholds in this subclause are compared with the signal level at each receiving antenna.

32.3.12.6.2 CCA sensitivity for signals occupying the primary 10 MHz channel

An NGV STA with a W MHz operating channel width shall detect, with $> 90\%$ probability, the start of a PPDU that occupies at least the primary 10 MHz channel in an otherwise idle W MHz operating channel width, and issue a `PHY CCA.indication(BUSY, {primary})` primitive within a period of `aCCATime` (see 17.4.4) if one of the following conditions is met:

- The start of a `NON_NGV_10` PPDU as defined in 17.3.10.6.
- The start of a 20 MHz non-NGV duplicate or NGV PPDU for which the power measured within the primary 10 MHz channel is above -85 dBm.

The CCA signal shall be held busy (`PHY-CCA.indication(BUSY, {primary})` primitive) for the duration of the PPDU.

The receiver shall issue a `PHY-CCA.indication(BUSY, {primary})` primitive for any signal that exceeds a threshold of -65 dBm in the primary 10 MHz channel within a period of `aCCATime` after the signal arrives at the receiver's antenna(s); then the receiver shall not issue a `PHY-CCA.indication(BUSY, {secondary})`, or `PHYCCA.indication(IDLE)` primitive while the threshold continues to be exceeded.

32.3.12.6.3 CCA sensitivity for signals not occupying the primary 10 MHz channel

The PHY shall issue a PHY-CCA.indication(BUSY, {secondary}) primitive if the conditions for issuing a PHY-CCA.indication(BUSY, {primary}) primitive are not present and one of the following conditions are present in an otherwise idle 20 MHz operating channel width:

- Any signal within the secondary 10 MHz channel at or above a threshold of -65 dBm within a period of aCCATime after the signal arrives at the receiver's antenna(s); then the PHY shall not issue a PHY-CCA.indication(IDLE) primitive while the threshold continues to be exceeded.
- A 10 MHz NON_NGV, or NGV PPDU detected in the secondary 10 MHz channel at or above -85 dBm with $> 90\%$ probability within a period aCCAMidTime (see 32.4.4).

32.3.13 NGV transmit procedure

There are two paths for the transmit PHY procedure:

- The first path, for which typical transmit procedures are shown in Figure 32-13, is selected if the FORMAT parameter of the PHY-TXSTART.request(TXVECTOR) primitive is NGV. These transmit procedures do not describe the operation of optional features, such as SU-MIMO.
- The second path is selected if the FORMAT parameter of the PHY-TXSTART.request(TXVECTOR) primitive is NON_NGV_10. Transmit parameter N_PPDU_REP is set via the PHY service interface using the PHY-TXSTART.request(TXVECTOR) primitive, as described in Table 32-1 to indicate the number of repetitions of NON_NGV_10 PPDU transmission.

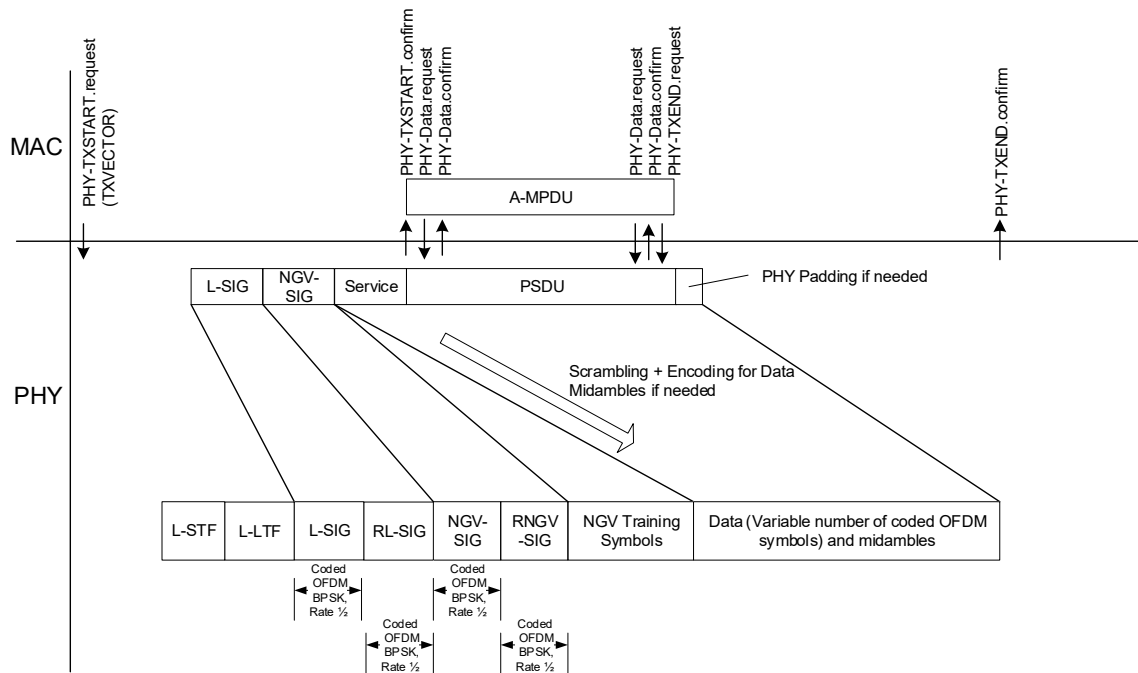


Figure 32-13—PHY transmit procedure for NGV transmission

In the second path, the transmission follows the transmit procedure in Clause 17 except that the signal referred to in Clause 17 is instead generated with the Duration/ID field to report transmitter is an NGV STA as defined in 31.2.1. If N_PPDU_REP is nonzero, the first NON_NGV_10 PPDU may be stored in the memory, and the same PPDU is transmitted N_PPDU_REP times after the initial transmission, with each repetition occurring after a SIFS following the previous NON_NGV_10 PPDU transmission.

In both paths, in order to transmit data, the MAC generates a PHY-TXSTART.request primitive, that causes the PHY entity to enter the transmit state. Further, the PHY is set to operate at the appropriate frequency through station management via PLME, as specified in 32.4. Other transmit parameters, such as NGV-MCS, Coding types, and transmit power, are set via the PHY SAP using the PHY-TXSTART.request(TXVECTOR) primitive, as described in Table 32-1. The remainder of the clause applies to the first path.

Transmission of the PPDU shall be initiated by the PHY after receiving the PHY-TXSTART.request(TXVECTOR) primitive. The TXVECTOR parameters for the PHY-TXSTART.request primitive are specified in Table 32-1.

Transmission of the PHY preamble shall start immediately provided that TIME_OF_DEPARTURE_REQUESTED is true, based on the parameters passed in the PHY-TXSTART.request primitive.

After the PHY preamble transmission is started, the PHY entity immediately initiates data scrambling and data encoding. The encoding method for the Data field is based on the CH_BANDWIDTH, NUM_SS, and NGV-MCS parameters of the TXVECTOR, as described in 32.3.2.

The SERVICE field and PSDU are scrambled and encoded as described in 32.3.3. The data shall be exchanged between the MAC and the PHY through a series of PHY-DATA.request(DATA) primitives issued by the MAC, and PHY-DATA.confirm primitives issued by the PHY. PHY padding bits are appended to the PSDU to make the number of bits in the coded SERVICE field and PSDU an integer multiple of the number of coded bits per OFDM symbol. Midambles are inserted every M OFDM symbols if present, where M is indicated by the Midamble Periodicity field in the NGV-SIG field.

Transmission can be prematurely terminated by the MAC through the PHY-TXEND.request primitive. PSDU transmission is terminated by receiving a PHY-TXEND.request primitive. Each PHY-TXEND.request primitive is acknowledged with a PHY-TXEND.confirm primitive from the PHY. Normal termination occurs after the transmission of the final bit of the last PSDU octet, according to the number of OFDM symbols indicated by N_{SYM} (see 21.3.10.5.4).

In the PHY, the GI is inserted in every data OFDM symbol as a countermeasure against delay spread.

When the PPDU transmission is completed the PHY entity enters the receive state.

A typical state machine implementation of the transmit PHY is provided in Figure 32-14. Request (.request) and confirmation (.confirm) primitives are issued as shown. This state machine does not describe the operation of optional features, such as SU-MIMO.

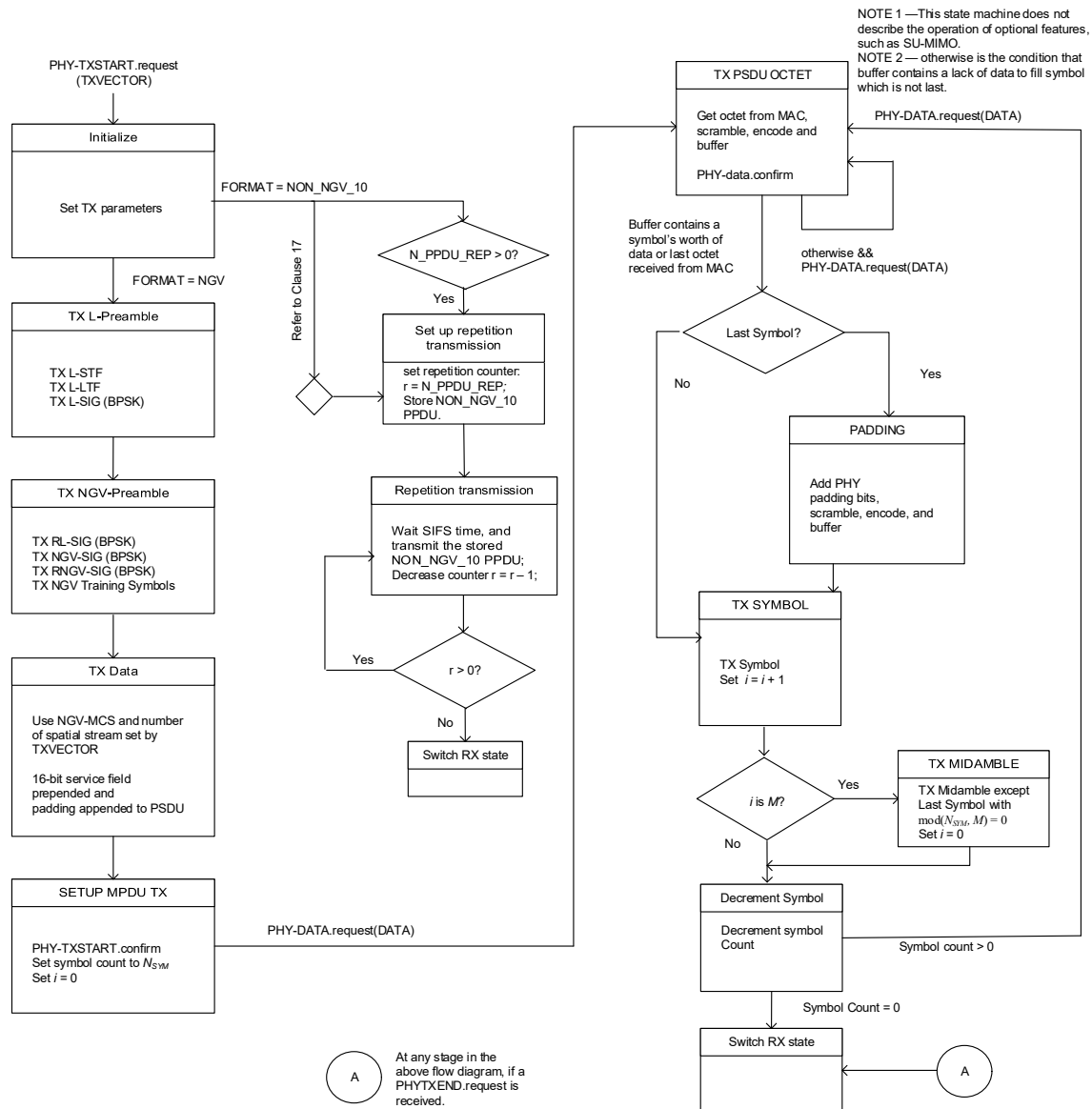


Figure 32-14—PHY transmit state machine

32.3.14 NGV receive procedure

A typical PHY receive procedure is shown in Figure 32-15 for NGV format. A typical state machine implementation of the receive PHY is given in Figure 32-16. This receive procedure and state machine do not describe the operation of optional features, such as SU-MIMO. If the detected format indicates a NON_NGV_10 PPDU:

- If combined detection of NON_NGV_10 PPDU with repetition is not considered, refer to the receive procedure and state machine in Clause 17.
- Else the receiver checks whether the currently received PPDU is qualified as a repetition of previously received NON_NGV_10 PPDU. The repetition may be determined by checking whether the gap between the time of arrival of the currently detected PPDU and the ending time of the

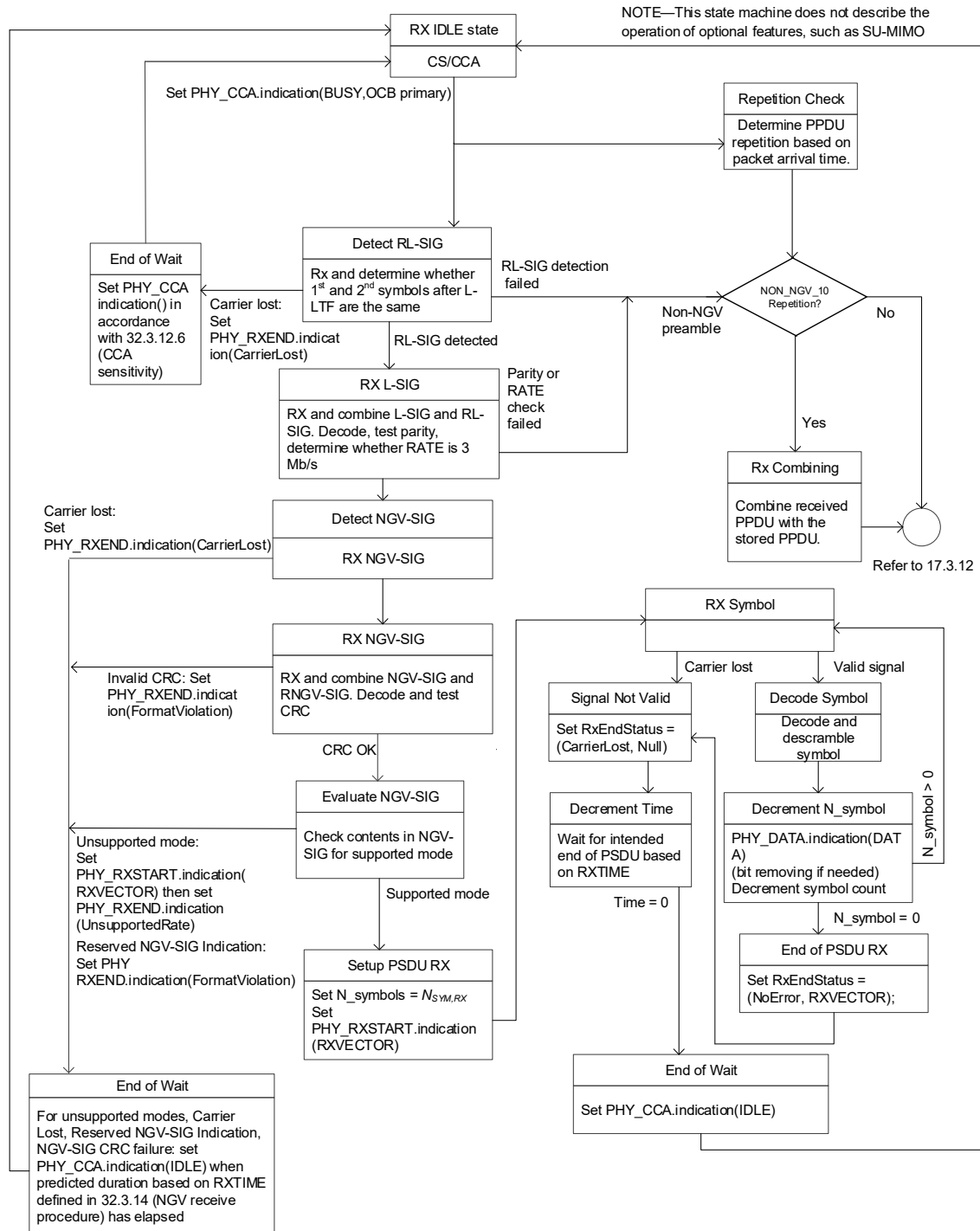


Figure 32-16—PHY receive state machine

The PHY shall not issue a PHY-RXSTART.indication primitive in response to a PPDU that does not overlap the OCB primary 10 MHz channel.

The PHY includes the most recently measured RSSI value in the PHY-RXSTART.indication(RXVECTOR) primitive issued to the MAC.

After the PHY-CCA.indication(BUSY, channel-list) primitive is issued, the PHY entity shall begin receiving the training symbols and searching for the preambles for non-NGV and NGV PPDU. For detecting the NGV preamble, the PHY entity shall search for L-SIG and RL-SIG in order to set the maximum duration of the data stream. If RL-SIG is detected, the PHY entity should check the parity bit and RATE fields in L-SIG and RL-SIG. If RL-SIG detection fails, the check of the parity bit is invalid, or the RATE field is not equal to 3 Mb/s, a PHY-RXSTART.indication primitive is not issued, and instead the PHY should continue to detect the received signal using NON_NGV_10 Repetition check. If a valid parity bit and the RATE with 3 Mb/s are indicated in L-SIG and RL-SIG, the NGV PHY shall maintain the PHY-CCA.indication(BUSY, channel-list) primitive for the predicted duration of the transmitted PPDU, as defined by RXTIME in Equation (32-41), for all supported modes, unsupported modes, Reserved NGV-SIG Indication, and invalid SIG CRC. Reserved NGV-SIG Indication is defined as an NGV-SIG with the Reserved bits equal to 0 or any other NGV-SIG field bit combinations that do not correspond to modes of PHY operation defined in Clause 32. If the NGV-SIG indicates an unsupported mode, the PHY shall issue a PHY-RXEND.indication(UnsupportedRate) primitive. If the NGV-SIG indicates an invalid CRC or Reserved NGV-SIG indication, the PHY shall issue the error condition PHY-RXEND.inidcation(FormatViolation).primitive.

After receiving a valid L-SIG, RL-SIG, NGV-SIG, and RNGV-SIG indicating a supported mode, the PHY entity shall begin receiving the NGV training symbols.

The PHY entity shall check the PHY Version in the NGV-SIG field. If the PHY Version does not contain an intended value with 0 for NGV PHY in Table 32-10, the PHY entity shall issue a PHY-RXSTART.indication(RXVECTOR) and then issue a PHY-RXEND.indication(Filtered).

Following training fields, the Data field shall be received. The number of symbols in the Data field is determined by Equation (32-39).

$$N_{SYM,RX} = \begin{cases} N'_{SYM,RX}, & \text{if LDPC Extra Symbol} = 0 \\ N'_{SYM,RX} - 1, & \text{if LDPC Extra Symbol} = 1 \end{cases} \quad (32-39)$$

where

LDPC Extra Symbol is a field in the NGV-SIG field (see Table 32-10).

$$N'_{SYM,RX} = \left\lfloor \left(\frac{(L_LENGTH + 3) \times 8}{3} - T_{NGV-PREAMBLE} - N_{MA,RX} \times N_{NGV-LTF} \times T_{NGV-LTF} \right) / T_{SYM} \right\rfloor$$

where

L_LENGTH is the LENGTH field in L-SIG

$$T_{NGV-PREAMBLE} = T_{RL-SIG} + T_{NGV-SIG} + T_{RNGV-SIG} + T_{NGV-STF} + N_{NGV-LTF} \times T_{NGV-LTF}$$

T_{RL-SIG} , $T_{NGV-SIG}$, $T_{RNGV-SIG}$, $T_{NGV-STF}$, and $T_{NGV-LTF}$, are defined in Table 32-6

$N_{NGV-LTF}$ is defined in Table 32-7

$N_{MA,RX}$ may be computed by multiple methods that get the same results, an example method is provided in Equation (32-40)

$$N_{MA,RX} = \left\lfloor \frac{\frac{(L_LENGTH + 3) \times 8}{3} - T_{NGV-PREAMBLE} - T_{SYM}}{N_{NGV-LTF} \times T_{NGV-LTF} + M \times T_{SYM}} \right\rfloor \quad (32-40)$$

where M is the midamble periodicity as defined in Table 32-10.

If signal loss occurs during reception prior to completion of the PSDU reception, the error condition shall be reported to the MAC using a PHY-RXEND.indication(CarrierLost) primitive. After waiting for the intended end of the PSDU as determined by Equation (32-41), the PHY shall generate a PHY-CCA.indication(IDLE) primitive and return to RX IDLE state.

$$\text{RXTIME} = \left\lceil \frac{(\text{L_LENGTH} + 3)}{3} \right\rceil \times 8 + 40 \quad (32-41)$$

The received PSDU bits are assembled into octets, decoded, and presented to the MAC using a series of PHY-DATA.indication(DATA) primitive exchanges. Any final bits that cannot be assembled into a complete octet are considered pad bits and discarded. After the reception of the final bit of the last PSDU octet, and possible padding, the receiver shall be returned to the RX IDLE state, as shown in Figure 32-16. A PHY-RXEND.indication(NoError) primitive shall be issued on entry to the RX IDLE state.

32.3.15 Regulatory requirements

Wireless LANs (WLANs) implemented in accordance with this standard are subject to equipment certification and operating requirements established by regional and national regulatory administrations. The PHY specification establishes minimum technical requirements for interoperability, based upon established regulations at the time this standard was issued. These regulations are subject to revision or may be superseded. Requirements that are subject to local geographic regulations are annotated within the PHY specification. Regulatory requirements that do not affect interoperability are not addressed in this standard. Implementers are referred to the regulatory sources in Annex D for further information. Operation in countries within defined regulatory domains might be subject to additional or alternative national regulations.

32.3.16 NGV ranging NDP

The NGV ranging NDP is a variant of NGV PPDU but without the Data field. The format of an NGV ranging NDP is shown in Figure 32-17.

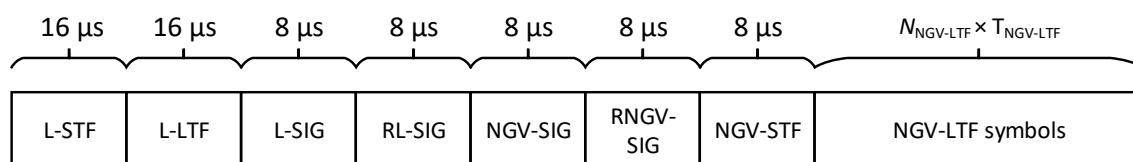


Figure 32-17—NGV ranging NDP format

The NGV ranging NDP has the following properties:

- The TXVECTOR parameter NUM_SS is used to encode the NSS field of the NGV-SIG and RNGV-SIG.
- The TXVECTOR parameter LTF_REP indicates the number of repetitions of the NGV-LTF symbols.
- The TXVECTOR parameter NGV-MCS shall be set to 0.

In an NGV ranging NDP, the NGV-LTF symbol format shall be NGV-LTF-2x.

The number of NGV-LTF symbols in an NGV ranging NDP is the product of the TXVECTOR parameter NUM_SS and (LTF_REP + 1), see Table 32-11. The number of LTF repetitions equals LTF_REP. When LTF repetition is used, LTF_REP is 1. Otherwise, LTF_REP is 0.

The construction of the NGV-LTFs in an NGV ranging NDP is done by performing the steps in 32.3.4.7 LTF_REP + 1 times. An example of NGV_LTF with two spatial streams and one LTF repetition is shown in Figure 32-18.

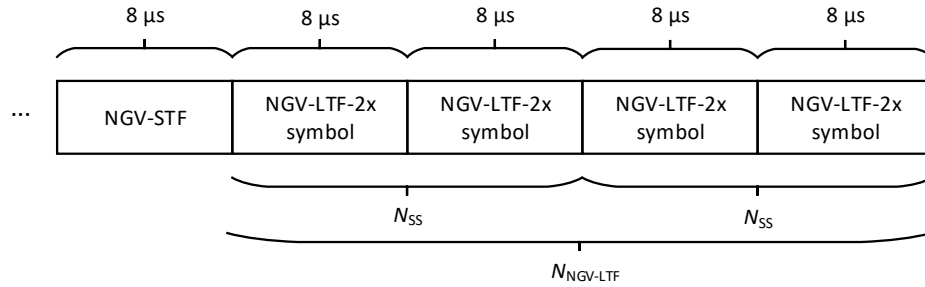


Figure 32-18—Example of NGV-LTF with $N_{SS} = 2$ and $LTF_REP = 1$

32.4 NGV PLME

32.4.1 PLME-SAP

Table 32-19 lists the MIB attributes that may be accessed by the PHY entities and the intralayer of higher level LMEs. These attributes are accessed via the PLME-GET, PLME-SET, PLME-RESET, and PLME-CHARACTERISTICS primitives defined in 6.5.

Table 32-19—NGV PHY MIB attributes

Managed object	Default value/range	Operational semantics
dot11PHYOperationTable		
dot11PHYType	ngv	Static
dot11PHYTxPowerTable		
dot11NumberSupportedPowerLevelsImplemented	Implementation dependent	Static
dot11TxPowerLevel1	Implementation dependent	Static
dot11TxPowerLevel2	Implementation dependent	Static
dot11TxPowerLevel3	Implementation dependent	Static
dot11TxPowerLevel4	Implementation dependent	Static
dot11TxPowerLevel5	Implementation dependent	Static
dot11TxPowerLevel6	Implementation dependent	Static
dot11TxPowerLevel7	Implementation dependent	Static
dot11TxPowerLevel8	Implementation dependent	Static
dot11CurrentTxPowerLevel	Implementation dependent	Static

Table 32-19—NGV PHY MIB attributes (*continued*)

Managed object	Default value/range	Operational semantics
dot11TxPowerLevelExtended	Implementation dependent	Static
dot11PHYNGVTable		
dot11CurrentChannelWidth	Implementation dependent	Dynamic
dot11CurrentPrimaryChannel	Implementation dependent	Dynamic
dot11CurrentSecondaryChannel	Implementation dependent	Dynamic
dot11NGVDYN20MAllowed	Implementation dependent	Static

32.4.2 PHY MIB

NGV PHY MIB attributes are defined in C.3 with specific values defined in Table 32-19. The “Operational semantics” column in Table 32-19 contains two types: Static and Dynamic.

- Static MIB attributes are fixed and cannot be modified for a given PHY implementation.
- Dynamic MIB attributes are interpreted according to the MAX-ACCESS field of the MIB attribute. When MAX-ACCESS is read-only, the MIB attribute value may be updated by the PLME and read from the MIB attribute by management entities. When MAX-ACCESS is read-write, the MIB attribute may be read and written by management entities but shall not be updated by the PLME.

32.4.3 TXTIME and PSDU_LENGTH calculation

The value of the TXTIME parameter returned by the PLME-TXTIME.confirm primitive shall be calculated for an NGV PPDU using Equation (32-42).

$$\text{TXTIME} = 40 + T_{RL-SIG} + T_{NGV-SIG} + T_{RNGV-SIG} + T_{NGV-STF} + N_{SYM} \times T_{SYM} + 8 \times \left\lceil \frac{N_{NGV-LTF} \times T_{NGV-LTF} + N_{MA} \times N_{NGV-LTF} \times T_{NGV-LTF}}{8} \right\rceil \quad (32-42)$$

where

T_{RL-SIG} , $T_{NGV-SIG}$, $T_{RNGV-SIG}$, $T_{NGV-STF}$, and T_{SYM} are defined in Table 32-6

$N_{NGV-LTF}$ is defined in Table 32-7

N_{MA} is defined in 32.3.9.9

For an NGV PPDU using LDPC encoding, the total number of data symbols in the Data field, N_{SYM} , is given in 21.3.10.5.4 and computed using Equation (19-41) in step d) of 19.3.11.7.5.

The value of the PSDU_LENGTH parameter returned in the PLME-TXTIME.confirm primitive for an NGV SU PPDU using LDPC encoding is calculated using Equation (32-43).

$$\text{PSDU_LENGTH} = \left\lceil \frac{N_{SYM,init} N_{DBPS} - N_{service}}{8} \right\rceil \quad (32-43)$$

where

$N_{SYM,init}$ is given by Equation (21-62) with $m_{STBC} = 1$.

32.4.4 NGV PHY

The static NGV PHY characteristics, provided through the PLME-CHARACTERISTICS service primitive, shall be as shown in Table 32-20 with 10 MHz channel spacing unless otherwise listed in Table 32-20. The definitions for these characteristics are given in 6.5.4.

Table 32-20—NGV PHY characteristics

Characteristics	Value
aSlotTime	If dot11OperatingClassesRequired is false, 13 μ s. If dot11OperatingClassesRequired is true, 13 μ s plus any coverage-class-dependent aAirPropagationTime (see Table 9-95).
aSIFSTime	32 μ s
aCCAMidTime	45 μ s
aPPDUMaxTime	10.968 ms
aPSDUMaxLength	244 800 octets (see NOTE 1)
aRxPHYStartDelay	64 μ s (see NOTE 2)
NOTE 1—This is the maximum length in octets for an NGV PPDU with a bandwidth of 20 MHz, NGV-MCS 9, and two spatial streams, and limited by 1360 possible data symbols in aPPDUMaxTime. This is the maximum PSDU length an NGV PHY could support assuming no restrictions in MAC. See 10.3.2 and 9.2.4.7.1 for additional restrictions on the maximum number of octets the MAC could support. NOTE 2—This value arises from the time to the end of RNGV-SIG (see Figure 32-5).	

32.5 Parameters for NGV-MCSs

The rate-dependent parameters for 10 MHz and 20 MHz are given in Table 32-21 to Table 32-24.

Table 32-21—NGV-MCSs for 10 MHz, $N_{SS} = 1$

NGV-MCS index	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s) 1.6 μ s GI NGV-LTF-2x
0	BPSK	1/2	1	52	52	26	3.3
1	QPSK	1/2	2	52	104	52	6.5
2	QPSK	3/4	2	52	104	78	9.8
3	16-QAM	1/2	4	52	208	104	13.0
4	16-QAM	3/4	4	52	208	156	19.5
5	64-QAM	2/3	6	52	312	208	26.0
6	64-QAM	3/4	6	52	312	234	29.3
7	64-QAM	5/6	6	52	312	260	32.5
8	256-QAM	3/4	8	52	416	312	39.0
9	Not valid						
10–14	Reserved						
15	BPSK-DCM	1/2	1	26	26	13	1.6

Table 32-22—NGV-MCSs for 10 MHz, $N_{SS} = 2$

NGV-MCS index	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s) 1.6 μ s GI NGV-LTF-2x
0	BPSK	1/2	1	52	104	52	6.5
1	QPSK	1/2	2	52	208	104	13.0
2	QPSK	3/4	2	52	208	156	19.5
3	16-QAM	1/2	4	52	416	208	26.0
4	16-QAM	3/4	4	52	416	312	39.0
5	64-QAM	2/3	6	52	624	416	52.0
6	64-QAM	3/4	6	52	624	468	58.5
7	64-QAM	5/6	6	52	624	520	65.0
8	256-QAM	3/4	8	52	832	624	78.0
9	Not valid						
10–15	Reserved						

Table 32-23—NGV-MCSs for 20 MHz, $N_{SS} = 1$

NGV-MCS index	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s) 1.6 μ s GI NGV-LTF-2x
0	BPSK	1/2	1	108	108	54	6.8
1	QPSK	1/2	2	108	216	108	13.5
2	QPSK	3/4	2	108	216	162	20.3
3	16-QAM	1/2	4	108	432	216	27.0
4	16-QAM	3/4	4	108	432	324	40.5
5	64-QAM	2/3	6	108	648	432	54.0
6	64-QAM	3/4	6	108	648	486	60.8
7	64-QAM	5/6	6	108	648	540	67.5
8	256-QAM	3/4	8	108	864	648	81.0
9	256-QAM	5/6	8	108	864	720	90.0
10–14	Reserved						
15	BPSK-DCM	1/2	1	54	54	27	3.4

Table 32-24—NGV-MCSs for 20 MHz, $N_{SS} = 2$

NGV-MCS index	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s) 1.6 μ s GI NGV-LTF-2x
0	BPSK	1/2	1	108	216	108	13.5
1	QPSK	1/2	2	108	432	216	27.0
2	QPSK	3/4	2	108	432	324	40.5
3	16-QAM	1/2	4	108	864	432	54.0
4	16-QAM	3/4	4	108	864	648	81.0
5	64-QAM	2/3	6	108	1296	864	106.0
6	64-QAM	3/4	6	108	1296	972	121.5
7	64-QAM	5/6	6	108	1296	1080	135.0
8	256-QAM	3/4	8	108	1728	1296	162.0
9	256-QAM	5/6	8	108	1728	1440	180.0
10–15	Reserved						

Annex B

(normative)

Protocol Implementation Conformance Statement (PICS) proforma

B.2 Abbreviations and special symbols

B.2.2 General abbreviations for Item and Support columns

Insert the following abbreviation in B.2.2 in alphabetical order:

NGV Next Generation V2X

B.4 PICS proforma—IEEE Std 802.11-2020⁷

*Change the rows for items *CFDMG, *CFOCB, *CFEDMG, and *CFNTB and insert two new rows at the end of the table in B.4.3 as follows (unchanged rows not shown):*

B.4.3 IUT configuration

Item	IUT configuration	References	Status	Support
...				
*CFDMG	Directional multi-gigabit (DMG) PHY	9.4.2.127	O.2 CFDMG:M CFEDMG:M CFTDD:M <u>CFNGV60:M</u>	Yes ☐ No ☐
...				
*CFOCB	Operation outside the context of a BSS (OCB)	11.18	O.1 CF5G9:M <u>CFNGV:M</u> <u>CFNGV60:M</u>	Yes ☐ No ☐
...				
*CFEDMG	Enhanced directional multi-gigabit (EDMG) PHY	9.4.2.265	CFEDMG:M <u>CFNGV60:O</u>	Yes ☐ No ☐
...				
*CFNTB	Support for FTM non-TB sounding	11.21.6.4.4	CFHE:O <u>CFNGV:O</u>	Yes ☐ No ☐
...				
*CFNGV	Next generation V2X (NGV) operation in the 5.9 GHz band	31.2	O	Yes ☐ No ☐

⁷Copyright release for PICS proforma: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

B.4.3 IUT configuration (*continued*)

Item	IUT configuration	References	Status	Support
*CFNGV60	Next generation V2X (NGV) operation in the 60 GHz band	31.3	O	Yes ☐ No ☐

B.4.24 DMG features

Change the row for item DMG-M15.1.2 in the table in B.4.24.1 as follows (unchanged rows not shown):

B.4.24.1 DMG MAC features

Item	Protocol capability	References	Status	Support
...				
DMG-M15.1.2	Beamformed link maintenance procedure	11.27.1	DMG-M15.1.1:O <u>NGVM1.2:M</u>	Yes ☐ No ☐ N/A ☐
...				

Insert B.4.38 after B.4.37 (inserted by IEEE Std 802.11az-2022) as follows:

B.4.38 NGV features

B.4.38.1 NGV MAC features

Item	Protocol capability	References	Status	Support
	Are the following MAC protocol features supported?			
NGVM1	Operating band			
NGVM1.1	NGV operation in the 5.9 GHz band	31.2	CFNGV:M	Yes ☐ No ☐ N/A ☐
NGVM1.2	NGV operation in the 60 GHz band	31.3	CFNGV60:M	Yes ☐ No ☐ N/A ☐
NGVM2	NGV fallback procedure			
NGVM2.1	20 MHz transmission fallback to 10 MHz	31.2.2	CFNGV:O	Yes ☐ No ☐ N/A ☐
NGVM3	DMG operation			
NGVM3.1	DMG OCB element	9.4.2.308	CFNGV60:M	Yes ☐ No ☐ N/A ☐
NGVM3.2	DMG beamforming outside the context of a BSS	31.3.1	CFNGV60:M	Yes ☐ No ☐ N/A ☐
NGVM4	NGV ranging			
NGVM4.1	Support for NGV ranging	31.4	CFNGV:O	Yes ☐ No ☐ N/A ☐
*NGVM4.2	Support for NGV ranging in the 5.9 GHz frequency band	31.4	(CFNGV AND CFNTB):O	Yes ☐ No ☐ N/A ☐

B.4.38.2 NGV PHY features

Item	Protocol capability	References	Status	Support
	Are the following PHY protocol features supported?			
NGVP1	Operating bandwidth			
*NGVP1.1	10 MHz operation in the 5.9 GHz band	32.5	CFNGV:M	Yes ☐ No ☐ N/A ☐
*NGVP1.2	20 MHz operation in the 5.9 GHz band	32.5	CFNGV:O	Yes ☐ No ☐ N/A ☐
NGVP2	Modulation and coding schemes (MCS)			
NGVP2.1	NGV-MCS with Index = 0–8, or 15 and 10 MHz and $N_{SS} = 1$	32.5	NGVP1.1:M	Yes ☐ No ☐ N/A ☐
NGVP2.2	NGV-MCS with Index = 0–8 and 10 MHz and $N_{SS} = 2$	32.5	NGVP1.1:O	Yes ☐ No ☐ N/A ☐
NGVP2.3	NGV-MCS with Index = 0–9, or 15 and 20 MHz and $N_{SS} = 1$	32.5	NGVP1.2:M	Yes ☐ No ☐ N/A ☐
NGVP2.4	NGV-MCS with Index = 0–9 and 20 MHz and $N_{SS} = 2$	32.5	NGVP1.2:O	Yes ☐ No ☐ N/A ☐
NGVP3	Transmit spectrum masks			
NGVP3.1	Spectrum mask, Class A (10 MHz channel spacing)	D.2.3	CFNGV:O	Yes ☐ No ☐ N/A ☐
NGVP3.2	Spectrum mask, Class B (10 MHz channel spacing)	D.2.3	CFNGV:O	Yes ☐ No ☐ N/A ☐
NGVP3.3	Spectrum mask, Class C (10 MHz channel spacing)	D.2.3	NGVP1.1:M	Yes ☐ No ☐ N/A ☐
NGVP3.4	Spectrum mask C2, Class C (20 MHz channel spacing)	D.2.3	NGVP1.2:M	Yes ☐ No ☐ N/A ☐
NGVP3.5	Spectrum mask, Class D (10 MHz channel spacing)	D.2.3	CFNGV:O	Yes ☐ No ☐ N/A ☐
NGVP4	NGV PPDU format			
NGVP4.1	Midambles	32.3	CFNGV:M	Yes ☐ No ☐ N/A ☐
NGVP4.2	LDPC support	32.3	CFNGV:M	Yes ☐ No ☐ N/A ☐
NGVP4.3	NGV ranging NDP	31.4	NGVM4.2:M	Yes ☐ No ☐ N/A ☐

B.4.38.3 NGV extended MAC service features

Item	Feature	References	Status	Support
	Are the following NGV extended MAC service features supported?			
NGVE1	NGV extended MAC service features			
NGVE1.1	NGV extended MAC service features— MLME; 5.9 GHz band (MLME-CANCELTX)	10.2.3.2	CFNGV:M	Yes ☐ No ☐ N/A ☐
NGVE1.2	NGV extended MAC service features— MLME; DMG (MLME-DMG-OCB-START, MLME-DMG-OCB-STOP, MLME-OCB-DMGDISCOVERY, MLME-OCB-LINKSTATUS)	11.1.4, 11.27	CFNGV60:M	Yes ☐ No ☐ N/A ☐

Annex C

(normative)

ASN.1 encoding of the MAC and PHY MIB

C.3 MIB detail

Change the item “dot11UnsolicitedBAAActivated” (inserted by IEEE Std 802.11ay-2021) at the end of the list in the object definition that begins with “Dot11StationConfigEntry ::=” in C.3 and insert five new items after it as follows:

dot11UnsolicitedBAAActivated	TruthValue,
dot11NONNGVRadioEnvironmentImplemented	TruthValue,
dot11NGVOptionImplemented	TruthValue,
dot11RadioEnvironmentMeasurementPeriod	Unsigned32,
dot11VirtualCSonOCBSecondaryImplemented	TruthValue,
dot11StationMeasurementPeriod	Unsigned32

Insert the following definitions in C.3 above the three lines containing “End of dot11StationConfig TABLE” in { dot11StationConfigEntry n } order:

```
dot11NONNGVRadioEnvironmentImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable. Its value is determined by device
        capabilities. A non-NGV STA supports the radio environment transmit vector
        and the radio environment status vector when this attribute is true."
    ::= { dot11StationConfigEntry 219 }
```

```
dot11NGVOptionImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable. Its value is determined by device
        capabilities. This attribute, when true, indicates that the STA supports
        NGV features. This STA also has dot11OCBActivated equal to true."
    ::= { dot11StationConfigEntry 203 }
```

```
dot11RadioEnvironmentMeasurementPeriod OBJECT-TYPE
    SYNTAX Unsigned32 (0..1000)
    UNITS "milliseconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable. It is written by an external management
        entity or the SME. Changes take effect as soon as practical in the
        implementations. This attribute indicates the amount of time an NGV STA
        conducts the radio measurement before reporting its radio environment.
        When equal to 0, measurements are not made."
    DEFVAL { 0 }
    ::= { dot11StationConfigEntry 204 }
```

```
dot11VirtualCSonOCBSecondaryImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable. Its value is determined by device
        capabilities. This attribute, when true, indicates that virtual CS
        operation in the OCB secondary channel is implemented."
    ::= { dot11StationConfigEntry 220 }

dot11StationMeasurementPeriod OBJECT-TYPE
    SYNTAX Unsigned32 (0..1000)
    UNITS "milliseconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable. It is written by an external management
        entity or the SME. Changes take effect as soon as practical in the
        implementations. This attribute indicates the amount of time an NGV STA
        conducts the STA measurement before reporting its radio environment. When
        equal to 0, measurements are not made."
    DEFVAL { 0 }
    ::= { dot11StationConfigEntry 221 }
```

Change the item “dot11DMGSTATxActivityReportActivated” (inserted by IEEE Std 802.11ay-2021) at the end of the list in the object definition that begins with “Dot11DMGSTAConfigEntry ::=” in C.3 and insert a new item after it as follows:

dot11DMGSTATxActivityReportActivated	TruthValue,
dot11DMGOCBActivated	TruthValue

Insert new text after the object definition that begins with “dot11DMGSTATxActivityReportActivated OBJECT-TYPE” (inserted by IEEE Std 802.11ay-2021) in C.3 as follows:

```
dot11DMGOCBActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by the SME or an external management entity.
        Changes take effect at the next occurrence of an MLME-START.request or
        MLME-DMG-OCB-START.request primitive.

        This attribute, when true, indicates that the STA supports communication
        outside the context of a BSS. This STA also has dot11OCBActivated equal to
        true."
    DEFVAL { false }
    ::= { dot11DMGSTAConfigEntry 17 }
```

Change the object definition that begins with “dot11PHYType OBJECT-TYPE” in C.3 as follows:

```
dot11PHYType OBJECT-TYPE
    SYNTAX INTEGER {
        dsss(2),
        ofdm(4),
        hrdsss(5),
        erp(6),
        ht(7),
        dm(8),
```

```

    vht(9),
    tvht(10),
    slg(11),
    cdmg(12),
    cmmg(13),
    he(14),
    edmg(15),
    ngv(17)}
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a status variable.
    It is written by the PHY.

    This is an 8-bit integer value that identifies the supported PHY type.
    Currently defined values and their corresponding PHY types are:

    DSSS 2.4 GHz = 2, OFDM = 4, HRDSSS = 5, ERP = 6, HT = 7, DMG = 8, VHT = 9,
    TVHT = 10, SlG = 11, CDMG = 12, CMMG = 13, HE = 14, EDMG = 15, NGV = 17."
::= { dot11PhyOperationEntry 1 }

```

Insert the following in C.3 after the three lines containing “End of dot11 TVHT Transmit Beamforming Config TABLE”:

```

-- *****
-- * dot11 Phy NGV TABLE
-- *****

dot11PhyNGVTable OBJECT-TYPE
    SYNTAX SEQUENCE OF Dot11PhyNGVEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "Entry of attributes for dot11PhyNGVTable. Implemented as a table indexed
        on ifIndex to allow for multiple instances on an Agent."
    ::= { dot11phy 37 }

dot11PhyNGVEntry OBJECT-TYPE
    SYNTAX Dot11PhyNGVEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "An entry in dot11PhyNGVEntryTable. ifIndex - Each IEEE Std 802.11
        interface is represented by an ifEntry. Interface tables in this MIB
        module are indexed by ifIndex."
    INDEX {ifIndex}
    ::= { dot11PhyNGVTable 1 }

Dot11PhyNGVEntry ::= SEQUENCE
{
    dot11NGVCurrentChannelWidth          INTEGER,
    dot11NGVCurrentPrimaryChannel        Unsigned32,
    dot11NGVCurrentSecondaryChannel      Unsigned32,
    dot11NGVDYN20MAAllowed               TruthValue
}

dot11NGVCurrentChannelWidth OBJECT-TYPE
    SYNTAX INTEGER { cbw10(0), cbw20(1) }
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a status variable.

```

Written by the PHY.

This attribute indicates the operating channel width."
DEFVAL { cbw10 }
::= { dot11PhyNGVEntry 1 }

dot11NGVCurrentPrimaryChannel OBJECT-TYPE

SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a status variable.
It is written by the PHY.

This attribute indicates the operating channel. If 10/20 MHz OCB is currently in use then this attribute indicates the primary channel."
::= { dot11PhyNGVEntry 2 }

dot11NGVCurrentSecondaryChannel OBJECT-TYPE

SYNTAX Unsigned32
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a status variable.
It is written by the PHY.

This attribute indicates the channel number of the secondary channel. If 10/20 MHz OCB is not currently in use, this attribute value shall be 0."
::= { dot11PhyNGVEntry 3 }

dot11NGVDYN20MAllowed OBJECT-TYPE

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the NGV STA implementation supports dynamic 20 MHz operation. This capability is disabled otherwise."
::= { dot11PhyNGVEntry 6 }

Change the item “dot11DMGSTATxActivityReportActivated” (inserted by IEEE Std 802.11ay-2021) at the end of the list in the object definition that begins with “dot11DMGComplianceGroup OBJECT-GROUP” in C.3 and insert a new item after it as follows:

dot11DMGSTATxActivityReportActivated,
dot11DMGOCBActivated

Insert new text after the object definition that begins with “dot11CDMGComplianceGroup1 OBJECT-GROUP” in C.3 as follows:

dot11NGVComplianceGroup OBJECT-GROUP

OBJECTS {
dot11NGVOptionImplemented,
dot11RadioEnvironmentMeasurementPeriod,
dot11VirtualCSonOCBSecondaryImplemented,
dot11NONNGVRadioEnvironmentImplemented,
dot11StationMeasurementPeriod }
STATUS current
DESCRIPTION

```
"Attributes that configure the NGV Group for IEEE Std 802.11."  
::= { dot11Groups 125 }
```

```
dot11PhyNGVComplianceGroup OBJECT-GROUP  
OBJECTS {  
    dot11NGVCurrentChannelWidth,  
    dot11NGVCurrentPrimaryChannel,  
    dot11NGVCurrentSecondaryChannel,  
    dot11NGVDYN20MAllowed }  
STATUS current  
DESCRIPTION  
    "Attributes that configure the NGV PHY."  
::= { dot11Groups 126 }
```

Change the beginning of “dot11Compliance MODULE-COMPLIANCE” in the “Compliance Statements” part of C.3 as follows:

```
dot11Compliance MODULE-COMPLIANCE  
    STATUS current  
    DESCRIPTION  
        "The compliance statement for SNMPv2 entities that implement the IEEE  
        802.11 MIB."  
    MODULE -- this module  
    MANDATORY-GROUPS {  
        dot11SMTbase16,  
        dot11MACbase5,  
        dot11CountersGroup5,  
        dot11SmtAuthenticationAlgorithms,  
        dot11ResourceTypeID,  
        dot11PhyOperationComplianceGroup2,  
        dot11EDMGComplianceGroup }  
  
    GROUP dot11PhyDSSSSComplianceGroup  
    DESCRIPTION  
        "Implementation of this group is required when object dot11PHYType is  
        dsss.  
        This group is mutually exclusive to the following groups:  
        dot11PhyOFDMComplianceGroup3  
        dot11PhyHRDSSSSComplianceGroup  
        dot11PhyERPComplianceGroup  
        dot11PhyHTComplianceGroup  
        dot11DMGComplianceGroup  
        dot11PhyVHTComplianceGroup  
        dot11PhyTVHTComplianceGroup  
        dot11PhyS1GComplianceGroup  
        dot11CDMGComplianceGroup1  
        dot11CMMGComplianceGroup  
        dot11PhyHEComplianceGroup  
        dot11EDMGComplianceGroup  
        dot11PhyNGVComplianceGroup"  
  
    GROUP dot11PhyOFDMComplianceGroup3  
    DESCRIPTION  
        "Implementation of this group is required when object dot11PHYType is  
        ofdm.  
        This group is mutually exclusive to the following groups:  
        dot11PhyDSSSSComplianceGroup  
        dot11PhyHRDSSSSComplianceGroup  
        dot11PhyERPComplianceGroup  
        dot11PhyHTComplianceGroup  
        dot11DMGComplianceGroup  
        dot11PhyVHTComplianceGroup  
        dot11PhyTVHTComplianceGroup
```

dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11EDMGComplianceGroup
dot11PhyNGVComplianceGroup"

GROUP dot11PhyHRDSSSComplianceGroup

DESCRIPTION

"Implementation of this group is required when object dot11PHYType is hrdsss.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11EDMGComplianceGroup
dot11PhyNGVComplianceGroup"

GROUP dot11PhyERPComplianceGroup

DESCRIPTION

"Implementation of this group is required when object dot11PHYType is erp.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11EDMGComplianceGroup
dot11PhyNGVComplianceGroup"

GROUP dot11PhyHTComplianceGroup

DESCRIPTION

"Implementation of this group is required when object dot11PHYType is ht.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11EDMGComplianceGroup
dot11PhyNGVComplianceGroup"

GROUP dot11PhyVHTComplianceGroup

DESCRIPTION

"Implementation of this group is required when object dot11PHYType is vht.
This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11DMGComplianceGroup
dot11PhyHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11EDMGComplianceGroup
dot11PhyNGVComplianceGroup"

GROUP dot11PhyTVHTComplianceGroup

DESCRIPTION

"Implementation of this group is required when object dot11PHYType is
tvht.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11EDMGComplianceGroup
dot11PhyNGVComplianceGroup"

GROUP dot11PhyS1GComplianceGroup

DESCRIPTION

"Implementation of this group is required when object dot11PHYType is slg.
This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11PhyNGVComplianceGroup"

GROUP dot11PhyHEComplianceGroup

DESCRIPTION

"Implementation of this group is required when object dot11PHYType has the
value of HE.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup

```
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyNGVComplianceGroup"
```

GROUP dot11PhyNGVComplianceGroup
DESCRIPTION

"Implementation of this group is required when the object dot11PHYType has the value of ngv.
 This group is mutually exclusive to the following groups:
dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11HTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup
dot11EDMGComplianceGroup"

Change the object definition that begins with “GROUP dot11EDMGComplianceGroup” (inserted by IEEE Std 802.11ay-2021) in C.3 as follows:

GROUP dot11EDMGComplianceGroup
 DESCRIPTION

"Implementation of this group is required when the object dot11PHYType has the value of EDMG.
 This group is mutually exclusive to the following groups:
 dot11PhyDSSSComplianceGroup
 dot11PhyOFDMComplianceGroup3
 dot11PhyHRDSSSComplianceGroup
 dot11PhyERPComplianceGroup
 dot11PhyHTComplianceGroup
 dot11PhyVHTComplianceGroup
 dot11PhyTVHTComplianceGroup
dot11PhyNGVComplianceGroup"

Change the object definition that begins with “GROUP dot11DMGComplianceGroup” (as amended by IEEE Std 802.11ax-2021) in C.3 as follows:

GROUP dot11DMGComplianceGroup
 DESCRIPTION

"Implementation of this group is required when the object dot11PHYType is dmg.
 This group is mutually exclusive to the following groups:
 dot11PhyDSSSComplianceGroup
 dot11PhyOFDMComplianceGroup3
 dot11PhyHRDSSSComplianceGroup
 dot11PhyERPComplianceGroup
 dot11PhyHTComplianceGroup
 dot11PhyVHTComplianceGroup
 dot11PhyTVHTComplianceGroup

```
dot11PhyS1GComplianceGroup  
dot11CDMGComplianceGroup1  
dot11CMMGComplianceGroup  
dot11PhyHEComplianceGroup  
dot11PhyNGVComplianceGroup"
```

Change the object definition that begins with “GROUP dot11CDMGComplianceGroup1” in C.3 as follows:

```
GROUP dot11CDMGComplianceGroup1  
DESCRIPTION  
    "Implementation of this group is required when the object  
    dot11PHYType has the value of CDMG.  
    This group is mutually exclusive to the following groups:  
    dot11PhyDSSSComplianceGroup  
    dot11PhyOFDMComplianceGroup3  
    dot11PhyHRDSSSComplianceGroup  
    dot11PhyERPComplianceGroup  
    dot11PhyHTComplianceGroup  
    dot11PhyVHTComplianceGroup  
    dot11PhyTVHTComplianceGroup  
    dot11DMGComplianceGroup  
    dot11CMMGComplianceGroup  
    dot11PhyNGVComplianceGroup"
```

Change the object definition that begins with “GROUP dot11CMMGComplianceGroup” in C.3 as follows:

```
GROUP dot11CMMGComplianceGroup  
DESCRIPTION  
    "Implementation of this group is required when the object  
    dot11PHYType has the value of CMMG.  
    This group is mutually exclusive to the following groups:  
    dot11PhyDSSSComplianceGroup  
    dot11PhyOFDMComplianceGroup3  
    dot11PhyHRDSSSComplianceGroup  
    dot11PhyERPComplianceGroup  
    dot11PhyHTComplianceGroup  
    dot11PhyVHTComplianceGroup  
    dot11PhyTVHTComplianceGroup  
    dot11DMGComplianceGroup  
    dot11CDMGComplianceGroup  
    dot11PhyNGVComplianceGroup"
```

Insert the following text into “dot11Compliance MODULE-COMPLIANCE” before the line “-- OPTIONAL-GROUPS {“ in C.3:

```
GROUP dot11NGVComplianceGroup  
DESCRIPTION  
    "NGV Compliance Group"
```

Insert the following text after “Compliance Statements - WUR” (inserted by IEEE Std 802.11ba-2021) in C.3:

```
-- *****
-- * Compliance Statements - NGV
-- *****

dot11NGVCompliance MODULE-COMPLIANCE
    STATUS current
    DESCRIPTION
        "This object class provides the objects from IEEE 802.11
        MIB required to manage Next Generation V2X functionality."
    MODULE - this module
    MANDATORY-GROUPS {
        dot11NGVComplianceGroup,
        dot11PhyNGVComplianceGroup }
-- OPTIONAL-GROUPS { }
    ::= { dot11Compliances 27 }
```

Annex D

(normative)

Regulatory references

D.2 Radio performance specifications

D.2.3 Transmit spectrum mask

Change the second paragraph of D.2.3 as follows:

For operation in the 5.85–5.925 GHz band the transmitted spectrum shall be as follows:

- a) For any STA using 5 MHz channel spacing, the transmitted spectral density shall have a 0 dBr bandwidth not exceeding 4.5 MHz and shall not exceed the spectrum mask created using the permitted power spectral density levels listed in Table D-5 for the transmit power class of the STA.
- b) For any STA using 10 MHz channel spacing, the transmitted spectral density shall have a 0 dBr bandwidth not exceeding 9 MHz and shall not exceed the spectrum mask created using the permitted power spectral density levels listed in Table D-6 for the transmit power class of the STA.
- c) For any STA using 20 MHz channel spacing and transmitting a PPDU that is not a 20 MHz NGV PPDU or 20 MHz non-NGV duplicate PPDU, the transmitted spectral density shall have a 0 dBr bandwidth not exceeding 18 MHz and shall not exceed the spectrum mask created using the permitted power spectral density levels listed in Table D-7 for the transmit power class of the STA.

NOTE—The conditions under which conformance to transmit spectral mask C2 is required are specified in 32.3.11.1.

Annex P

(informative)

Location and Time Difference accuracy test

P.3 Differential Distance Computation using Fine Timing Measurement frames

Change the two equations in P.3 to give them equation numbers as follows:

$$D_{SR} = c \times (T_{SO} - T_{RO}) \quad \text{(P-1)}$$

$$D_{SR} = c \times (t_{c1} - t_{c2} - T - (t_1 - t_4)) \quad \text{(P-2)}$$

Insert the following paragraph and Figure P-2 at the end of P.3 as follows:

In Figure P-2 the observing or passive STA (PSTA) that supports NGV non-TB ranging is able to listen to the NGV non-TB ranging measurement exchange between ISTA and RSTA. The time of flight of a line of sight transmission between the ISTA and RSTA is denoted as T . At the PSTA, the TOAs of the I2R NDP and the R2I NDP are respectively t_{c1} and t_{c2} . At the ISTA, the TOD of the I2R NDP and the TOA of the R2I NDP are t_1 and t_4 , respectively. D_{SR} can now be computed according to Equation (P-2). The PSTA might obtain t_1 and t_4 by receiving the I2R LMR frame or by higher layer information exchange.

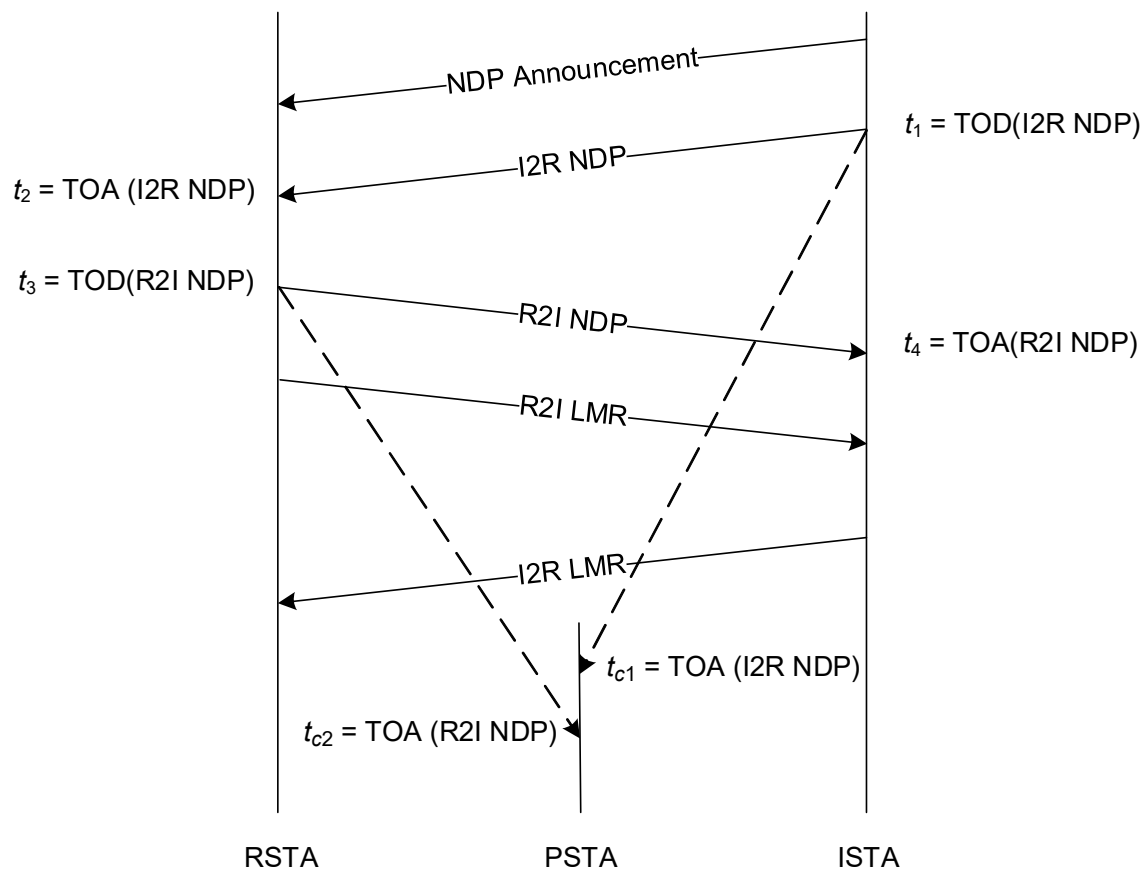


Figure P-2—Parameters recorded by PSTA when monitoring non-TB ranging measurement exchange

RAISING THE WORLD'S STANDARDS

Connect with us on:

-  **Twitter:** twitter.com/ieeesa
-  **Facebook:** facebook.com/ieeesa
-  **LinkedIn:** linkedin.com/groups/1791118
-  **Beyond Standards blog:** beyondstandards.ieee.org
-  **YouTube:** youtube.com/ieeesa

standards.ieee.org
Phone: +1 732 981 0060